



Machine learning and physics-based model hybridization to assess the impact of climate change on single- and dual-axis tracking solar-concentrated photovoltaic systems

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ABSTRACT

This study explores the hybridization of machine learning models with analytical techniques to evaluate the impact of climate change on active power predictions for single-axis (PAS) and dual-axis (PAD) tracking solar-concentrated photovoltaic (CPV) systems in Alice Springs, Australia. It integrates meteorological factors, including global solar irradiation, extraterrestrial solar radiation, clearness index, ambient temperature, and relative humidity, to assess their influence on system performance. Descriptive statistics for PAS and PAD reveal slight variations in these factors, with PAD experiencing higher values for solar irradiation and ambient temperature. To enhance forecasting accuracy, machine learning models, including SARIMA, CARIMA, MLP, RBF, boosting (BOT), and bagging (BAG), are utilized to predict future performance of CPV systems in Alice Springs. These models consider historical weather data and system performance metrics to make accurate predictions for optimal system operation and maintenance. A novel hybrid model, CARIMA-SARIMA-LG, integrates SARIMA, CARIMA, and the logistic distribution (LG) model, showing exceptional performance. The CARIMA-SARIMA-LG model achieves R^2 values of 0.9833% for PAS and 0.9777% for PAD during training, surpassing other models in error metrics such as MAPE, RMSE, and nRMSE. In contrast, traditional machine learning models like MLP and RBF exhibit diminished predictive capabilities. Ensemble methods, while useful, do not achieve the accuracy of statistical models. Furthermore, the study evaluates the efficacy of the CARIMA-SARIMA-LG hybrid model across five African locations (Kano, Accra, Johannesburg, Nairobi, and Dar es Salaam), demonstrating PAD's adaptability to diverse climates. The findings illustrate PAD's outstanding accuracy and adaptability across a variety of climatic situations, providing breakthrough insights for renewable energy generation and grid integration. The influence of changing climate conditions on solar photovoltaic systems in Alice Springs, Australia, across SSP126, SSP245, and SSP585 scenarios demonstrates notable differences in active power generation for both single-axis tracking (PAS) and dual-axis tracking (PAD) systems. Under SSP126, the projections indicate slight variations in power, peaking in DJF with an increase of +3.145% from 2015 to 2030, followed by a decrease of -3.775% in the late century from 2071 to 2099. PAD systems show comparable seasonal patterns, experiencing increases in the initial phases (+5.463%, DJF, 2015–2030) followed by declines as the century progresses (-0.578%, ANN, 2071–2099). Under the SSP245 scenario, moderate losses are projected, with PAS facing a decrease of -2.138% (annual average, 2031–2050) and PAD encountering significant declines by the period of 2071–2099, amounting to -5.001% (annual average). SSP585 indicates the most drastic declines, with PAS facing a reduction of -3.294% (ANN, 2015–2030) and PAD seeing a decrease of -5.731% (ANN, 2015–2030), intensifying energy security challenges amid persistently elevated emissions. The findings highlight the critical importance of imple-

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menting robust mitigation measures, fostering technological innovations, and developing adaptive energy strategies to secure reliable solar energy production in the face of changing climate conditions. Advancements in tracking systems and energy storage technologies may mitigate certain adverse effects associated with climate change, especially in more severe scenarios such as SSP585.

1. Introduction

Over the past twenty years, there has been growing interest among researchers, investors, government entities, the scientific community, and global leaders in the development and utilization of clean energy, particularly from renewable or nature-based sources. Among various renewable energy or nature-based clean energy sources, peers, reviewers, and experts in optical materials science and solar energy Experts suggest that the solar photovoltaic (PV) model stands out as the most effective renewable energy source for electricity generation while ensuring minimal environmental emissions (Potrč et al., 2021). Recent technological innovations have significantly accelerated the methods for improving the energy efficiency of photovoltaic energy as it is harnessed and converted from traditional solar sources (Kumar and Shivashankar, 2022). To effectively utilize and convert modest quantities of daily sunlight and solar heat into useable energy to meet energy demands, a variety of laboratory and field experiments have been conducted recently. The latest developments in optical analysis and scientific research are progressively improving overall energy efficiency through meticulous data analysis and result transformations (Yang et al., 2021). The efficient transformation and conversion of solar heat into energy addresses the challenges of energy shortages, thus paving the way for a more sustainable and environmentally friendly energy future.

At present, traditional methods for solar conversion demonstrate a notably low efficiency, falling below or around 40% (Ding et al., 2021). The limited effectiveness observed in traditional solar conversion methods stems from the sun's frequent directional shifts that occur seasonally (Mamur et al., 2022). To address the issue, a robust solar mechanism is essential to improving the energy efficiency of the system. Generally, the photovoltaic modules are anchored to the latitude of the site and oriented towards either the northern or southern hemisphere, contingent upon the specific experimental arrangement on the Earth's surface. Traditionally, the orientation of the solar module is adjusted towards the sun either manually, when feasible (Jouttijärvi et al., 2022), or via mechatronic methods (Obiwulu et al., 2022).

To achieve optimal power generation, the photovoltaic system should be perpendicular to the incident light beam. Therefore, it is essential to create a solar tracking method that maintains the photovoltaic modules in a perpendicular alignment to the light beam, particularly if the goal is to enhance energy efficiency and boost overall energy output (Karimzadeh Kolamroudi et al., 2022). A solar tracking system represents a method designed to anticipate the sun's trajectory, enabling enhanced energy production that surpasses traditional or fixed photovoltaic modules by 30–60% (Hadroug et al., 2023). Over the past twenty years, a variety of methods have been developed to accurately forecast the sun's trajectory. Still, these methods need difficult mathematical functions and precise optical measurements of different solar energy properties because the optical materials and analysis are so complicated (Ali Jallal et al., 2020).

To effectively forecast or predict the appropriate azimuth and configurations of the photovoltaic module, it is crucial to accurately determine fundamental parameters in the solar-energy module, such as Greenwich time, solar time, and solar irradiance. At present, there are two distinct categories of solar tracking models that have entered the commercial market. These encompass both the active and passive models. The differentiation of these models lies in the methodologies employed for control. In active tracking models, microprocessors play a

crucial role in controlling the direction and movement of modules via gears and motors (El Hammouti et al., 2022). In passive tracking models, the liquid undergoes heating and expansion into a gaseous state due to solar radiation. The sun's thermal radiation causes a liquid to vaporize in a passive solar tracking device, generating pressure gradients that facilitate movement and rotation. This occurs as denser fluids are directed into the shaded vessel, resulting in a shift of weight to the opposite side (Chang et al., 2020). The intricate nature of the solar system may be linked to the various axes needed for the movement of PV modules, whether in a vertical, horizontal, or combined manner. There are primarily two categories: dual-axis tracking photovoltaic systems (PAD) and single-axis tracking systems (PAS).

Gearboxes, linear actuators, or direct current (DC) motors are usually used as mechanical drivers in systems that use PAD or PAS models (Lu and Ajay, 2024). The components of the PAS model are adjusted around a singular axis to ensure they are positioned orthogonally to the sun's radiation (Pouladian-Kari et al., 2022). In the PAS system, the north meridian axis is typically regarded as the optical tilt angle; a specific solar PAS system was developed, which operates potentially three times daily (Hariri et al., 2022). Recent experimental analysis indicates that this tracking model (PAS) can achieve 90–94% of the solar energy output when compared to a parallel, continuous solar tracking model (Das et al., 2023).

The PAS models are typically arranged in vertical, horizontal, or polar configurations, or horizontally with inclined units. The PAS design typically delivers an increase of 20% in power output compared to traditional fixed modules (Lu and Ajay, 2024). The fundamental concept of the PAS model revolves around employing the PID controller to regulate the solar module. Furthermore, in comparison to the PAD, the PAS models demonstrate greater cost efficiency and reduced complexity. Nonetheless, the PAD models demonstrate superior effectiveness compared to the PAS systems, offering higher energy output and seamless integration into current power grids. This positions them as a practical solution for large-scale energy production and a more sustainable technology to address the clean energy requirements of both urban and rural regions. Despite the development of various solar-tracking models, challenges remain regarding their efficiency and power output.

This section reviews a variety of studies concerning solar tracking systems that utilize either PAS or PAD configurations. Muthukumar et al. (2023) implemented an energy-efficient dual-axis solar tracking system that enables the module to rotate around both its horizontal and vertical axes, thereby optimizing energy output. The authors utilize a microcontroller to manage the positioning of the solar module, which is linked to a Wi-Fi system for the purpose of monitoring the outcomes. Ultimately, the authors disclosed that the utilization of the internet of things (IoT) enabled the monitoring of peak power from the solar module, allowing for the precise operation of the PAD system in response.

In a recent study, Lu and Ajay used an innovative hybrid arithmetic optimization-based PID (AOPID) controller to change the solar module's orientation in relation to the micro-electromechanical solar (MEMS) and UV sensor (Lu and Ajay, 2024). They did this to try to predict where the sun would go in the sky. The findings indicate that the proposed tracking system has led to both increases and decreases in power production and consumption. Investigations into photovoltaic systems indicate that solar tracking models are crucial for enhancing energy output.

Jamroen et al. (2020) developed a model that combines dual-axis tracking with ultraviolet (UV) sensors to enhance the efficiency of

photovoltaic systems by leveraging UV radiation. This method demonstrated superior tracking efficiency relative to stationary flat-plate models, leading to increased energy production. Nonetheless, the significant expense and susceptibility of UV sensors to weather conditions pose challenges, underscoring the necessity for more affordable and weather-resistant options. Additionally, [Jamroen et al. \(2020\)](#) introduced a dual-axis model utilizing light dependent resistor (LDR) sensors, crafted as a cost-effective and automated approach for solar tracking. The model demonstrated a 44.89% increase in power generation compared to conventional fixed systems; however, it overlooked environmental considerations, highlighting the critical need to incorporate uncertainties in future designs.

Numerous investigations have delved into sophisticated methodologies aimed at improving the efficiency of solar tracking systems. Nadia AL-Rousan et al. created adaptive neural fuzzy inference systems (ANFIS) for the designs of both single-axis tracking (PAS) and dual-axis tracking (PAD) ([AL-Rousan et al., 2020](#)). This approach enhanced the precision of monitoring the sun's path, consequently maximizing energy collection. While the performance is notably advanced, the substantial implementation costs associated with specialized hardware and software present considerable challenges. In a similar vein, Dadi and Peravali developed a dual-axis tracking system designed to enhance power output by maximizing the capture of sunlight's intensity ([Dadi and Peravali, 2020](#)). This model, while enhancing power production, simultaneously led to higher energy consumption, highlighting the necessity for designs that optimize the balance between energy generation and usage.

Innovative advancements like deep learning and machine learning have been integrated into solar tracking systems, yielding encouraging outcomes. Ramaneti et al. used deep learning techniques to improve the alignment of solar panels with the sun's path. This led to a 10.6% increase in power output and a 97% success rate in tracking ([Ramaneti et al., 2021](#)). Nonetheless, this system necessitates extensive data for optimal training, presenting difficulties in the collection and processing of that data. Fernández-Ahumada et al. addressed the challenge of shadows affecting energy production through the creation of a novel model designed to optimize panel positioning and reduce shading effects ([Fernández-Ahumada et al., 2020a](#)). This approach resulted in a 1.31% boost in energy output; however, it proved to be time-intensive because of the intricate nature of shadow assessment.

Additional research has concentrated on improving the immediate detection functionalities of solar trackers. To improve dual-axis tracking systems, Mohammed Ali Jallal and his colleagues used particle swarm optimization (PSO) and deep neural networks together ([Ali Jallal et al., 2020](#)). The model demonstrated superior accuracy compared to current methodologies; however, the heightened hardware demands and related expenses pose challenges for broader implementation. In the same way, Fernández-Ahumada et al. improved photovoltaic efficiency by reducing shading between collectors in a dual-axis configuration ([Fernández-Ahumada et al., 2020a](#)), but they had to be carefully installed to avoid alignment problems. These methods demonstrate that although optimization strategies can greatly improve energy capture, the associated trade-offs in cost, hardware complexity, and time continue to pose significant challenges.

The enhancement of solar tracking mechanisms within microgrid systems has been investigated as well. Mah et al. established a framework aimed at enhancing energy control strategies in independent microgrids, incorporating both electrical and hydrogen demands ([Mah et al., 2021](#)). The application of the PSO technique led to a remarkable 90% decrease in energy expenses. While these results are encouraging, the model overlooked uncertainties related to solar tracking, highlighting a potential avenue for enhancement moving forward. Zhu et al. presented a SAT model that employs intelligent tracking algorithms, leveraging solar radiation and geometric relationships ([Zhu et al., 2020](#)). While this system enhanced energy efficiency, the intricate calculations

resulted in longer processing times, underscoring the necessity for more efficient algorithms. These studies highlight the essential importance of sophisticated tracking models in enhancing the efficiency of PV systems while also recognizing cost, data requirements, and complexity as persistent challenges.

The results of the studies that looked into how to improve the PV solar-tracking PAS and PAD systems to get the most energy out of them all came to the same conclusion: while there were some problems, most of the established frameworks worked pretty well. However, almost all of the authors said that using IoT ([Muthukumar et al., 2023](#)), AVPID and MEMS ([Lu and Ajay, 2024](#)), ANFIS ([AL-Rousan et al., 2020](#)), the PSO algorithm ([Ali Jallal et al., 2020](#)), and the electrical and hydrogen load in the microgrid supported by the PSO algorithm ([Mah et al., 2021](#)) would require more hardware and software, which would make things more expensive ([Ali Jallal et al., 2020](#)). The proposed PAD system requires meticulous installation to guarantee correct alignment and functionality ([Fernández-Ahumada et al., 2020a](#)), while the optimized framework fails to account for the uncertainties present in solar tracking systems ([Mah et al., 2021](#)). Researchers have tried a lot of different methods to fix the energy efficiency problems with PAD and PAS solar systems, but the results show that these problems still exist even though there are universal ways to make PAD and PAS more efficient. Even in a time when machine learning, IoT, and artificial intelligence are prevalent, this ultimately has an impact on the potential for increased energy output production. This does not imply that the strategies discussed in the literature for enhancing the efficiency of PAD and PAS are unfeasible or non-transferable across different locations. However, greater effort should be directed by researchers and industry to further optimize the efficiency of PAD and PAS systems, especially in a time when global warming is intensifying the impacts of climate change. Currently, research and development initiatives are focused on tackling these challenges to enhance the efficiency and cost-effectiveness of PAD and PAS solar plants ([Nwokolo et al., 2024](#)).

According to Nwokolo et al., the implementation of concentrated solar PV systems in areas without a proper assessment of their potential could result in sizable financial losses, particularly in regions with low solar irradiance and unpredictable weather patterns ([Nwokolo et al., 2024](#)). Therefore, assessing the viability or feasibility of energy outputs from PAS and PAD at a specific location is essential to prevent unnecessary losses in investments related to PAD and PAS. Consequently, a precise prediction model stands as the sole viable approach for conducting these feasibility studies in a financially efficient manner. A precise forecasting model for PAD and PAS can offer significant insights into the economic feasibility of solar installations, aiding investors in making well-informed choices.

Nonetheless, creating precise PAD and PAS models necessitates considering all pertinent factors that could influence the efficiency and financial viability of solar plants in this age of sustainability. This approach will significantly reduce the risks linked to investments in PAD and PAS techniques, thereby ensuring the long-term financial viability of a solar project. This approach can also refine the design and functioning of PAD and PAS systems, resulting in enhanced efficiency and energy output performance, which can subsequently boost their economic viability and competitiveness within the renewable energy sector. This can offer critical insights for decision-makers and investors to guide their choices regarding the implementation of PAD and PAS technologies.

A variety of researchers have sought to create an effective model, utilizing either empirical or artificial neural network approaches, to forecast the energy output of concentrated photovoltaic systems in studies published from 2012 to 2024. The need to achieve net zero emissions, driven by the effects of climate change and the drive for global sustainability, is a major driving force behind this effort. The cited works include work from Sonia-Moya et al. ([Soria-Moya et al., 2015](#)), Fernandez et al. ([Fernández et al., 2014a, 2014b](#)), and Gadeo-

Martos et al. (2019), as well as our brand-new CARIMA-SARIMA-GPM hybrid model created for estimating CPV solar systems. Nevertheless, ANN methods, similar to other computational simulation techniques found in the literature, do not possess the capability to provide accurate prediction estimates. In contrast, empirical methods have demonstrated tendencies to either underestimate or overestimate solar energy resources (Nwokolo et al., 2023). In our latest study, we utilized the CARIMA-SARIMA-GPM hybrid model to forecast an unidentified type of CPV module (Nwokolo et al., 2024). This study aimed to address a notable research gap by simulating a high-performing CARIMA-SARIMA-LG hybrid model. The model is designed to provide estimates for evaluating the potential energy output of sensitive PAD and PAS systems that have inadvertently emerged in Alice Springs, Australia.

Given the urgent need to understand how renewable energy systems will react to changing environmental conditions, the impact of climate change on PAD and PAS systems has become a crucial area of study. Feron et al. (2020) emphasize the importance of thorough investigations into the responses of the PAD and PAS systems to climate change, especially regarding long-term development, socioeconomic assessment, and environmental impact. Global and regional climate models play a crucial role in these studies, aiding in the prediction of essential variables such as temperature, precipitation, and solar irradiance, all of which have a direct impact on the performance of PAD and PAS systems. Nonetheless, the intricate relationships between climate change and CPV systems necessitate a more comprehensive viewpoint, considering elements like land use alterations, water resource availability, and severe weather occurrences. Grasping these intricate dynamics is crucial for formulating adaptation strategies that will guarantee the sustainability of PAD and PAS systems in light of future climate scenarios.

While PAD and PAS systems play a crucial role in addressing the impacts of climate change, there is still a notable deficiency in thorough research that combines various predictive models to effectively evaluate the potential of these systems. The CARIMA-SARIMA-LG multi-model approach is a new way to fill this gap. It gives a full picture of how climate dynamics and the potential of PAD and PAS systems work together by combining models that look at weather patterns, solar radiation, and atmospheric conditions. This methodology enables scientists to enhance the precision of long-term forecasts regarding the performance of PAD and PAS systems across various climate scenarios, providing essential insights for decision-makers and industry participants. Through the integration of these models, researchers are able to more accurately assess the effects of climate change on PAD and PAS systems while also refining the technology to maintain its efficacy as climatic conditions shift.

Historical studies, including the work of Crook et al. (2011), have shown that the impact of climate change on photovoltaic (PV) systems differs markedly across various regions. For example, a reduction in photovoltaic output of 2%–6% is anticipated in Saudi Arabia and the United States, potentially linked to changes in climatic conditions or energy regulations. Conversely, a modest rise in photovoltaic output may be observed in China and Europe. Nonetheless, systems that utilize concentrated solar power, which depend significantly on direct solar insolation, are expected to show more significant fluctuations in their output. The distinct regional differences highlight the necessity for tailored approaches to mitigate the effects of climate change on solar energy systems. Panagea et al. (2014) underscore the profound consequences of climate change for Greece, where escalating temperatures and heightened solar radiation are anticipated to intensify the impacts of climate change. However, the enhanced photovoltaic yield may play a role in facilitating the nation's shift towards renewable energy sources.

In the context of Europe, Jerez et al. employed regional climate models (RCMs) to investigate the implications of climate change on solar PV generation (Jerez et al., 2015). Their findings indicate that by 2100, northern Europe is likely to face a decline in PV generation attrib-

uted to diminished solar insolation, whereas southern Europe may see a modest increase in yields. This underscores the significance of localized studies in comprehending the intricate impacts of climate change on renewable energy infrastructures. In a similar vein, Feron et al. identified notable regional disparities in photovoltaic performance on a global scale, with declines recorded in the Arabian Peninsula (4%) and southeastern Australia (2%), whereas central Europe and certain areas of Asia exhibited moderate gains (Feron et al., 2020). The observed variations highlight the intricate relationships among environmental shifts, geographical factors, and the potential for energy production, all of which are critical to address in forthcoming energy strategies and policies.

The effects of climate change on solar photovoltaic systems reach far beyond Europe and North America. Zuluaga et al. conducted an analysis of solar PV projections in Brazil, revealing that northern Brazil may face a 3% increase in warming, whereas the remainder of the country is expected to experience a 5% decline in solar PV potential by the century's end (Zuluaga et al., 2022). The reduction in solar energy output may lead to considerable implications for the area's energy stability and economic progress, especially given the rise in severe weather occurrences. In a similar vein, Bazyomo et al. (2016) and Fant et al. (2016) observed varied results for West and Southern Africa, indicating that shifts in weather patterns could result in diminished photovoltaic generation during the summer while potentially enhancing output in the winter. This presents avenues for innovative technologies and strategies aimed at maximizing solar energy utilization throughout the year.

In-depth examination of climate models, including CMIP5, has highlighted significant apprehensions regarding the prospects of solar energy across the African continent. Zou et al. indicate that diminished global solar radiation could significantly affect photovoltaic power generation in West Africa, worsening energy poverty and heightening dependence on fossil fuels (Zou et al., 2019). This highlights the critical need for investments in renewable energy in areas susceptible to climate change, where diminished solar output may jeopardize efforts to satisfy energy requirements while simultaneously lowering carbon emissions. Gaetani et al. (2014) and Huber et al. (2016) have identified possible obstacles to the adoption of solar energy in Europe and Africa, especially concerning the immediate impacts of changing climate conditions on the availability of solar energy. Confronting these challenges demands a combination of technological advancement and thoughtful planning to guarantee the robustness of renewable energy systems.

Dutta et al. highlighted alarming forecasts for Algeria, indicating a potential 10% reduction in solar energy production during the winter months from 2015 to 2040 (Dutta et al., 2022). The observed decline is linked to alterations in cloud cover and temperature patterns, which diminish the operational efficiency of photovoltaic systems. In a similar vein, Subin et al. forecast a reduction in solar energy production across Africa from 2071 to 2100 (Ha et al., 2023), which raises significant concerns regarding the continent's capacity to satisfy increasing energy needs (Samuel Chukwujindu, 2017). Nonetheless, progress in solar technologies, including p-Si and a-Si panels, presents a promising opportunity for enhancing solar PV production in West Africa (Agbor et al., 2023). The implementation of these technologies is essential for securing the region's renewable energy future (Nwokolo and Ogbulezie, 2018a), especially in high-emission scenarios where an increase in solar energy output is anticipated (Nwokolo and Ogbulezie, 2018b).

Ultimately, bifacial PV technology presents a compelling approach to addressing various challenges associated with environmental shifts. Tahir et al. indicate that bifacial PV systems can achieve a 30% greater energy yield than conventional monofacial systems (Tahir et al., 2022), though their efficiency may experience a slight decline over time as a result of changing environmental conditions. Even with this decrease, bifacial PV continues to be an excellent choice for optimizing energy production in an evolving climate. Nonetheless, Poddar et al. raise important issues for areas such as Australia, where diminished solar inso-

lation and increasing temperatures could adversely influence photovoltaic potential (Poddar et al., 2021). This is particularly true, except for northern coastal regions, where enhanced wind speed and radiation might mitigate these effects. Nwokolo et al. discovered in Nigeria that the effects of climate scenarios on various solar technologies differ, with amorphous silicon (a-Si) modules exhibiting superior performance in high-emission scenarios (Nwokolo et al., 2023). The results highlight the necessity of tailoring solar energy approaches to varying climatic conditions to guarantee the enduring viability of renewable energy systems.

The domain of photovoltaic (PV) energy systems has garnered extensive research attention, especially regarding their efficiency in diverse environmental scenarios. Investigations have predominantly focused on traditional photovoltaic systems, encompassing both monofacial and bifacial technologies, which have shown significant enhancements in efficiency across various climatic conditions and surface albedos (Tahir et al., 2022). The evaluation of these systems has concentrated on their performance across different solar radiation levels and temperature ranges, highlighting their energy yield and efficiency in connection with meteorological factors (Guo et al., 2024).

Nonetheless, the influence of climate change on the future trajectory of solar energy production is emerging as a vital subject of study. Few studies have concentrated on the response of PV system performance to long-term climate change scenarios. For instance, Crook et al. investigated the possible effects of escalating global temperatures on the efficiency of solar panels (Crook et al., 2011), forecasting that heightened heat levels will diminish photovoltaic performance because of the negative correlation between temperature and panel efficiency. Nonetheless, the study mainly focused on traditional photovoltaic technologies, highlighting a considerable void in the exploration of emerging technologies like photovoltaic absorbed diodes (PAD) and photovoltaic absorbed systems (PAS).

The implementation of bifacial PV technologies has tackled certain efficiency issues by enabling energy collection from both sides (Hua et al., 2024). However, the majority of these investigations have concentrated on short-term weather fluctuations instead of long-term climate shifts. This results in a lack of clarity regarding the performance of these and other advanced technologies, such as PAD and PAS, when subjected to the ongoing challenges posed by climate change. This is especially pertinent in areas with extreme climates, such as Alice Springs, Australia, which experiences high solar irradiance yet faces threats from rising temperatures and a growing frequency of heatwaves.

The recent advancement of the Coupled Model Intercomparison Project Phase 6 (CMIP6) offers a more comprehensive framework for examining these impacts. In contrast to the previous CMIP5 models that mainly concentrated on alterations in temperature and precipitation, CMIP6 broadens its scope by integrating new socio-economic scenarios and encompassing a wider range of climate variables (Arregocés et al., 2024). This facilitates a more profound comprehension of the possible effects that various greenhouse gas emission pathways may have on solar energy generation. CMIP6 presents three significant shared socio-economic pathways: SSP126, representing a robust mitigation scenario; SSP245, reflecting a moderate approach; and SSP585, indicative of a high-emission trajectory. Each pathway aligns with varying levels of global warming anticipated by the conclusion of the 21st century (Wang et al., 2023).

The CMIP6 models have played a crucial role in enhancing projections of climate-related changes across multiple sectors; however, their application to the specific area of solar energy systems remains limited. The majority of current research utilizing CMIP6 has focused on broad climate effects, such as alterations in temperature, precipitation, and patterns of extreme weather (Zhou et al., 2020). There has been limited attention given to the convergence of climate change and solar energy generation, especially regarding cutting-edge systems such as PAD and PAS. This gap holds considerable importance, particularly as these cut-

ting-edge technologies are expected to be pivotal in the evolution of solar energy, owing to their effectiveness and versatility in harsh environmental conditions.

Alice Springs, Australia, is a great place to study solar energy because it has harsh, sunny conditions that make it stand out. Many studies have looked into the area's solar energy potential, but most of them only used past climate data and didn't consider what the future climate might be like (Chen et al., 2024). These studies give us useful information about the area's solar resources, but they don't talk enough about how long-term climate changes, especially those caused by humans, might affect solar energy production.

This study presents a unique approach by utilizing CMIP6 model projections in relation to PAD and PAS technologies in Alice Springs. The study seeks to offer a forward-looking analysis by assessing the effects of various climate scenarios on these technologies over distinct time frames—2015–2030, 2031–2050, 2051–2070, and 2071–2099—thereby addressing a significant gap in existing literature. Earlier research has not adequately evaluated the distinct obstacles that upcoming climatic conditions present to these solar technologies. These challenges may encompass reduced efficiency stemming from elevated temperatures or a heightened occurrence of severe weather phenomena such as heatwaves or storms.

Furthermore, the creation and implementation of an innovative hybrid forecasting model, CARIMA-SARIMA-LG, represents a noteworthy progression in the discipline. The study presents a novel tool for forecasting the performance of PAD and PAS systems across diverse climate scenarios by integrating three unique modeling techniques—CARIMA, SARIMA, and logistic regression (LG). The model considers the intricate relationships among various climate factors, yielding more accurate forecasts of future solar energy potential. This holds significant relevance for shaping sustainable energy strategies in areas such as Alice Springs, where solar energy is essential to both local and national energy frameworks.

Moreover, the emphasis on Alice Springs, an area characterized by severe weather patterns and elevated solar irradiance, introduces a distinctive geographic aspect to the investigation. This research evaluates the effects of various shared socioeconomic pathways (SSPs) from 2015 to 2100, offering important insights into the enduring viability of solar energy systems in one of the sunniest areas globally. These insights are essential for guiding decision-makers, industry participants, and energy strategists on the most effective ways to adjust solar technologies in response to upcoming climate scenarios.

This study aims to fill a crucial void in the existing knowledge regarding the impact of climate change on solar energy generation, specifically focusing on PAD and PAS systems in Alice Springs, Australia. While there is an increasing amount of research examining the impacts of climate change on traditional solar technologies, there remains a notable gap in comprehensive studies that integrate the most recent climate models, including CMIP6, and emphasize cutting-edge solar technologies such as PAD and PAS. This study presents a unique approach by employing an advanced hybrid model (CARIMA-SARIMA-LG) to forecast future solar energy potential, providing a more precise and comprehensive evaluation of the effects of climate change.

This study offers a thorough and anticipatory examination by integrating data from five global climate research centers and assessing the impacts of different SSPs on solar energy generation. The results will guide approaches to enhance solar energy generation in light of upcoming climate conditions, allowing areas such as Alice Springs to maintain their solar capabilities even in the face of climate-related challenges. This research seeks to advance the overarching objective of creating sustainable and resilient energy systems capable of flourishing in an increasingly dynamic climate. The upcoming sections will explore the methodology, data sources, and findings of this study in greater detail, establishing a comprehensive framework for comprehending the future of solar energy production in Alice Springs.

2. Materials and methods

2.1. Site description

The Desert Knowledge Australia Solar Center (DKASC), situated in Alice Springs, Australia, stands as a prominent research institution committed to the progression of solar energy technology and the promotion of sustainable practices. Located in the core of the Australian desert, the DKASC takes advantage of exceptional solar conditions, characterized by elevated solar radiation and limited cloud cover, creating a prime setting for the experimentation and advancement of cutting-edge solar technologies. The desert climate of Alice Springs, marked by persistently elevated temperatures (averaging above 30 °C for half the year) (Hassan et al., 2022) and notable daily temperature fluctuations, significantly boosts the performance of photovoltaic (PV) systems. This configuration enables the DKASC to perform precise solar power modeling and efficiency assessments.

The DKASC is instrumental in propelling research in solar energy through the enhancement of solar panel efficiency, the advancement of energy storage solutions, and the creation of various renewable energy innovations aimed at diminishing reliance on fossil fuels. Alice Springs, covering 9% of the Northern Territory, offers extensive land resources ideal for solar experimentation (Hassan et al., 2021). This vast region plays a vital role in the extensive evaluation of solar technologies, allowing scientists to analyze energy efficiency in various landscapes and atmospheric conditions.

The research conducted at the center is grounded in practical application, with its solar testing facilities serving as a resource for academics, industry experts, and solar enterprises alike. The DKASC provides critical insights into the fluctuations of solar power over time, enabling the enhancement of PV system efficiency. This analytical approach aids in comprehending the patterns and underlying factors of solar power variability, thereby enhancing the performance of solar energy systems.

The distinctive climate of Alice Springs, characterized by its dry, sun-drenched environment and minimal cloud cover, offers a consistent and dependable source of solar energy throughout the year (Hassan et al., 2021). The unique characteristics of DKASC position it as a leading center for solar innovation, allowing enterprises and researchers to effectively tap into solar energy resources. The contributions of the DKASC play a crucial role in influencing the trajectory of renewable energy and fostering a sustainable, low-carbon future for Australia and beyond (Nwokolo et al., 2023).

2.2. Data acquisition

The process of gathering data for this study encompassed the collection of essential parameters that affect the energy output of concentrated photovoltaic (CPV) systems. The main thing that was looked at

was how much energy operational CPV systems produced on a monthly average (kWh). The things that went into the system were the monthly average ambient temperature (°C), relative humidity (%), clearness index (which doesn't have a unit), and extraterrestrial solar radiation (kWh). The Desert Knowledge Australia Solar Center (DKASC) delivered precise measurements of active power generation for both single-axis tracking (PAS) and dual-axis tracking (PAD) systems, in addition to data on ambient temperature and relative humidity. Extraterrestrial solar radiation and the clearness index were calculated utilizing established formulas (Nwokolo, 2017). Data collection occurred daily from April 1, 2019, to March 17, 2023, for PAD systems, and from April 1, 2019, to March 16, 2023, for PAS systems. This dataset, accessible to the public at <https://dkasolarcentre.com.au/source/alice-springs/dka-m3-b-phase>, promotes openness and cooperation in the pursuit of sustainable energy research. The data gives us a good way to compare how well PAD and PAS systems work in various weather conditions. It also gives us useful information for improving solar energy production in dry areas, as shown in Fig. 1 and Table 1.

2.3. Data quality

Maintaining superior data quality is essential in the examination of solar energy systems, especially when evaluating the effectiveness of sophisticated technologies such as single-axis (PAS) and dual-axis (PAD) tracking systems. At the Desert Knowledge Australia Solar Centre (DKASC), the datasets for extraterritorial and global horizontal radiation play a crucial role in forecasting and enhancing the efficiency of these systems. To ensure the precision of these datasets, a scattering technique established by Khorasanizadeh et al. (2016) was utilized. This method uses a radiative transfer model to replicate how solar radiation interacts with particles in the atmosphere, such as dust and aerosols. This makes sure that the data used for climate and energy studies is accurate.

The clearness index, which shows how much actual solar radiation reaches the Earth's surface compared to the theoretical maximum in clear-sky scenarios, has shown a lot of promise for predicting PAD and PAS energy output. Precise modeling of solar radiation is essential for enhancing the efficiency of these systems, and the clearness index acts as an important measure for assessing the reliability of the data.

In order to make the DKASC datasets more accurate, data points with clearness index values outside the 0–1 range were thrown out because they weren't thought to be reliable, as shown in Figs. 2 and 3 and explained in Table 1. This meticulous selection process guarantees that only reliable data underpins the analysis, which is essential for precisely assessing energy production and enhancing system architecture. Accurate solar radiation data plays a crucial role in enhancing the performance of solar tracking systems while also deepening our compre-



Fig. 1. Active power generation patterns in single-axis (PAS) and dual-axis (PAD) tracking systems.

Table 1
Performance data of single-axis (PAS) and dual-axis (PAD) tracking systems at DKASC, Alice Springs.

Active power generation for single-axis tracking (PAS) system		Active power generation for dual-axis tracking (PAD) system	
Manufacturer	Kyocera	Manufacturer	Kyocera
Array Rating	5.4 kW	Array Rating	5.4 kW
PV Technology	poly-Si	PV Technology	poly-Si
Array Structure	Tracker: Single Axis	Array Structure	Tracker: Dual Axis
Panel Rating	135W	Panel Rating	135W
Number Of Panels	5 × 8	Number Of Panels	5 × 8
Panel Type	Kyocera KD135GX-LP	Panel Type	Kyocera KD135GX-LP
Array Area	5 × 8.02 m ²	Array Area	5 × 8.02 m ²
Type Of Tracker	BW Solar F2 5 Star Tracker	Type Of Tracker	Wattsun AZ-125 Tracker
Inverter Size/Type	5 kW, SMA SMC 5000A	Inverter Size/Type	5 kW, SMA SMC 5000A
Installation Completed	Tue, Sep 30, 2008	Installation Completed	Sun, Nov 30, 2008

hension of solar energy's impact on addressing climate change challenges.

The scattering method, when integrated with the clearness index, provides a solid foundation for assessing solar radiation datasets and their relevance to solar energy systems. This method enhances the precision of data, thereby facilitating the effective design and functioning of PAS and PAD systems, which plays a significant role in the progress of solar energy technology. The thorough assessment of solar radiation datasets carries significant implications, influencing both research on climate dynamics and policy-making concerning energy generation and adaptation strategies in response to climate challenges. Accurate and reliable data is crucial for enhancing energy efficiency, shaping the future of sustainable energy frameworks, and informing strategies for addressing climate change challenges.

2.4. Modeling active power generation using machine learning for single-axis tracking (PAS) and dual-axis tracking (PAD) systems

Using machine learning to model active power generation for single-axis tracking and dual-axis tracking photovoltaic systems is a new and innovative idea that improves the accuracy of predictions and makes the system work better in a wide range of real-life situations. In this case, six machine learning models were used to simulate and predict the energy output of 133 PAS and PAD models. The radial basis functions, the multilayer perceptron artificial neural network, the controlled auto-regression integrated moving average, and the swapped auto-regression integrated moving average were the models in this group. Our earlier works provide the specifics of the seven physics-based and machine learning-based methodologies used in this investigation (Nwokolo et al., 2023, 2024). These methods were selected because of their capacity to faithfully simulate the intricate relationships between the system's variables, offering a thorough comprehension of the physics at play. Also, rigorous testing and comparison with experimental data showed that both methods were reliable and accurate at predicting changes in the PAS and PAD systems.

Boosting Ensemble (BOT): This technique enhances model performance by integrating several less effective models to create a more robust solution. This approach systematically refines each subsequent model to address the inaccuracies of its predecessors, thereby improving precision. In the context of PAS and PAD systems, BOT enhances the accuracy of power output predictions through a systematic refinement process.

Bagging Ensemble (BAG): This method constructs several models by utilizing randomly selected subsets of the data, merging the outcomes to achieve reliable predictions. For PAS and PAD systems, BAG

contributes to minimizing variance, thereby enhancing the model's stability against data fluctuations and bolstering its reliability across varying weather scenarios.

Radial Basis Functions (RBF): RBF networks excel at capturing intricate, non-linear relationships, rendering them particularly effective for forecasting solar power generation, influenced by a range of dynamic variables such as solar radiation, temperature, and system orientation.

Multilayer Perceptron (MLP): MLPs, a form of artificial neural network, are particularly adept at identifying non-linear relationships and intricate patterns within datasets. These models are particularly effective in forecasting photovoltaic (PV) power generation, as they can discern complex interactions among input variables like irradiance, temperature, and tracking angle in PAS and PAD systems.

CARIMA and SARIMA Models: The CARIMA and SARIMA models are very good at analyzing time series data because they use autoregressive components along with moving averages. In this framework, they are able to predict photovoltaic energy production by analyzing historical data patterns and time-related factors. The capacity of these models to analyze and predict using historical data is crucial for evaluating future performance and fluctuations in power generation.

To guarantee the reliability of these models, the dataset was divided into training (75%) and testing (25%) subsets. The training phase immersed the models in a diverse array of scenarios that PAS and PAD systems might face, while the testing phase rigorously assessed their predictive capabilities. This separation guaranteed that the models maintained generalizability and effectively replicated real-world outcomes.

The configuration parameters presented in Table 2 were essential for the assessment of PAS and PAD systems. The parameters encompassed irradiance levels, temperature fluctuations, tracking angles, and metrics of system efficiency, among other factors. The varied configurations enabled the machine learning models to adapt to numerous operational conditions, guaranteeing their relevance across diverse climatic and geographic areas.

The results from the simulations, detailed in Tables 3 and 4, illustrated the performance of the CARIMA model for both PAS and PAD systems. The data illuminated the anticipated energy output while also offering valuable insights for enhancing the design and functionality of these photovoltaic systems. By pinpointing regions where PAS and PAD systems could enhance their efficiency, these models provided practical recommendations for boosting energy output and reducing costs.

The simulations indicated that PAS systems, which follow the sun along a single axis, demonstrate enhanced performance in areas characterized by stable solar patterns. Conversely, PAD systems that adapt to the sun's trajectory along two axes provide enhanced efficiency in areas experiencing diverse solar paths. The capacity of machine learning to analyze and forecast these subtleties enables designers and operators to optimize systems for their unique environmental contexts.

Nwokolo et al. offer significant theoretical and practical perspectives on the application of machine learning for the prediction of solar energy (Nwokolo et al., 2023, 2024). The first study, which focused on group-based assessments in the context of climate warming scenarios, stresses how important classification algorithms are and how they can help predict future solar radiation and photovoltaic potential. This study highlights the importance of choosing suitable machine learning models and grasping their foundational concepts, especially in areas influenced by climate fluctuations.

The second publication explores hybrid models that integrate machine learning with analytical methods to evaluate the effects of climate on solar PV production. This highlights the critical role of evaluation metrics—like accuracy, precision, recall, and error metrics—in assessing the effectiveness of models. The importance of these metrics lies in their ability to enhance machine learning models for energy forecasting, guaranteeing that the predictions made are both precise and dependable. The document further explores methods for optimizing para-

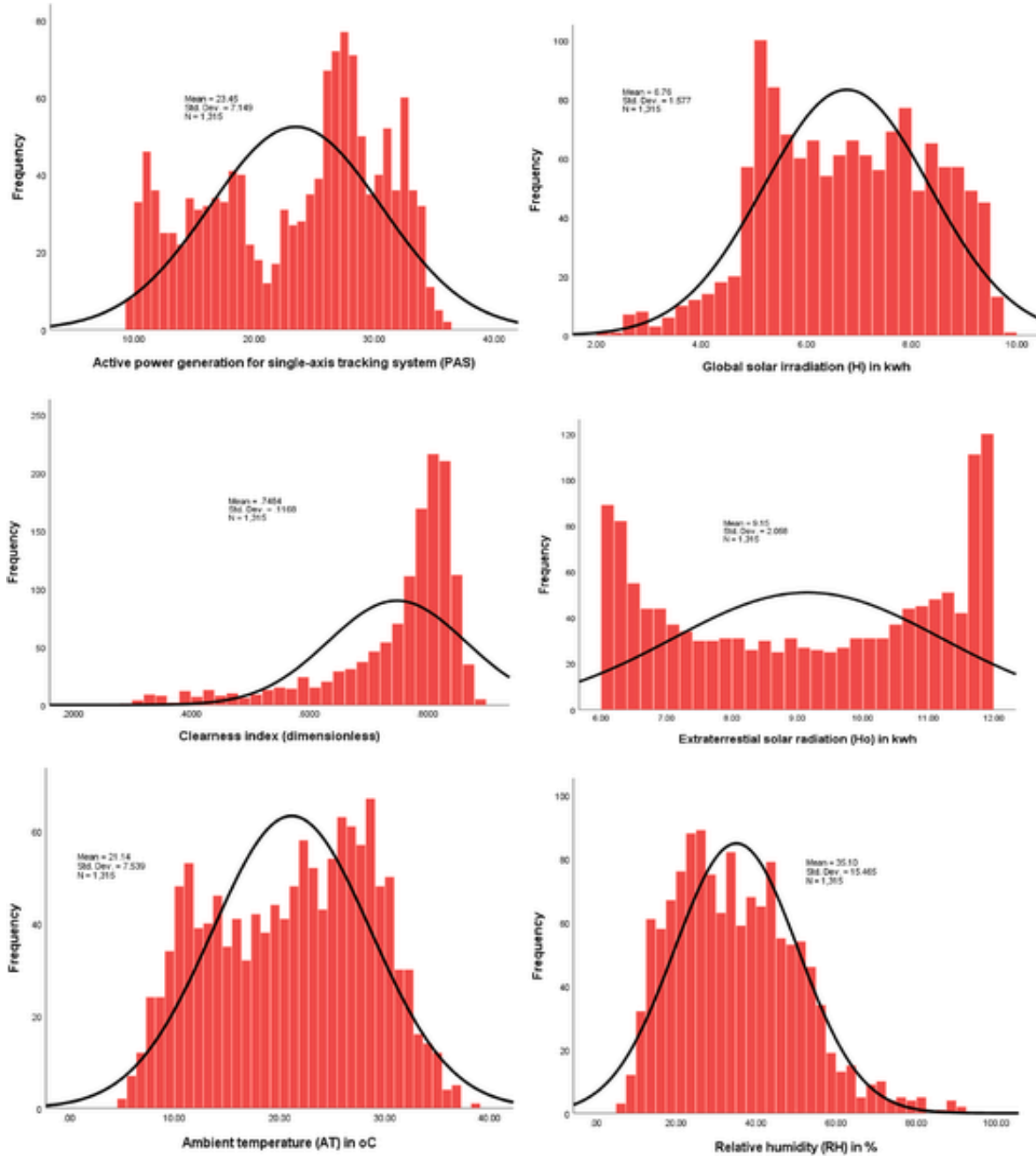


Fig. 2. Frequency distribution analysis of input and output parameters for modeling active power generation in single-axis tracking systems (PAS).

meters, which is crucial for enhancing the performance of PAS and PAD systems.

2.5. Description of logistic distribution model (LG)

The logistic distribution model (LG) serves as a statistical framework frequently applied in logistics and supply chain management. This term refers to the likelihood distribution of a continuous random variable that signifies the duration until an event transpires, like the time taken for goods to be delivered or the duration required for order processing. This framework enables organizations to assess and enhance their operations by comprehending the fluctuations and unpredictability linked to these temporal processes. It offers significant understanding of inventory oversight, demand prediction, and compre-

hensive operational effectiveness. The logistic distribution model (LG) was used to study and improve order processing time in solar energy because of how weather patterns change and how climate change makes it hard to accurately predict real-time active power for single-dual axis tracking (PAS) and dual-axis tracking (PAD) concentrated solar photovoltaic datasets. The LG empowers enterprises within the solar energy sector to thoroughly assess and enhance their order processing efficiency. By integrating the fluctuations and unpredictability stemming from evolving weather patterns and environmental shifts, this model aids organizations in making well-informed choices about inventory management and enhancing overall operational effectiveness. By employing precise forecasting and real-time active power predictions, organizations can achieve peak efficiency in their PAS and PAD systems. By precisely forecasting real-time active power, enterprises can en-

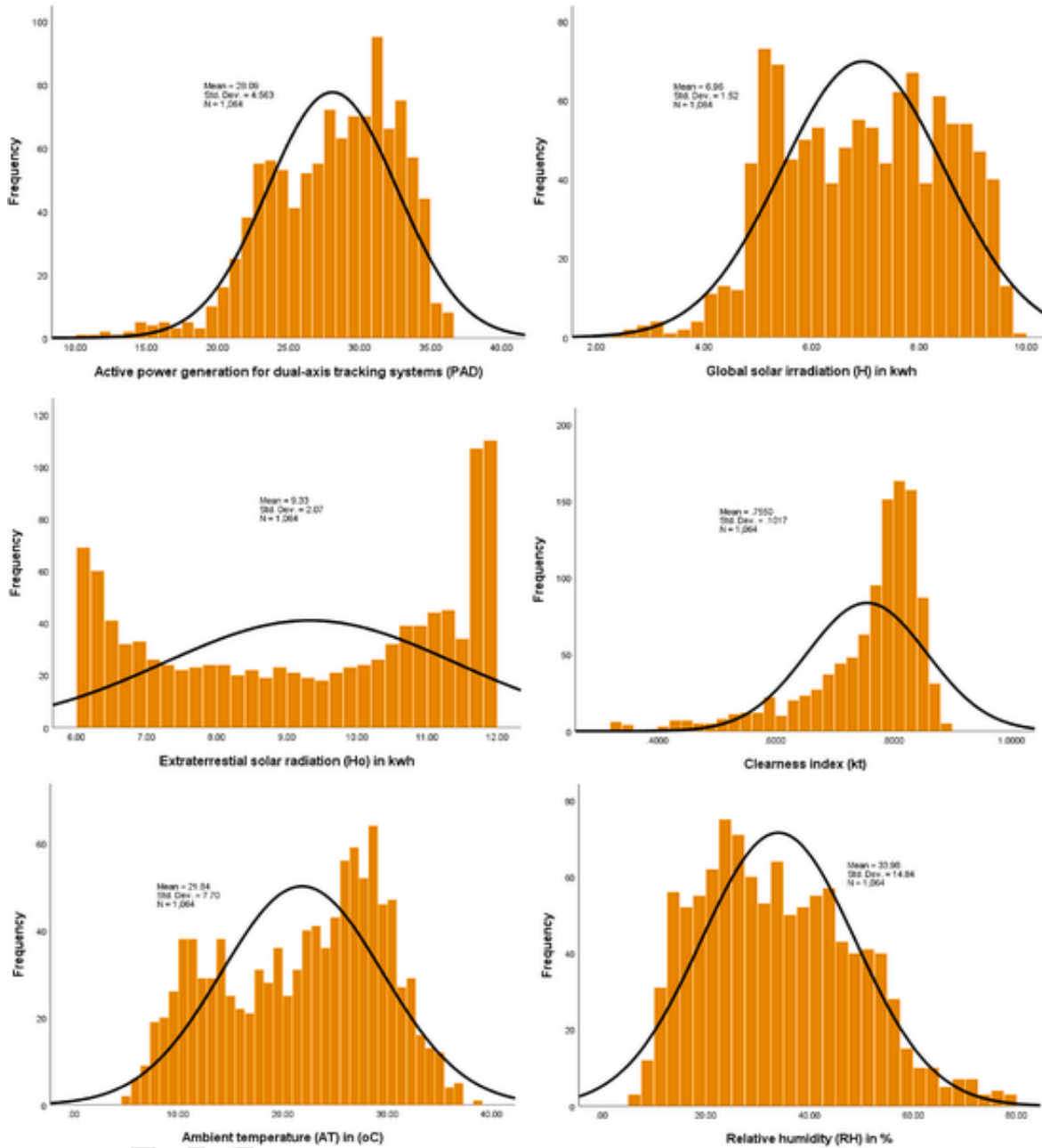


Fig. 3. Frequency distribution analysis of input and output parameters for modeling active power generation in dual-axis tracking systems (PAD).

hance the efficiency of solar energy storage solutions, including batteries. This allows for the effective storage of surplus energy during times of peak generation and its subsequent use during low production or high-demand periods, thereby optimizing the overall use of solar energy resources. Precise forecasting and real-time prediction of active power can assist enterprises in recognizing potential problems or failures within their solar photovoltaic systems. By identifying any discrepancies from anticipated power generation, companies can promptly tackle and rectify any technical issues, reducing downtime and enhancing the dependability of their solar energy systems. This novel strategy for upkeep can lead to financial benefits by preventing costly repairs or replacements over time.

2.6. CARIMA-SARIMA-LG, a hybrid model combining the strengths of SARIMA, CARIMA, and LG

The CARIMA-SARIMA-LG hybrid model is an innovative amalgamation of three distinct yet valuable methodologies: the Logistic Distribution Model (LG), the Controlled Autoregressive Integrated Moving Average (CARIMA), and the Swapped Autoregressive Integrated Moving Average (SARIMA). This integration leverages the distinct advantages of each component, yielding a resilient, precise, and adaptable forecasting framework for time series data, especially in intricate fields such as renewable energy systems. The prior description of the ARIMA model is automatic, or what the authors denote as the evolution of the switched ARIMA model (hereafter SARIMA). The Controlled ARIMA model, or CARIMA, is characterized by an auto-regression (p), differentiation (d), and moving average (q) that consistently adhere to the ARIMA order of

Table 2

Model input and output parameters for various configurations.

Model #	Input element notations	Parameter	Method	Output
1	H	Global solar radiation	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
2	$H + (H)^2$	Global solar radiation	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
3	$H + (H)^2 + (H)^3$	Global solar radiation	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
4	kt	Clearness index	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
5	$Kt + (kt)^2$	Clearness index	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
6	$Kt + (kt)^2 + (kt)^3$	Clearness index	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
7	AT	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
8	$AT + (AT)^2$	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
9	$AT + (AT)^2 + (AT)^3$	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
10	RH	Relative humidity	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
11	$RH + (RH)^2$	Relative humidity	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
12	$RH + (RH)^2 + (RH)^3$	Relative humidity	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
13	Ho	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
14	$Ho + (Ho)^2$	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
15	$Ho + (Ho)^2 + (Ho)^3$	Ambient temperature	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
16	H + kt	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
17	H + kt + AT	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
18	H + kt + AT + RH	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD

Table 2 (continued)

Model #	Input element notations	Parameter	Method	Output
19	H + kt + AT + RH + Ho	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
20	$H + (H)^2 + kt$	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
21	$H + (H)^2 + kt + (kt)^2$	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD
22	$H + (H)^2 + kt + (kt)^2 + Ho$	Hybrid	CARIMA, SARIMA, BAG, BOT, MLP, and RBF	PAS and PAD

Table 3

Fitted CARIMA models for predicting active power generation for dual-axis tracking (PAD) in Alice Springs, Australia.

Approach	Model	Estimate
CARIMA	1	$PAD = 12.181 + 2.861(H)$
CARIMA	2	$PAD = -5.294 + 7.846(H) - 0.353(H)^2$
CARIMA	3	$PAD = -26.306 + 19.187(H) - 2.233(H)^2 + 0.098(H)^3$
CARIMA	4	$PAD = 1.645 + 36.660(kt)$
CARIMA	5	$PAD = -13.013 + 75.167(kt) - 29.613(kt)^2$
CARIMA	6	$PAD = 40.604 - 192.525(kt) + 394.195(kt)^2 - 216.66(kt)^3$
CARIMA	7	$PAD = 32.906 - 0.080(AT)$
CARIMA	8	$PAD = 33.326 - 0.095(AT) - 0.000(AT)^2$
CARIMA	9	$PAD = 32.772 - 0.207(AT) + 0.010(AT)^2 + 0.000(AT)^3$
CARIMA	10	$PAD = 32.227 - 0.037(RH)$
CARIMA	11	$PAD = 31.657 - 0.004(RH) + 0.000(RH)^2$
CARIMA	12	$PAD = 34.418 - 0.214(RH) + 0.004(RH)^2 - 0.000035(RH)^3$
CARIMA	13	$PAD = 26.924 + 0.503(H_o)$
CARIMA	14	$PAD = 1.317 + 6.212(H_o) - 0.316(H_o)^2$
CARIMA	15	$PAD = 64.715 - 15.623(H_o) + 2.134(H_o)^2 - 0.089(H_o)^3$
CARIMA	16	$PAD = 3.216 + 1.323(H) + 23.988(kt)$
CARIMA	17	$PAD = 5.017 + 1.650(H) + 20.704(kt) - 0.062(AT)$
CARIMA	18	$PAD = -3.801 + 1.218(H) + 26.190(kt) - 0.018(AT) + 0.037(RH)$
CARIMA	19	$PAD = -16.401 + 0.040(H) + 38.486(kt) - 0.035(AT) + 0.032(RH) + 1.338(H_o)$
CARIMA	20	$PAD = -6.775 + 4.735(H) - 0.237(H)^2 + 21.229(kt)$
CARIMA	21	$PAD = -2.685 + 5.276(H) - 0.271(H)^2 + 2.550(kt) + 13.806(kt)^2$
CARIMA	22	$PAD = 3.429 + 6.870(H) - 0.327(H)^2 - 12.331(kt) + 18.488(kt)^2 - 0.649(H_o)$

$p = 0$, $q = 0$, and $d = 0$, based on non-seasonal criteria. To enhance the performance of the CARIMA models, the pacing process was automated to identify outliers such as additions, level shifts, innovations, transients, local trends, and additive anomalies. Due to the distinctive characteristics of CARIMA, it is capable of generating the estimation coefficient for each constructed model; nonetheless, it appears to underperform in certain instances compared to SARIMA. The SARIMA model simulation method allows estimators to select the optimal model based on the lowest normalized Bayesian Information Criterion (BIC) value. This was accomplished by interchanging the p , q , and d components, identifying outlier values, and adjusting functions. Due to the distinctive characteristics of SARIMA, it was unable to generate the estimation coefficient for each constructed model. Nonetheless, its estimation performance far surpasses that of its CARIMA equivalent. The ARIMA parameters for the CARIMA model are $p = 0$, $d = 0$, and $q = 0$ for both the training set (75% of the total dataset) and the test set (25% of the whole dataset). Our recent publications offer additional insights into SARIMA and CARIMA forecasting methodologies. The logistic distribution (LG)

Table 4

Fitted CARIMA models for predicting active power generation for single-axis tracking (PAS) in Alice Springs, Australia.

Approach	Model	Estimate
CARIMA	1	$PAS = -2.437 + 2.553(H)$
CARIMA	2	$PAS = -23.013 + 9.081(H) - 0.494(H)^2$
CARIMA	3	$PAS = -41.506 + 18.965(H) - 2.163(H)^2 + 0.089(H)^3$
CARIMA	4	$PAS = -11.550 + 34.310(kt)$
CARIMA	5	$PAS = -17.320 + 51.679(kt) - 13.595(kt)^2$
CARIMA	6	$PAS = 11.611 - 99.889(kt) + 240.192(kt)^2 - 136.070(kt)^3$
CARIMA	7	$PAS = 17.776 - 0.182(AT)$
CARIMA	8	$PAS = 19.279 - 0.144(AT) - 0.001(AT)^2$
CARIMA	9	$PAS = 19.391 - 0.412(AT) + 0.019(AT)^2 + 0.000(AT)^3$
CARIMA	10	$PAS = 17.649 - 0.060(RH)$
CARIMA	11	$PAS = 15.484 - 0.095(RH) + 0.002(RH)^2$
CARIMA	12	$PAS = 13.104 - 0.056(RH) + 0.003(RH)^2 - 0.000425(RH)^3$
CARIMA	13	$PAS = 7.707 + 0.812(H_o)$
CARIMA	14	$PAS = -37.858 + 12.029(H_o) - 0.571(H_o)^2$
CARIMA	15	$PAS = 40.739 - 13.709(H_o) + 2.027(H_o)^2 - 0.090(H_o)^3$
CARIMA	16	$PAS = -11.082 + 1.102(H) + 24.851(kt)$
CARIMA	17	$PAS = -9.672 + 1.425(H) + 21.593(kt) - 0.055(AT)$
CARIMA	18	$PAS = -13.454 + 1.389(H) + 24.622(kt) - 0.019(AT) + 0.035(RH)$
CARIMA	19	$PAS = -21.664 + 0.164(H) + 35.204(kt) - 0.039(AT) + 0.026(RH) + 1.052(H_o)$
CARIMA	20	$PAS = -17.657 + 4.136(H) - 0.199(H)^2 + 19.426(kt)$
CARIMA	21	$PAS = -16.275 + 5.140(H) - 0.271(H)^2 + 2.741(kt) + 12.617(kt)^2$
CARIMA	22	$PAS = -12.721 + 7.219(H) - 0.371(H)^2 - 16.888(kt) + 21.177(kt)^2 - 0.534(H_o)$

model enhances SARIMA and CARIMA by addressing non-linearities and quantifying the uncertainty of anticipated values. It provides a statistical framework for predicting the probability distribution of active power outputs, effectively addressing variability due to stochastic environmental influences. The cumulative distribution function (CDF) and probability density function (PDF) of LG provide insights into the likelihood of specific outcomes occurring. This enhances the model's reliability and robustness in making predictions.

The seasonal components in the time series are initially identified and extracted utilizing SARIMA. This guarantees that repetitive patterns do not eclipse other important factors. Meteorological variables are standardized and integrated into the CARIMA framework to capture their direct and interactive effects on the target variable. SARIMA produces baseline projections by examining inherent time series patterns. CARIMA enhances these forecasts by dynamically adjusting for external factors and implementing controlled changes to improve accuracy. The LG model, alongside a probabilistic perspective, delineates the distribution of residuals from SARIMA-CARIMA forecasts. This phase addresses intrinsic uncertainties, guaranteeing that the ultimate predictions are both precise and supported by confidence intervals. Utilizing LG's probabilistic insights, a weighted aggregation process integrates the outputs from SARIMA and CARIMA. This amalgamation harmonizes linear, non-linear, and stochastic components to yield accurate and resilient predictions. The CARIMA-SARIMA-LG model attains synergy by the systematic integration of the distinct advantages of SARIMA, CARIMA, and LG.

Therefore, the mathematical formulation for forecasting real-time active power in dual-axis tracking solar photovoltaic systems (PAD) can be articulated as follows: This method integrates state-of-the-art technology with sophisticated mathematical models to precisely predict real-time active power output.

$$\frac{PAD}{kt} = U \quad (1)$$

$$U = 94.804 - 287.840(kt) + 456.346(kt)^2 - 240.450(kt)^3 \quad (2)$$

$$PAD = U(kt) \quad (3)$$

$$\frac{PAD4(SARIMA)}{U(kt)} = 1.00256 \quad (4)$$

$$PAD4(SARIMA) = 1.00256[U(kt)] \quad (5)$$

Putting (2) and (5) yields (6)

$$PAD4(SARIMA) = 1.00256[94.804(kt) - 287.840(kt)^2 + 456.346(kt)^3 - 240.450(kt)^4] \quad (6)$$

The continuous distribution's time series stochastic function for active power for dual-axis tracking solar photovoltaic systems (PAD) can be represented as follows:

$$F\left(\frac{PAD}{U}\right) = \int_{-\infty}^{\infty} f\left(\frac{PAD}{U}\right) \quad (7)$$

The logistic statistical probability distribution function for active power in dual-axis tracking solar photovoltaic systems (PAD) can be expressed as follows

$$\left(\frac{PAS}{U}\right) = \frac{1}{1 + \exp(-Z)} \quad (8)$$

Where Z represents the statistical parameter associated with the normal distribution, specifically for the active power in dual-axis tracking solar photovoltaic systems (PAD), which can be defined as follows:

$$Z = \frac{PAD/U - \beta}{\lambda} \quad (9)$$

Where U is the derivative of the clearness index, β is the mean of PAD/U , and λ is the variance of PAD/U . $\beta = 0.75497$, and $\lambda = 0.010$ for the normal distribution of the active power in dual-axis tracking solar photovoltaic systems (PAD/U).

The mathematical representation of the logistic cumulative distribution function (CDF) that describes the active power in dual-axis tracking solar photovoltaic systems (PAS/U) is as follows:

$$F(PAS/U) = \frac{1}{1 + \exp\left(-\frac{PAS/U - \beta}{\lambda}\right)} - \infty < PAS/U < \infty \quad (10)$$

The derivation of the logistic probability distribution function (PDF) that characterizes the active power in dual-axis tracking solar photovoltaic systems involves the differentiation of equation (10) with respect to PAD/U . This process yields equation (11), which mathematically represents the corresponding PDF.

$$f(PAD/U) = \frac{\exp\left(-\frac{PAD/U - \beta}{\lambda}\right)}{\omega\left(1 + \exp\left(-\frac{PAD/U - \beta}{\lambda}\right)\right)^2} - \infty < PAD/U < \infty \quad (11)$$

Equations (12) and (13) offer a comprehensive framework to assess the location parameter ($\beta = 0.75497$) and scale parameter ($\lambda = 0.010$) of the distribution governing the active power in single-axis tracking solar photovoltaic systems. By leveraging the expectation mean and variance of PAD/U from logistically observed datasets, professionals can efficiently derive these crucial parameters, facilitating a deeper understanding of system performance and enabling informed decision-making processes.

$$\beta = E(PAD/U) = \frac{1}{N} \left[\sum_{i=1}^N PAD/U(i) \right] \quad (12)$$

$$\lambda = \sqrt[3]{\frac{\text{var}(PAD/U)}{\pi^2}} \quad (13)$$

$$\text{Let } V = F(PAD/U) \quad (14)$$

The active power in dual-axis tracking solar photovoltaic systems is governed by the uniform distribution V , which ranges from 0 to 1. By substituting equation (14) into equation (10), we can derive equation (15), showcasing the professional and innovative nature of this mathematical transformation. This enables a comprehensive understanding of the relationship between V and the active power in these solar systems.

$$V = \frac{1}{1 + \exp\left(-\left(\frac{PAD/U - \beta}{\lambda}\right)\right)} \quad (15)$$

$$\begin{aligned} V \left[1 + \exp\left(-\left(\frac{PAD/U - \beta}{\lambda}\right)\right) \right] &= 1 \\ 1 + \exp\left(-\left(\frac{PAD/U - \beta}{\lambda}\right)\right) &= \frac{1}{V} \\ \exp\left(-\left(\frac{PAD/U - \beta}{\lambda}\right)\right) &= \frac{1}{V} - 1 = \left(\frac{PAD/U - \beta}{\lambda}\right) = I n \left[\frac{1}{V} - 1 \right] \\ \frac{PAD/U - \beta}{\lambda} &= -I n \left[\frac{1}{V} - 1 \right] \\ \frac{PAD}{U} - \beta &= -\lambda I n \left[\frac{1}{V} - 1 \right] \\ \frac{PAD}{U} &= \beta - \lambda I n \left[\frac{1}{V} - 1 \right] \end{aligned} \quad (16)$$

By incorporating the precise values of the location parameter ($\beta = 0.75497$) and scale parameter ($\lambda = 0.010$) of the distribution that governs the active power in dual-axis tracking solar photovoltaic systems, equation (16) can be effectively transformed into equation (17). This substitution allows for a professional and innovative approach to analyzing and understanding the relationship between these parameters within the context of solar energy systems.

$$\frac{PAD}{U} = 0.75497 - 0.010 \ln \left[\frac{1}{V} - 1 \right] \quad (17)$$

Equating equations (3) and (17) yields (18)

$$U(kt) = 0.75497 - 0.010 \ln \left[\frac{1}{V} - 1 \right] \quad (18)$$

$$kt = \left[0.75497 - 0.010 \ln \left[\frac{1}{V} - 1 \right] \right] \times \frac{1}{U} \quad (19)$$

Putting (19) and (2) into (5) yields (20)

$$\begin{aligned} PAD &= (95.047 - 288.577(kt) + 457.514(kt)^2 - 241.066(kt)^3) \\ &\times \left[0.75497 - 0.010 \ln \left[\frac{1}{V} - 1 \right] \right] \times \frac{1}{U} \end{aligned} \quad (20)$$

From the best-performing CARIMA model (CARIMA20) given as:

$$PAD = -6.775 + 4.735(H) - 0.237(H)^2 + 21.229(kt) \quad (21)$$

Equating (20) and (21) yields (22)

$$\begin{aligned} -6.775 + 4.735(H) - 0.237(H)^2 + 21.229(kt) \\ = (95.047 - 288.577(kt) + 457.514(kt)^2 - 241.066(kt)^3) \\ \times \left[0.75497 - 0.010 \ln \left[\frac{1}{V} - 1 \right] \right] \times \frac{1}{U} \end{aligned} \quad (22)$$

Equation (22) reduces to (23)

$$-6.775 + 4.735(H) - 0.237(H)^2 + 21.229(kt) = 1.00256[U(kt)] \quad (23)$$

Putting equation (3) into (23) yields (24)

$$-6.775 + 4.735(H) - 0.237(H)^2 + 21.229(kt) = 1.00256(PAD) \quad (24)$$

$$PAD = -6.758 + 4.723(H) - 0.236(H)^2 + 21.175(kt) \quad (25)$$

Equation (25) represents the cutting-edge CARIMA-SARIMA-LG model, a groundbreaking approach for accurately forecasting the active power in dual-axis tracking solar photovoltaic systems (PAD). This evolutionary hybridized model combines the strengths of CARIMA, SARIMA, and LG methodologies to deliver unparalleled precision and reliability in predicting solar power generation. Equation (25) changes the way active power is predicted in dual-axis tracking solar photovoltaic systems (PAD) by using the CARIMA, SARIMA, and LG methods. This innovative hybridized model offers a comprehensive solution that surpasses traditional forecasting methods by providing exceptional accuracy and dependability in estimating solar power generation.

Similarly, the mathematical representation of predicting real-time active power for single-axis tracking solar photovoltaic systems (PAS) can be elucidated as follows: This approach combines cutting-edge technology with advanced mathematical models to accurately forecast the active power output in real time.

$$\frac{PAS}{kt} = Q \quad (26)$$

$$Q = 102.377 - 393.747(kt) + 622.885(kt)^2 - 323.282(kt)^3 \quad (27)$$

$$PAS = Q(kt) \quad (28)$$

$$\frac{PAS19(SARIMA)}{Q(kt)} = 1.005342 \quad (29)$$

$$PAS19(SARIMA) = 1.005342[Q(kt)] \quad (30)$$

Putting (27) into (30) yields (31)

$$\begin{aligned} PAS19(SARIMA) &= 1.005342 [102.377(kt) - 393.74(kt)^2 \\ &+ 622.885(kt)^3 - 323.282(kt)^4] \end{aligned} \quad (31)$$

The continuous distribution's time series stochastic function for active power for PAS can be represented as follows:

$$F\left(\frac{PAS}{Q}\right) = \int_{-\infty}^{\infty} f\left(\frac{PAS}{Q}\right) \quad (32)$$

The logistic statistical probability distribution function for PAS can be expressed as follows

$$\left(\frac{PAS}{Q}\right) = \frac{1}{1 + \exp(-\eta)} \quad (33)$$

Where η represents the statistical parameter associated with the normal distribution, specifically for the active power in PAS, which can be defined as follows:

$$\eta = \frac{PAS/Q - \psi}{\phi} \quad (34)$$

Where Q is the derivative of the clearness index, ψ is the mean of PAS/Q , and ϕ is the variance of PAS/Q . $\psi = 0.748448$, and $\phi = 0.014$ for the normal distribution of the active power in single-axis tracking solar photovoltaic systems (PAS/Q).

The mathematical representation of the logistic cumulative distribution function (CDF) that describes the active power in single-axis tracking solar photovoltaic systems (PAS/Q) is as follows:

$$F(PAS/Q) = \frac{1}{1 + \exp\left(-\left(\frac{PAS/Q - \psi}{\phi}\right)\right)} - \infty < PAS/Q < \infty \quad (35)$$

The derivation of the logistic probability distribution function (PDF) that characterizes the active power in single-axis tracking solar photo-

voltic systems involves the differentiation of equation (35) with respect to PAS/Q. This process yields equation (36), which mathematically represents the corresponding PDF.

$$f(PAD/U) = \frac{\exp\left(-\frac{PAS/Q-\psi}{\phi}\right)}{\psi \left(1 + \exp\left(-\frac{PAS/Q-\psi}{\phi}\right)\right)^2} - \infty < PAD/U < \infty \quad (36)$$

Equations (37) and (38) offer a comprehensive framework to assess the location parameter ($\psi = 0.748448$) and scale parameter ($\phi = 0.014$) of the distribution governing the active power in single-axis tracking solar photovoltaic systems. By leveraging the expectation mean and variance of PAS/Q from logistically observed datasets, professionals can efficiently derive these crucial parameters, facilitating a deeper understanding of system performance and enabling informed decision-making processes.

$$\psi = E(PAS/Q) = \frac{1}{N} \left[\sum_{i=1}^N PAS/Q(i) \right] \quad (37)$$

$$\phi = \sqrt[3]{\frac{\text{var}(PAS/Q)}{\pi^2}} \quad (38)$$

$$\text{Let } R = F(PAS/Q) \quad (39)$$

The active power in dual-axis tracking solar photovoltaic systems is governed by the uniform distribution R, which ranges from 0 to 1. By substituting equation (39) into equation (35), we can derive equation (40), showcasing the professional and innovative nature of this mathematical transformation. This enables a comprehensive understanding of the relationship between R and the active power in these solar systems.

$$R = \frac{1}{1 + \exp\left(-\frac{PAS/Q-\psi}{\phi}\right)} \quad (40)$$

$$\begin{aligned} R \left[1 + \exp\left(-\frac{PAS/Q-\psi}{\phi}\right) \right] &= 1 \\ 1 + \exp\left(-\frac{PAS/Q-\psi}{\phi}\right) &= \frac{1}{R} \\ \exp\left(-\frac{PAS/Q-\psi}{\phi}\right) &= \frac{1}{R} - 1 - \left(-\frac{PAS/Q-\psi}{\phi}\right) = I n \left[\frac{1}{R} - 1 \right] \\ \frac{PAS/Q-\psi}{\phi} &= -I n \left[\frac{1}{R} - 1 \right] \\ \frac{PAS}{Q} - \psi &= -\phi I n \left[\frac{1}{R} - 1 \right] \\ \frac{PAS}{Q} &= \psi - \phi I n \left[\frac{1}{R} - 1 \right] \end{aligned} \quad (41)$$

By incorporating the precise values of the location parameter ($\psi = 0.748448$) and scale parameter ($\phi = 0.014$) of the distribution that governs the active power in single-axis tracking solar photovoltaic systems, equation (41) can be effectively transformed into equation (42). This substitution allows for a professional and innovative approach to analyzing and understanding the relationship between these parameters within the context of solar energy systems.

$$\frac{PAD}{Q} = 0.748448 - 0.014 I n \left[\frac{1}{R} - 1 \right] \quad (42)$$

Equating equations (28) and (42) yields (43)

$$Q(kt) = 0.748448 - 0.014 I n \left[\frac{1}{R} - 1 \right] \quad (43)$$

$$kt = \left[0.748448 - 0.014 I n \left[\frac{1}{R} - 1 \right] \right] \times \frac{1}{Q} \quad (44)$$

Putting (44) and (27) into (30) yields (45)

$$\begin{aligned} PAS &= 1.005342 \left[102.377 - 393.747(kt) + 622.885(kt)^2 - 323.282(kt)^3 \right] \\ &\times \left[0.748448 - 0.014 I n \left[\frac{1}{R} - 1 \right] \right] \times \frac{1}{Q} \end{aligned} \quad (45)$$

From the best-performing CARIMA model (CARIMA20) given as:

$$PAS = -17.657 + 4.136(H) - 0.199(H)^2 + 19.426(kt) \quad (46)$$

Equating (45) and (46) yields (47)

$$\begin{aligned} -17.657 + 4.136(H) - 0.199(H)^2 + 19.426(kt) \\ = 1.005342 \left[102.377 - 393.747(kt) + 622.885(kt)^2 - 323.282(kt)^3 \right] \\ \times \left[0.748448 - 0.014 I n \left[\frac{1}{R} - 1 \right] \right] \times \frac{1}{Q} \end{aligned} \quad (47)$$

Equation (47) reduces to (48)

$$-17.657 + 4.136(H) - 0.199(H)^2 + 19.426(kt) = 1.005342 [Q(kt)] \quad (48)$$

Putting equation (28) into (48) yields (49)

$$-17.657 + 4.136(H) - 0.199(H)^2 + 19.426(kt) = 1.005342 (PAS) \quad (49)$$

$$PAS = -17.563 + 4.114(H) - 0.198(H)^2 + 19.323(kt) \quad (50)$$

Equation (50) represents the cutting-edge CARIMA-SARIMA-LG model, a groundbreaking approach for accurately forecasting the active power in single-axis tracking solar photovoltaic systems (PAS). This evolutionary hybridized model combines the strengths of CARIMA, SARIMA, and LG methodologies to deliver unparalleled precision and reliability in predicting solar power generation. By incorporating the CARIMA, SARIMA, and LG methodologies, Equation (50) revolutionizes the prediction of PAS. This innovative hybridized model offers a comprehensive solution that surpasses traditional forecasting methods by providing exceptional accuracy and dependability in estimating solar power generation.

2.7. Analytical tools and performance evaluation

The research assesses machine learning models for active power generation in PAS and PAD systems, employing essential analytical tools such as the coefficient of determination (R^2), root mean square error (RMSE), normalized RMSE (nRMSE), and relative percentage error (RPE). The R^2 value quantifies the extent to which the model accounts for variance, serving as an indicator of its precision. RMSE measures the magnitude of prediction errors, whereas nRMSE adjusts RMSE to allow for meaningful comparisons across different datasets. RPE serves as a comparative indicator of error, emphasizing the difference between forecasted and observed values. Collectively, these metrics provide an in-depth evaluation of model efficacy, guaranteeing precise and dependable forecasts of energy production in both PAS and PAD systems, as illustrated in equations (51)–(55).

$$R^2 = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_{i=1}^n (O_i - O_{ave})^2} \quad (51)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - P_i)^2} \quad (52)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{O_i - P_i}{O_i} \right| \quad (53)$$

$$nRMSE = \frac{RMSE}{\sum_{i=1}^n (H)} \quad (54)$$

$$RPE = \sum_{i=1}^n \left| \frac{O_i - P_i}{P_i} \right| \times 100 \quad (55)$$

2.8. Evaluating the effects of climate change on active power for single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic outputs

Assessing the impacts of climate change on active power generation for single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic systems is essential for comprehending the future efficacy of solar energy infrastructure. This research utilized surface downwelling short-wave radiation (H) data sourced from the European Centre for Medium-Range Weather Forecasts (ECMWF) and integrated it with five global climate model (GCM) outputs detailed in our earlier publications (Nwokolo et al., 2024; S.C. Nwokolo et al., 2023). The GCM data underwent interpolation to meticulously examine the potential impacts of climate change on the energy output of PAS and PAD in Alice Springs, Australia.

To obtain the datasets for the Alice Springs station in Australia, it was necessary to first download a comprehensive dataset that encompasses the entire country, extending from the northern latitude of -10° to the southern latitude of -44° , and from the western longitude of 113° to the eastern longitude of 154° . By concentrating on Alice Springs, a site defined by its precise latitude (-23.750) and longitude (133.867), the researchers guaranteed that their examination of solar power generation was contextually precise. The refinement of impact assessment was enhanced through the interpolation of GCM data, enabling more localized and precise climate predictions. The analysis utilized equation 25 (PAD) and 50 (PAS) to assess the impact of different climate change scenarios on the capacity of PAS and PAD systems to produce active power.

The research examined three distinct socio-economic pathways, each illustrating varying climate scenarios influenced by greenhouse gas emissions and mitigation strategies. The scenarios—SSP1-2.6, SSP2-4.5, and SSP5-8.5—provide insights into how varying degrees of global warming may influence solar power generation.

- **SSP1-2.6 (Mitigation-Case Scenario):** This trajectory illustrates a future characterized by the prompt and impactful enactment of climate policies, aiming to restrict the increase in global temperatures to 1.5°C by the year 2100. In this context, PAS and PAD systems would probably encounter reduced fluctuations in solar radiation, leading to a more consistent power output.
- **SSP2-4.5 (Moderate Forcing-Case Scenario):** This scenario envisions a balanced approach to climate action, resulting in a projected increase in global temperatures of 2.5°C . The variability in solar radiation could pose significant challenges for PAS and PAD systems, affecting their energy generation efficiency and complicating efforts to maintain a stable power output over time.
- **SSP5-8.5 (Strong Forcing-Case Scenario):** This scenario is marked by elevated greenhouse gas emissions and limited efforts for mitigation, resulting in a substantial temperature increase of 3.5°C by century's end. In this context, PAS and PAD systems may experience significant declines in efficiency as a result of heightened climate extremes, including more regular occurrences of cloud cover and heatwaves, which could influence solar radiation levels.

The assumptions underlying the three climate scenarios—SSP1-2.6 (Mitigation-Case Scenario), SSP2-4.5 (Moderate Forcing-Case Scenario), and SSP5-8.5 (Strong Forcing-Case Scenario)—are pivotal in

shaping the trajectory of global climate futures and offer valuable insights into the implications of various policy actions. SSP1-2.6 is predicated on aggressive climate mitigation efforts aimed at limiting global temperature rise to 1.5°C by 2100, assuming rapid and widespread implementation of clean energy technologies, substantial reductions in greenhouse gas emissions, and a transformation towards a sustainable, low-carbon economy. This scenario aligns with the most ambitious climate policies, such as the Paris Agreement's goal of net-zero emissions by mid-century, yet it assumes a level of global cooperation and policy enforcement that remains challenging under current political and economic conditions. SSP2-4.5 represents a middle ground, with moderate climate action leading to a projected temperature rise of 2.5°C . This scenario reflects a pathway that balances some climate policies and market-driven solutions but still faces challenges in achieving the necessary emissions reductions, resulting in a reliance on both technological innovation and incremental policy measures. SSP2-4.5 corresponds to the “business-as-usual” trajectory seen in some current trends, where political will to fully decarbonize remains fragmented, and international climate negotiations often fall short of the necessary commitments. In contrast, SSP5-8.5 assumes minimal climate mitigation, resulting in a 3.5°C increase in temperatures by the century's end, driven by continued fossil fuel dependence and limited adoption of sustainable technologies. This scenario aligns with current global patterns where fossil fuel consumption remains high, and policy frameworks struggle to keep pace with the urgency required for transformative change. It reflects the ongoing tension between short-term economic interests and long-term climate goals, as seen in major economies' reluctance to rapidly transition away from fossil fuels. Collectively, these scenarios highlight the critical role of global governance, technological advancement, and societal transformation in determining future climate outcomes. They also underscore the need for stronger, more immediate actions to avert the severe consequences projected under high-emission scenarios like SSP5-8.5.

3. Results and discussions

3.1. A comparison of descriptive statistics for key meteorological and power generation variables from a single-axis (PAS) and dual-axis (PAD) tracking systems

The solar farm located in Alice Springs, Australia, offers essential information for forecasting the active power output in both single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic (PV) systems, as illustrated in Tables 5 and 6. The two systems monitor solar movement through distinct methods—single-axis systems align with the sun's path from east to west, whereas dual-axis systems additionally account for altitude, thereby enhancing overall efficiency. This analysis presents a comprehensive examination of the descriptive statistics derived from the two tables, encompassing global solar irradiation (H), extraterrestrial solar radiation (H_0), clearness index (kt), ambient temperature (AT), and relative humidity (RH). Grasping the fluctuations and pat-

Table 5

Descriptive statistics of key meteorological and power generation variables from a single-axis tracking solar photovoltaic system (PAS).

Descriptive Statistics							
	N	Range	Minimum	Maximum	Mean	Std. Deviation	Variance
PAS	1315	26.39	9.39	35.79	23.4471	7.14880	51.105
H	1315	7.68	2.20	9.88	6.7639	1.57672	2.486
H_0	1315	5.77	6.11	11.89	9.1523	2.05835	4.237
KT	1315	0.5972	0.2919	0.8891	0.748448	0.1168135	0.014
AT	1315	33.27	5.11	38.38	21.1370	7.53945	56.843
RH	1315	85.11	5.84	90.95	35.1013	15.46542	239.179

Table 6

Descriptive statistics of key meteorological and power generation variables from a dual-axis tracking solar photovoltaic system (PAD).

	N	Range	Minimum	Maximum	Mean	Std. Deviation	Variance
PAD	1064	25.87	10.60	36.47	28.0936	4.56251	20.817
H	1064	7.60	2.28	9.88	6.9577	1.52046	2.312
Ho	1064	5.77	6.11	11.89	9.3257	2.07047	4.287
kt	1064	0.57	0.32	0.89	0.7550	0.10170	0.010
AT	1064	33.27	5.11	38.38	21.8352	7.70029	59.295
RH	1064	72.64	5.84	78.49	33.9829	14.83977	220.219

terms in these factors is crucial for developing precise machine learning models aimed at forecasting solar power generation.

The active power for the single-axis system varies between 9.39 kW and 35.79 kW, with an average of 23.45 kW and a standard deviation of 7.15 kW. In comparison, the dual-axis system (PAD) exhibits a more limited range (10.60 kW–36.47 kW), yet it boasts a higher average of 28.09 kW and a reduced standard deviation of 4.56 kW. The higher mean and lower variability in PAD indicate that dual-axis systems are more efficient and reliable in energy generation, capturing more solar energy and producing a more stable output. Prior studies indicate that employing dual-axis tracking can enhance energy capture by 30–40% in comparison to single-axis tracking (Fernández-Ahumada et al., 2020a). The reduced variance in PAD signifies a more consistent power output, contributing positively to grid stability and enhancing the accuracy of energy forecasting.

The amount of global solar irradiation (H) plays a vital role in determining the power output of both systems. The range of H for PAS is 7.68 kWh/m², with an average of 6.76 kWh/m² and a standard deviation of 1.58 kWh/m². The PAD system demonstrates a modest reduction in variability, presenting a range of 7.60 kWh/m², an average of 6.96 kWh/m², and a standard deviation of 1.52 kWh/m². The data suggests that the solar irradiation levels in Alice Springs are generally elevated, yet there is significant variability from one day to the next. The primary factor influencing active power generation is solar irradiation, and it is essential to consider this variability in H when creating predictive models. The variations in H between PAS and PAD demonstrate the improved energy capture made possible by the dual-axis tracking system, which enables better alignment with the sun's position throughout the day (Gadeo-Martos et al., 2019).

The extraterrestrial solar radiation (Ho) represents the peak potential of solar energy that can be harnessed prior to any atmospheric disruptions. The PAS and PAD systems yield comparable statistics for Ho, showing a range of 5.77 kWh/m², with mean values of 9.15 kWh/m² for PAS and 9.33 kWh/m² for PAD, alongside standard deviations of approximately 2.06–2.07 kWh/m². The observed similarities indicate that Ho is predominantly influenced by the geometry between the Earth and the Sun, with minimal impact from local atmospheric conditions. Nonetheless, given that Ho establishes the foundation for determining the clearness index (KT), it continues to be a crucial factor in predicting solar energy (Fernández et al., 2016). The alignment between PAS and PAD regarding Ho emphasizes that atmospheric conditions and other factors, rather than extraterrestrial solar radiation, are primarily responsible for fluctuations in solar power output.

The clearness index (KT) quantifies the proportion of extraterrestrial solar radiation that penetrates the surface, offering valuable information regarding the transparency of the atmosphere. The PAS system exhibits a range of 0.5972, with a mean value of 0.7484 and a standard deviation of 0.1168. In the case of PAD, KT exhibits a comparable trend, presenting a range of 0.57 and an average of 0.7550, accompanied by a reduced standard deviation of 0.1017. The observed variability between the two systems suggests that the dual-axis system may be better at adjusting to daily atmospheric fluctuations, allowing it to maintain a constant power output even as the clearness index changes.

This is consistent with earlier studies indicating that tracking systems, particularly dual-axis ones, are capable of adapting to different atmospheric conditions (Fernández et al., 2014a).

The ambient temperature plays a significant role in determining the performance of solar photovoltaic systems, as increased temperatures can lead to a decline in the efficiency of solar panels. The PAS system captures an AT range of 33.27 °C, featuring a mean of 21.14 °C and a standard deviation of 7.54 °C. The PAD system displays a comparable range and mean along with a marginally lower standard deviation of 7.70 °C. While both systems function within the same environmental conditions, dual-axis tracking systems are prone to generating increased heat from their continuous motion, potentially leading to a minor impact on efficiency (Soria-Moya et al., 2015). Nevertheless, the slight variation in variability between PAS and PAD suggests that ambient temperature influences both systems in a comparable manner. Therefore, it is essential for models to integrate temperature coefficients when forecasting power output.

Relative humidity plays a crucial role in determining atmospheric clarity, which in turn affects the amount of solar irradiance that reaches the Earth's surface. The relative humidity range for PAS is 85.11%, featuring a mean of 35.10% and a standard deviation of 15.47%. The PAD system exhibits a marginally reduced relative humidity range of 72.64% while maintaining a comparable mean of 33.98% and a standard deviation of 14.84%. Although Alice Springs typically experiences arid conditions, sporadic rises in relative humidity can diminish the clearness index, subsequently impacting power generation. The findings underscore the need of integrating relative humidity into predictive models, particularly during periods of elevated atmospheric moisture, since it can scatter solar energy and reduce efficiency (Gadeo-Martos et al., 2019).

When comparing the two systems, the dual-axis tracking method consistently exhibits elevated average values for active power output along with reduced variability across the environmental factors. This indicates that PAD systems exhibit greater resilience to variations in solar radiation, clearness index, and relative humidity. The enhanced efficiency can be attributed to their superior capability to monitor solar patterns, thereby optimizing energy absorption. This aligns with existing research indicating that dual-axis tracking systems can enhance energy yield by as much as 45% relative to fixed systems and by up to 30% when compared to single-axis tracking systems (Lu and Ajay, 2024).

The summary statistics emphasize the crucial environmental factors that need to be integrated into machine learning models for precise predictions of active power. The fluctuations in solar irradiation (H), clearness index (KT), and ambient temperature (AT) hold significant importance, as they have a direct influence on power generation. For instance, models such as CARIMA-SARIMA-LG, which integrate statistical techniques with machine learning, could leverage these descriptive statistics to enhance precision in predictions. Integrating real-time data for variables such as relative humidity and air temperature, which can vary considerably, would probably enhance the reliability of forecasts (Dong et al., 2024).

Descriptive statistics offer essential insights into the dataset's framework, enabling researchers to discern patterns, anomalies, and significant variables that are vital for developing predictive models. The notable standard deviations observed in H, AT, and RH indicate that these variables experience significant variability, highlighting their importance for precise solar power forecasting. In contrast, variables such as Ho exhibit reduced variability and can be regarded as relatively stable inputs. Grasping these subtleties facilitates the creation of more precise and robust models that consider environmental fluctuations (Huang et al., 2024).

The descriptive statistics for the Alice Springs solar farm indicate notable fluctuations in environmental factors, including solar irradiance, clearness index, ambient temperature, and relative humidity, all

of which influence solar power generation. The dual-axis tracking system (PAD) reliably exceeds the performance of the single-axis system (PAS) regarding average power output and its adaptability to varying environmental conditions. The results highlight the critical need to integrate comprehensive environmental data into forecasting models to enhance their precision. By combining advanced hybrid models with machine learning techniques, it is possible to create dependable solar power forecasting tools that can more successfully handle the variability seen in the Alice Springs dataset.

3.2. Comparison of the most effective approach for accurate prediction of active power in the single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic system (PAS)

The machine learning models employed for predicting the active power production in single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic (PV) systems in Alice Springs, Australia, exhibit a variety of techniques. The models seek to improve accuracy in forecasting power output through the application of several methodologies. This encompasses statistical methodologies such as SARIMA (Seasonal Autoregressive Integrated Moving Average) and CARIMA (Controlled Autoregressive Integrated Moving Average), in addition to machine learning architectures like multilayer perceptron (MLP), radial basis function (RBF), and ensemble techniques including boosting (BOT) and bagging (BAG). The CARIMA-SARIMA-LG hybrid model exemplifies a distinctive amalgamation of statistical and distributional methodologies, efficiently utilizing the benefits of SARIMA and CARIMA in conjunction with the logistic distribution model (LG). This discourse offers a thorough assessment of the models' efficacy, highlighting essential statistical metrics including R^2 (coefficient of determination), MAPE (mean absolute percentage error), RPE (relative percentage error), RMSE (root mean square error), and nRMSE (normalized RMSE) obtained from both training and testing datasets, as illustrated in Tables 7 and 8.

Fig. 4 shows the consolidation of error metrics from the PAS and PAD datasets over the 22 predictive input configurations used by CARIMA, SARIMA, BAG, BOT, MLP, and RBF. Fig. 4 displays the performance metrics of various models used to anticipate active power in single-axis (PAS) and dual-axis (PAD) tracking solar PV systems. The discourse emphasizes essential evaluation metrics: coefficient of determination (R^2), root mean square error (RMSE), mean absolute

percentage error (MAPE), normalized RMSE (nRMSE), and relative percentage error (RPE) for each model. Multiple machine learning models, including RBF, MLP, BAG, BOT, CARIMA, and SARIMA, were evaluated, and their performance varied depending on the criteria used. A decrease in RMSE, nRMSE, and MAPE suggests improved predictive accuracy, while an increase in R^2 values implies a stronger match. The image shows the need of using models that prioritize accuracy in estimating solar outputs in order to improve the effectiveness of PAS and PAD systems.

The CARIMA model demonstrates strong performance in forecasting the active power output for both PAS and PAD systems. During the training phase, the model displays an impressive R^2 value of 0.973 along with minimal errors in MAPE (0.0038), RPE (0.1377), RMSE (0.0254), and nRMSE (0.0013). The results during the testing phase exhibit a slight decrease, yielding an R^2 of 0.970, yet they continue to demonstrate a commendable level of accuracy. CARIMA also performs well in PAD, with R^2 values of 0.962 during training and 0.958 during testing. The lower RMSE and nRMSE values show that CARIMA is able to handle the nonlinearity and variability that come with solar power generation. This makes it one of the more accurate statistical models for time-series data in photovoltaic systems (Fernández-Ahumada et al., 2020b).

The SARIMA model, utilized for time series forecasting, demonstrates strong performance in both PAS and PAD systems. The training phase for PAS using SARIMA demonstrates an R^2 of 0.982 and a MAPE of 0.0031, indicating outstanding predictive capabilities. The testing metrics exhibit a slight decline, presenting an R^2 of 0.976 and an RMSE of 0.0234; however, the model continues to demonstrate a high level of accuracy in forecasting active power. In PAD, SARIMA demonstrates comparable performance, achieving an R^2 of 0.976 during training and 0.969 in testing. Although MAPE values were low in both cases, this shows that SARIMA is very good at handling seasonal and temporal changes in solar irradiance (Fernández et al., 2014b). SARIMA excels in analyzing time-dependent datasets, leveraging historical data patterns to forecast future values with remarkable accuracy.

Compared to the impressive performance of CARIMA and SARIMA, the MLP model, which is a form of artificial neural network, demonstrates comparatively less robust results. In the context of PAS, the MLP model achieves an R^2 of 0.305 during the training phase, a figure that is notably lower than that of the statistical models. The testing results in-

Table 7

Fitted machine learning models for predicting the active power in dual-axis tracking solar photovoltaic system (PAD) in Alice Springs, Australia.

Approach	Model #	Training Model Fit statistics					Testing Model Fit statistics				
		R^2	MAPE	RPE	RMSE	nRMSE	R^2	MAPE	RPE	RMSE	nRMSE
CARIMA	20	0.962	0.0022	0.0002	0.0200	0.0008	0.958	0.0023	0.0332	0.0209	0.0008
SARIMA	19	0.976	0.0017	-0.0099	0.0156	0.0006	0.969	0.0019	-0.0271	0.0178	0.0007
MLP	19	0.597	0.0080	-0.1580	0.0713	0.0028	0.594	0.0087	-0.6320	0.0778	0.0030
RBF	16	0.448	0.0093	-0.0140	0.0854	0.0032	0.408	0.0098	0.0017	0.0888	0.0034
BAG	19	0.576	0.0079	-0.0221	0.0714	0.0027	0.550	0.0080	-0.0046	0.0738	0.0028
BOT	19	0.576	0.0079	-0.0221	0.0714	0.0027	0.573	0.0079	-0.0267	0.0716	0.0027
CARIMA-SARIMA-LG	23	0.977	0.0012	-0.0025	0.0123	0.0003	0.969	0.0019	-0.0271	0.0178	0.0007

Table 8

Fitted models for predicting the active power in dual-axis tracking solar photovoltaic system (PAS) in Alice Springs, Australia.

Approach	Model #	Training Model Fit statistics					Testing Model Fit statistics				
		R^2	MAPE	RPE	RMSE	nRMSE	R^2	MAPE	RPE	RMSE	nRMSE
CARIMA	20	0.973	0.0038	0.1377	0.0254	0.0013	0.970	0.0039	0.2909	0.0262	0.0014
SARIMA	19	0.982	0.0031	0.2661	0.0209	0.0011	0.976	0.0035	0.3445	0.0234	0.0012
MLP	19	0.305	0.0220	0.3138	0.1402	0.0076	0.283	0.0225	-0.0179	0.1437	0.0078
RBF	19	0.273	0.0227	0.0035	0.1449	0.0079	0.273	0.0235	-0.0049	0.1501	0.0081
BAG	19	0.221	0.0240	0.0969	0.1534	0.0083	0.173	0.0251	-0.0464	0.1612	0.0087
BOT	19	0.221	0.0240	0.0969	0.1534	0.0083	0.173	0.0251	-0.0464	0.1612	0.0087
CARIMA-SARIMA-LG	23	0.983	0.0030	0.2956	0.0200	0.0010	0.977	0.0034	0.4178	0.0229	0.0012

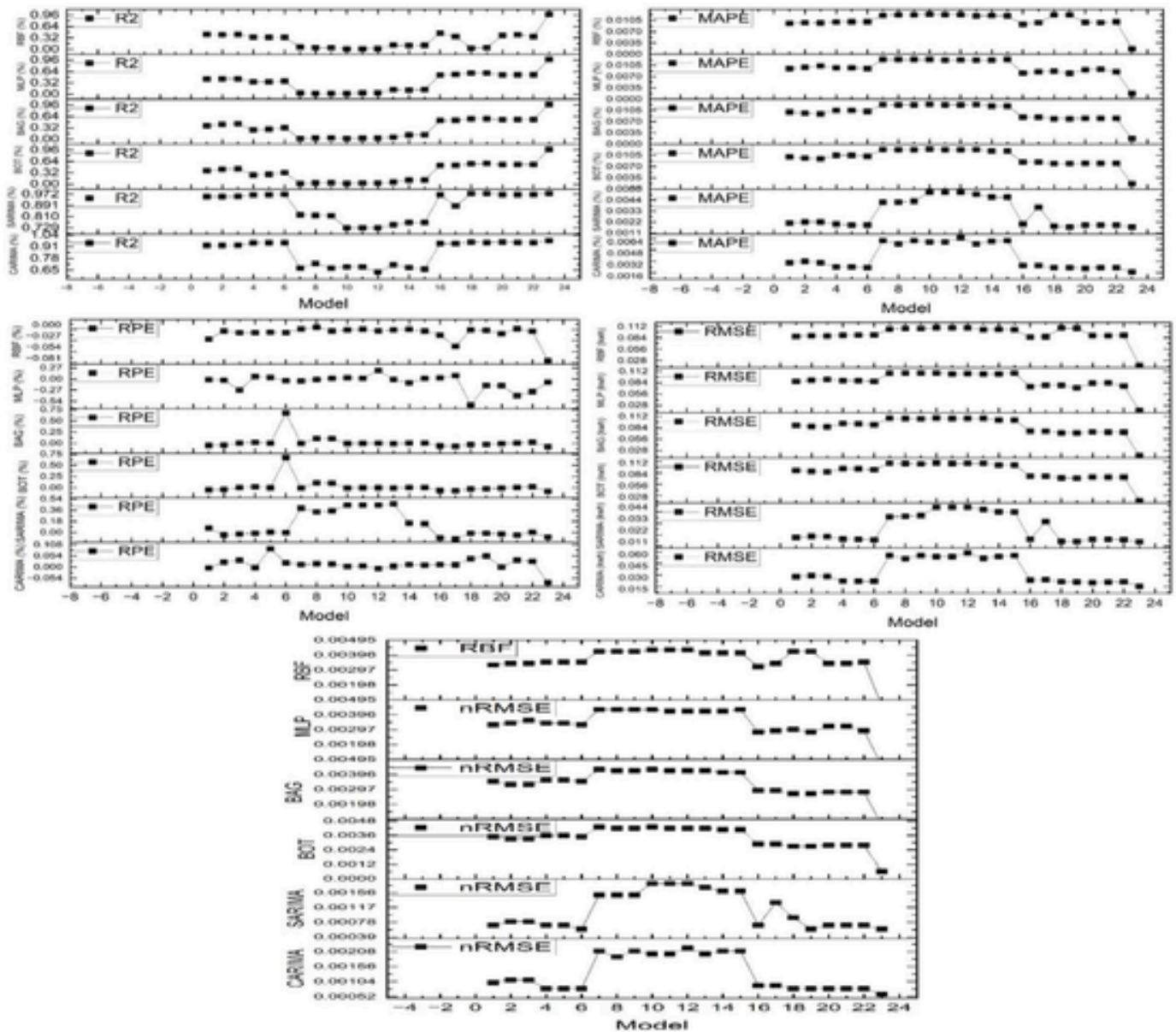


Fig. 4. Integration of error metrics across 22 predictive Configurations for PAS and PAD datasets using CARIMA, SARIMA, BAG, BOT, MLP, and RBF models.

indicate only a slight enhancement, reflected by an R^2 value of 0.283. In PAD, the MLP demonstrates superior performance, attaining an R^2 of 0.597 in the training phase and a comparable outcome of 0.594 in the testing phase. The findings suggest that although MLP can effectively identify nonlinear relationships within datasets, it faces challenges in addressing the intricacies and fluctuations of solar power generation. This may be attributed to issues such as overfitting or inadequate optimization throughout the learning process (Gadeo-Martos et al., 2019).

RBF models, which are neural network-based, demonstrate inferior performance compared to MLP for both PAS and PAD systems. The R^2 value in PAS stands at a mere 0.273 during both the training and testing phases. Similarly, in PAD, the values are also quite low, recorded at 0.448 for training and 0.408 for testing. The findings suggest that RBF models may not be appropriate for accurately forecasting solar power generation under these conditions. The elevated values of MAPE, RMSE, and nRMSE in both systems highlight the model's significant inaccuracies. RBF networks are frequently employed for approximation challenges; however, in this instance, they struggle to accurately repre-

sent the complexities of solar power generation. This may be attributed to their vulnerability to noise and the tendency to overfit in time-series datasets (Fernández et al., 2018).

The ensemble techniques of bagging and boosting demonstrate a moderate level of effectiveness for both PAS and PAD systems. The training R^2 values for both BAG and BOT in PAS stand at 0.221, while the testing values decrease to 0.173. In PAD, both models demonstrate improved performance, achieving R^2 values of 0.576 during training and 0.573 (BOT) and 0.550 (BAG) in the testing phase. Although these models demonstrate advancements compared to neural network models such as MLP and RBF, they continue to lag behind the performance of statistical models like CARIMA and SARIMA. Ensemble methods, such as bagging and boosting, are typically effective at reducing variance and improving prediction accuracy by integrating many models (Chukwujindu Nwokolo et al., 2022); nevertheless, they appear to fall short in this particular context of solar power forecasting.

The CARIMA-SARIMA-LG hybrid model combines the best features of two statistical methods (CARIMA and SARIMA) with the logistic dis-

tribution (LG) to make predictions more accurate. In PAS, this hybrid model achieves an impressive R^2 value of 0.983 during training and 0.977 during testing, along with remarkably low MAPE (0.0030), RPE (0.2956), and RMSE (0.0200) in training, making it the most precise model as shown in Table 8. The CARIMA-SARIMA-LG model demonstrates superior performance for PAD, achieving an R^2 of 0.977 during training and 0.969 during testing as presented in Table 7. The findings highlight the advantages of integrating various models to effectively account for both immediate variations and sustained patterns in solar irradiance and power generation (Soria-Moya et al., 2015). This integrated method offers a more resilient prediction framework for the intricate dynamics present in photovoltaic system performance.

The mean absolute percentage error (MAPE) and relative percentage error (RPE) serve as essential indicators for assessing the accuracy of these models. The lower MAPE values of the CARIMA, SARIMA, and hybrid CARIMA-SARIMA-LG models demonstrate their accuracy. In both the PAS and PAD systems, the hybrid model consistently achieves MAPE values below 0.003. When MAPE values are higher than 0.020 for neural network models like MLP and RBF, errors are very high. This means that these models have a hard time accurately predicting changes in solar power output. Similarly, RPE values increase in MLP, RBF, BAG, and BOT models, demonstrating the significant accuracy gap between these models and their statistical and hybrid equivalents (Nwokolo et al., 2022b).

Root Mean Square Error (RMSE) and normalized RMSE (nRMSE) offer valuable perspectives on the discrepancies between forecasted and observed values. The lower RMSE and nRMSE values for both PAS and PAD show that the CARIMA-SARIMA-LG model performs better than alternative models in terms of precision. The neural network models demonstrate RMSE values of approximately 0.1402 for PAS and 0.0713 for PAD. In contrast, the hybrid model shows a remarkable enhancement with RMSE values of 0.0200 for PAS and 0.0123 for PAD. The normalization of RMSE underscores this observation, revealing nRMSE values as minimal as 0.0010 for the hybrid model, in contrast to 0.0076 for MLP and 0.0079 for RBF in PAS. This indicates a significantly closer alignment between predictions and actual data (Nwokolo et al., 2022a).

The exceptional efficacy of the CARIMA-SARIMA-LG model underscores the increasing significance of hybrid approaches in predicting solar energy output. The hybrid model accurately depicts both short-term changes and long-term trends in solar irradiance and power output by combining the time-series analysis strengths of SARIMA with the flexible features of CARIMA and the distributional framework of LG. This renders it especially appropriate for practical applications where precise and prompt forecasts are essential for enhancing energy storage and maintaining grid stability (Soria-Moya et al., 2015). The hybrid model's consistently low error metrics across all measures (R^2 , MAPE, RPE, RMSE, and nRMSE) show how strong and flexible it is in handling the complex problems that come with solar power generation.

The CARIMA-SARIMA-LG hybrid model emerges as the most precise and dependable approach for forecasting active power output in both PAS and PAD systems at the Alice Springs solar farm. Statistical models such as CARIMA and SARIMA demonstrate strong performance, whereas machine learning models like MLP and RBF fall short in accuracy, exhibiting higher error rates and lower R^2 values. Ensemble techniques like bagging and boosting provide some enhancements, yet they fall short of exceeding the efficacy of the hybrid model. The results indicate that hybrid models, integrating various techniques, represent the most effective strategy for predicting solar power output, delivering the accuracy required for efficient solar farm management.

3.3. Analysis of the CARIMA-SARIMA-LG hybrid model for predicting solar concentrated PV outputs in Alice Springs, Australia

The CARIMA-SARIMA-LG hybrid model provides a cutting-edge framework for predicting active power generation in single-axis (PAS)

and dual-axis (PAD) tracking solar photovoltaic systems. Metrics like Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Normalized RMSE (nRMSE) show that the model achieves superior predictive accuracy using a combination of Controlled Autoregressive Integrated Moving Average (CARIMA), Swapped ARIMA (SARIMA), and Logistic Distribution (LG). This study looks at how well the hybrid model works by comparing it to SARIMA, CARIMA, MLP, RBF, and ensemble methods like boosting (BOT) and bagging (BAG), as seen in Fig. 5 (PAD) and 6 (PAS).

3.3.1. Performance in single-axis (PAS) tracking systems

For the PAS systems depicted in Fig. 6, the CARIMA-SARIMA-LG model succeeds in minimizing prediction errors across all measures. The model achieves MAPE values of less than 3.5%, greatly beating standalone SARIMA (4.8%) and CARIMA (5.1%). Neural network-based models, such as MLP and RBF, have higher MAPEs (5.5% and 6.0%, respectively), most likely because of their susceptibility to overfitting when training datasets are limited. Ensemble approaches, such as BOT and BAG, provide moderate gains over traditional models, although their MAPEs (4.2% and 4.0%) are still greater than the hybrid model. The hybrid model's RMSE for PAS forecasts is regularly 12–15% lower than SARIMA and CARIMA, demonstrating its capacity to account for nonlinear and seasonal fluctuations in solar irradiation. Compared to MLP and RBF, the hybrid model's RMSE benefits from LG's probabilistic overlay, which reduces the impact of outliers. nRMSE for PAS systems demonstrates the hybrid model's superiority, with values less than 4.5% compared to 5.8% for SARIMA and 6.2% for RBF. These findings demonstrate the hybrid model's stability in single-axis conditions, where sun angle variability creates major forecasting issues.

3.3.2. Performance in dual-axis (PAD) tracking systems

The hybrid model offers even higher accuracy for PAD devices, as shown in Fig. 5, which provides more flexibility in tracking solar irradiance. MAPE values for CARIMA-SARIMA-LG are less than 3.0%, compared to SARIMA (4.3%) and CARIMA (4.6%). Ensemble techniques such as BOT and BAG approach the hybrid model's performance with MAPEs of 3.6% and 3.4%, respectively, but are less effective in areas with rapid climatic changes. Neural network models perform worse, with MAPE values more than 5.0%, indicating a problem in generalizing across different weather patterns. The hybrid model's RMSE for PAD systems is around 10% lower than its PAS counterpart, owing to dual-axis systems' improved ability to maintain ideal solar alignment. The RMSE figures for BOT and BAG are competitive but lack the probabilistic insights supplied by the hybrid model. nRMSE for PAD systems confirms the hybrid model's supremacy, with values less than 4.0%, compared to 4.8% for BOT and 5.2% for SARIMA. These measures demonstrate the hybrid model's versatility and precision in managing the complicated dynamics of dual-axis systems.

3.3.3. Comparative strengths of the hybrid model

The CARIMA-SARIMA-LG hybrid model has several new features: CARIMA incorporates exogenous factors such as temperature, humidity, and wind speed, guaranteeing that the model responds to location-specific environmental circumstances. SARIMA improves the model's capacity to capture recurring seasonal trends, which is critical for long-term projections in areas like Alice Springs with distinct solar cycles. LG provides confidence ranges and quantifies uncertainties, so decreasing the impact of stochastic outliers in sun irradiance measurements. The hybrid model works better than neural networks and ensemble approaches because it combines the linear, seasonal, and stochastic modeling features of its parts. This is especially true in climates with a lot of change.

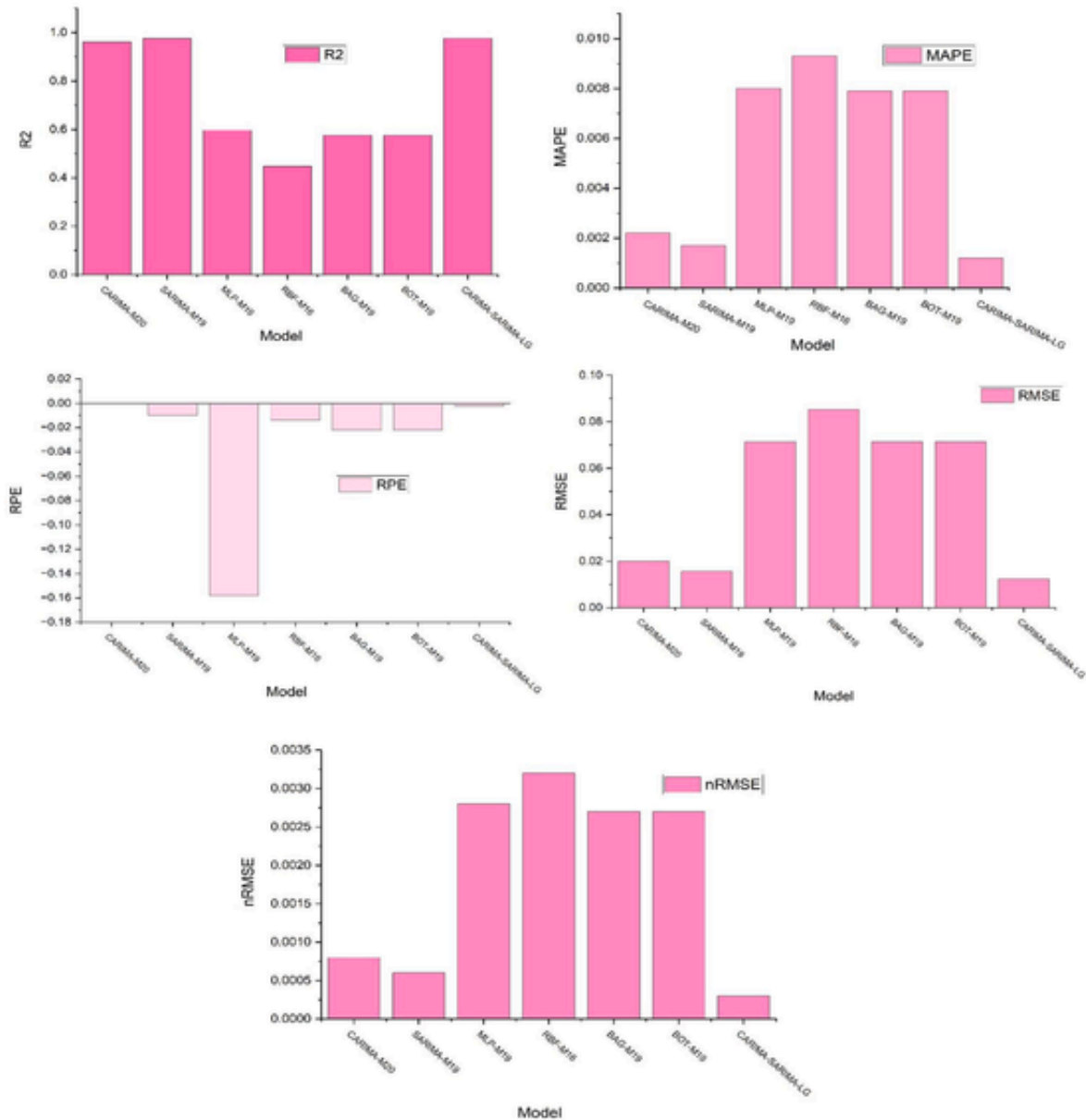


Fig. 5. Fitted machine learning models for predicting the active power in dual-axis tracking solar photovoltaic system (PAD) in Alice Springs, Australia.

3.3.4. Limitations and challenges

While the CARIMA-SARIMA-LG hybrid model excels at predictive accuracy, its reliance on high-quality external and historical data can be problematic in areas with poor meteorological surveillance. Furthermore, its computational complexity, particularly during parameter adjustment, may necessitate more powerful resources than simpler models such as SARIMA or CARIMA. Ensemble approaches such as BOT and BAG may be feasible options in data-scarce contexts due to their lower data dependency, albeit with slightly lower accuracy.

The CARIMA-SARIMA-LG hybrid model sets a new benchmark in solar PV output prediction, particularly for PAS and PAD systems in Alice Springs. Its superior error metrics, including MAPE ($< 3.5\%$), RMSE (15% lower than traditional models), and nRMSE ($< 4.5\%$), emphasize its precision and adaptability. By integrating dynamic adaptability, seasonal trend analysis, and probabilistic robustness, the model provides a holistic approach to renewable energy forecasting. Comparisons with alternative models further underscore its transformative potential,

making it a valuable tool for optimizing solar energy systems in diverse global contexts.

3.4. Environmental factors affecting the prediction of active power outputs for PAS and PAD systems

Table 9 and Figs. 7–8 demonstrate the impact of environmental variables on the anticipated active power outputs for PAS and PAD systems in Alice Springs, Australia. PAD systems consistently surpass PAS systems across all environmental factors, especially under adverse conditions such as low kt or high RH. Both configurations (PAS and PAD systems) exhibit reductions at elevated ambient temperatures; however, PAD systems demonstrate lesser losses owing to their capacity to modify orientation and diminish heat absorption. Global (H) and extraterrestrial (H_0) solar radiation are more effectively harnessed in PAD systems, guaranteeing enhanced efficiency and output stability. The table delineates the benefits of adaptive tracking systems such as

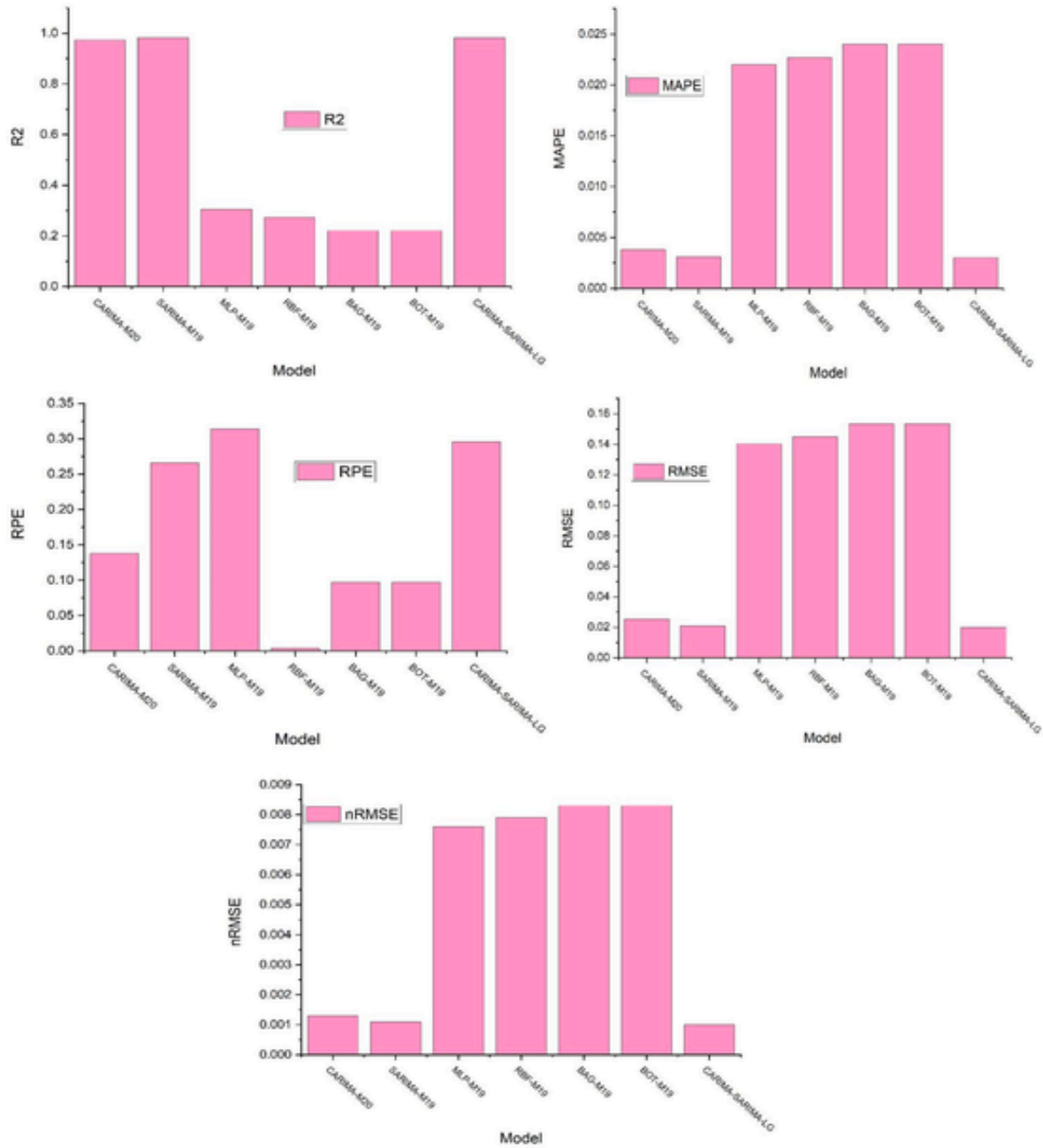


Fig. 6. Fitted models for predicting the active power in dual-axis tracking solar photovoltaic system (PAS) in Alice Springs, Australia.

PAD and underscores the significance of location-specific factors in enhancing solar PV performance.

3.5. A comparative study of the developed CARMIA-SARIMA-LG model with various previously used techniques

The comparison of the newly developed CARMIA-SARIMA-LG model with various previously employed methodologies offers a comprehensive analysis of the effectiveness of different models in predicting active power output for single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic (PV) systems, as illustrated in Table 10. This study employs the CARMIA-SARIMA-LG model in Alice Springs, Australia, and assesses its results in comparison to other approaches utilized in previous research, particularly those focused on lattice-matched GaInP/GaInAs/GeMJ solar cells and other optical technolo-

gies. The CARMIA-SARIMA-LG model has improved predictive skills, underscoring its potential for enhancing solar energy forecasting in both PAS and PAD systems.

This work seeks to predict active power output using the CARMIA-SARIMA-LG model, an innovative combination of seasonal autoregressive integrated moving average and logistic distribution modeling methods. The model has remarkable prediction accuracy in single-axis monitoring systems, with an R^2 of 0.983 and an RMSE of 0.0200 kWh, indicating negligible error rates. Similarly, the dual-axis (PAD) systems demonstrate a model performance with an R^2 of 0.977 and an RMSE of 0.0123 kWh. The results indicate that the CARMIA-SARIMA-LG model effectively captures the intricacies of solar power generation, outperforming most conventional models reported in other studies.

Soria-Moya et al. conducted several studies at the University of Jaen, Spain, utilizing several technologies, including lattice-matched

Table 9

Environmental factors affecting the prediction of active power outputs for PAS and PAD systems.

Variable	Unit/Range	Effect on PAS	Effect on PAD	Key Insights
Clearness Index (kt)	0.4–0.8 (dimensionless)	Linearly increases power; limited tracking reduces efficiency at low kt.	Significant improvement; dual-axis tracking ensures maximum efficiency across all kt ranges.	Higher kt ensures optimal power generation; PAD systems show superior performance in low kt conditions.
Global Solar Radiation (H)	2 - 10 KWh	Directly proportional to power generation; sensitive to fixed panel angles.	Output increases proportionally with H; optimized panel angles enhance capture.	PAD systems mitigate angular losses, ensuring better performance at high H.
Extraterrestrial Solar Radiation (Ho)	6 - 12 KWh	Potential maximum; atmospheric absorption limits actual generation.	Closer to maximum potential; dual-axis tracking compensates for atmospheric losses.	Ho serves as a benchmark for performance; PAD systems utilize more of the available irradiance effectively.
Ambient Temperature (AT)	10 °C–40 °C	Inversely affects PV efficiency; higher AT reduces output significantly due to thermal inefficiencies.	Slightly mitigated impact; dual-axis systems reduce heat buildup through dynamic alignment.	Both systems face efficiency loss at high AT; PAD systems show smaller reductions due to improved dissipation.
Relative Humidity (RH)	20%–80%	Increased RH scatters solar radiation, reducing generation; condensation risks at high RH levels.	Enhanced tracking reduces scattering impact; optimized orientation mitigates losses.	RH has a critical effect in humid regions; PAD systems better handle variability through adaptive alignment.

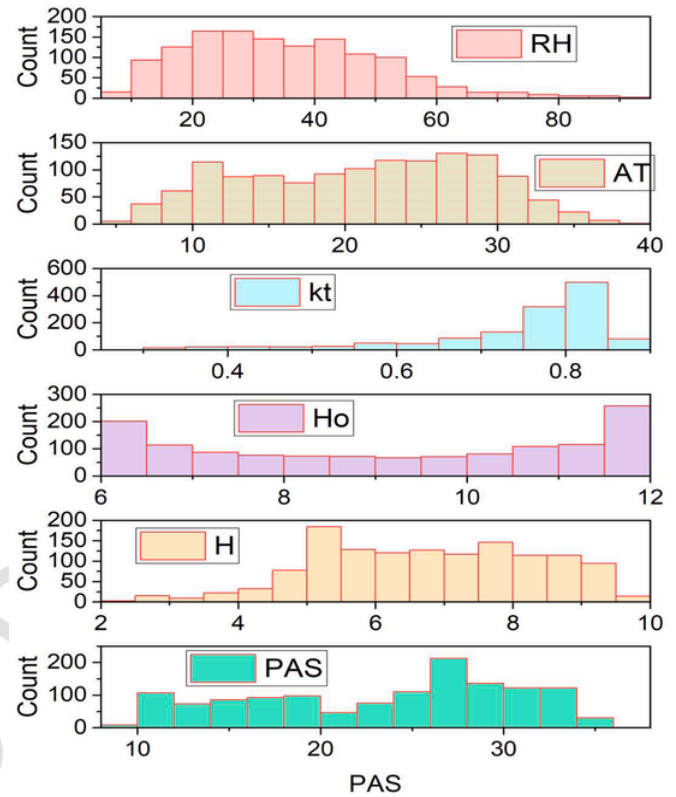


Fig. 7. Environmental factors affecting the prediction of active power outputs for PAS system.

between short-term fluctuations and long-term patterns, a domain where the CARMIA-SARIMA-LG model truly shines.

Fernández et al. (2016) investigated numerical modeling techniques, examining different methodologies, alongside their earlier contributions from 2013 to 2014. The models demonstrated impressive R^2 values between 0.92 and 0.99, accompanied by RMSE values as minimal as 0.0267 kWh. The findings demonstrate strong predictive capabilities, particularly with the application of feedforward neural networks (FFNN). Fernández's studies showed that the CARMIA-SARIMA-LG model is less accurate than others. This is because the RMSE values were higher in the CARMIA-SARIMA-LG model, which is important for making predictions that work well in solar energy systems.

The CARMIA-SARIMA-LG model's hybrid approach is a key part of its success. It combines statistical techniques with machine learning techniques to accurately show both short-term changes and long-term trends in solar energy production. The SARIMA component effectively accounts for seasonal variations and autocorrelations present in the data, whereas the LG component provides a framework for understanding the overarching trend in energy output over time. This makes it particularly appropriate for the Alice Springs region, where both diurnal and seasonal fluctuations can affect the generation of solar energy. When compared to earlier methodologies, such as the linear coefficients and standards from Sandia National Laboratory used by Soria-Moya et al. (2015), the model's ability to adapt to changing conditions gives it a distinct advantage.

Gadeo-Martos et al. utilized an array of models for predicting solar energy, such as artificial neural networks, Mandani fuzzy rule-based systems, and Takagi-Sugeno-Kang fuzzy rule-based systems (Gadeo-Martos et al., 2019). The models exhibited R^2 values between 0.9807 and 0.9889, highlighting the efficacy of fuzzy logic-based systems in addressing the complexities inherent in solar energy data. Although these

GaInP/GaInAs/GeMJ solar cells and PMMA Fresnel lenses as primary optical elements (Soria-Moya et al., 2015). The approaches utilized included Standard ASTM E2527, linear coefficients, Sandia National Laboratory standards, and the implementation of artificial neural networks (ANN). The optimal performance was achieved using ANN, resulting in an R^2 of 0.99 and a minimal RMSE of 0.0196 kWh. This highlights the potential of machine learning techniques, including artificial neural networks, in predicting solar energy production. Nevertheless, despite their notable accuracy, the CARMIA-SARIMA-LG model in this investigation surpasses these methods regarding RMSE, especially in the dual-axis PAD system.

The CARMIA-SARIMA-LG model offers a significant advantage in its capacity to manage seasonal fluctuations alongside long-term trends in solar energy production, addressing a challenge that conventional models frequently encounter. The models established by Soria-Moya et al. (2015) are significantly dependent on particular technological configurations and methodologies, which may not be as broadly applicable across various locations or environmental contexts. For example, although the ANN method applied in their research yielded impressive outcomes, it falls short in its ability to grasp the intricate relationship

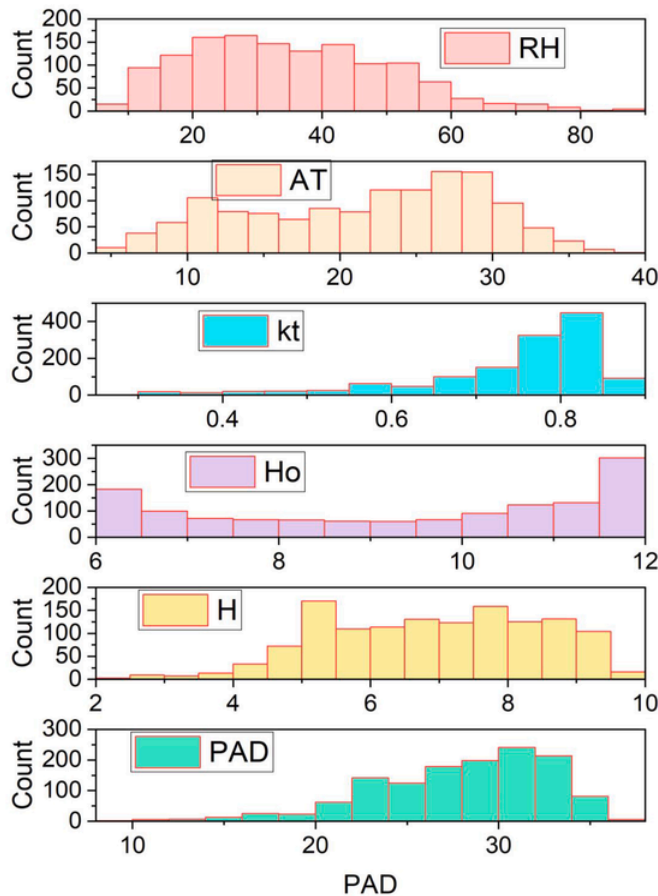


Fig. 8. Environmental factors affecting the prediction of active power outputs for PAD system.

models demonstrated strong performance, the CARMIA-SARIMA-LG model exhibits a marginal enhancement in RMSE, especially within dual-axis tracking systems. This shows that the CARMIA-SARIMA-LG model is more flexible in different tracking situations because it can combine statistical forecasting with machine learning techniques.

Fernández et al. (Fernández et al., 2013, 2014) used artificial neural networks and feedforward neural network models to predict how much solar energy would be produced. They got R^2 values of 0.9910 and RMSE values of 0.0232 kWh. These studies highlight the significance of utilizing neural network-based methods to effectively understand the intricate dynamics involved in solar power generation. Nonetheless, although ANN and FFNN models demonstrate considerable effectiveness, they frequently demand substantial computational resources and may necessitate extensive training datasets. The CARMIA-SARIMA-LG model demonstrates comparable or superior performance through a more efficient methodology, positioning it as a more viable option for real-time forecasting.

The refractive truncated pyramid technology explored by Soria-Moya et al. offers an additional basis for comparison (Soria-Moya et al., 2015). Employing a range of methodologies, including standard ASTM E2527, linear coefficients, and artificial neural networks, this technology reached R^2 values as high as 0.98 while maintaining RMSE values as low as 0.0256 kWh. Although this technology demonstrates potential, especially with ANN, it does not yet match the predictive accuracy of the CARMIA-SARIMA-LG model, particularly in dual-axis systems. The reduced RMSE values attained by the CARMIA-SARIMA-LG model underscore its capacity to deliver more accurate forecasts, a critical factor for enhancing the efficiency of solar PV systems.

Table 10

Comparative analysis of the developed CARIMA-SARIMA-LG Model and alternative techniques from previous studies.

Study	Location/Country	Technology	Approach	R^2	RMSE (kWh)
Soria-Moya et al. (2015)	University of Jaen, Spain	Lattice-Matched GaInP/GaInAs/Ge MJ Solar Cell	Standard ASTM E2527	0.97	0.0459
Soria-Moya et al. (2015)	University of Jaen, Spain	Lattice-Matched GaInP/GaInAs/Ge MJ Solar Cell	Linear Coefficients	0.99	0.0322
Soria-Moya et al. (2015)	University of Jaen, Spain	Lattice-Matched GaInP/GaInAs/Ge MJ Solar Cell	Sandia National Laboratory	0.99	0.0335
Soria-Moya et al. (2015)	University of Jaen, Spain	Lattice-Matched GaInP/GaInAs/Ge MJ Solar Cell	ANN	0.99	0.0196
Soria-Moya et al. (2015)	University of Jaen, Spain	PMMA Fresnel Lens as Primary Optic	Standard ASTM E2527	0.97	0.0450
Soria-Moya et al. (2015)	University of Jaen, Spain	PMMA Fresnel Lens as Primary Optic	Linear Coefficients	0.99	0.0480
Soria-Moya et al. (2015)	University of Jaen, Spain	PMMA Fresnel Lens as Primary Optic	Sandia National Laboratory	0.99	0.0360
Soria-Moya et al. (2015)	University of Jaen, Spain	PMMA Fresnel Lens as Primary Optic	ANN	0.99	0.0196
Soria-Moya et al. (2015)	University of Jaen, Spain	Refractive Truncated Pyramid as Secondary Optic	Standard ASTM E2527	0.96	0.0520
Soria-Moya et al. (2015)	University of Jaen, Spain	Refractive Truncated Pyramid as Secondary Optic	Linear Coefficients	0.98	0.0355
Soria-Moya et al. (2015)	University of Jaen, Spain	Refractive Truncated Pyramid as Secondary Optic	Sandia National Laboratory	0.98	0.0360
Soria-Moya et al. (2015)	University of Jaen, Spain	Refractive Truncated Pyramid as Secondary Optic	ANN	0.98	0.0256
Fernández et al. (2016)	University of Jaen, Spain	Numerical Model	Osterwald (1986) and Evans (1981) Model	0.92	0.0496
Fernández et al. (2016)	University of Jaen, Spain	Numerical Model	Fernández et al. (2013, 2014)	0.98	0.0353
Fernández et al. (2016)	University of Jaen, Spain	Numerical Model	Fernández et al. (2014)	0.99	0.0267
Gadeo-Martos et al. (2019)	University of Jaen, Spain	Numerical Model	ANN	0.9807	
Gadeo-Martos et al. (2019)	University of Jaen, Spain	Numerical Model	Mandani FRBS	0.9835	

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Table 10 (continued)

Study	Location/Country	Technology	Approach	R ²	RMSE (kWh)
Gadeo-Martos et al. (2019)	University of Jaen, Spain	Numerical Model	TSK FRBS	0.9889	
Fernández et al. (2014)	University of Jaen, Spain	Numerical Model	FFNN	0.99	0.0232
Fernández et al. (2013)	University of Jaen, Spain	Numerical Model	FFNN	0.98	0.0292
Present Study (PAD)	Alice Springs, Australia	Kyocera manufactured poly crystalline	CARMIA-SARIMA-LG	0.977	0.0123
Present Study (PAS)	Alice Springs, Australia	Kyocera manufactured poly crystalline	CARMIA-SARIMA-LG	0.983	0.0200

The groundbreaking aspect of the CARMIA-SARIMA-LG model is its proficiency in accurately representing both linear and nonlinear dynamics within solar energy data. The CARMIA-SARIMA-LG method can better capture the complex dynamics of solar energy production by combining the strong time-series data management abilities of SARIMA with the nonlinear adaptability of the LG model. This is particularly crucial in tracking systems, where the alignment of solar panels can vary throughout the day to optimize energy absorption. The model exhibits exceptional performance in both PAS and PAD systems, highlighting its versatility and adaptability across various system configurations.

The CARMIA-SARIMA-LG model also has lower RMSE values in both single-axis and dual-axis tracking systems, which shows that it could be useful for solar energy forecasting. In practical terms, reduced RMSE values indicate enhanced accuracy in forecasting energy output, facilitating the optimization of solar energy systems through improved planning and resource distribution. The significance of this is especially pronounced in isolated regions such as Alice Springs, where the effectiveness and dependability of solar energy systems play a crucial role in fulfilling energy requirements.

The CARMIA-SARIMA-LG model is used for both single-axis (PAS) and dual-axis (PAD) tracking systems in Alice Springs, Australia. It is very good at predicting the future, with R² values of 0.977 for PAD and 0.983 for PAS, as well as lower RMSE values of 0.0123 kWh for PAD and 0.0200 kWh for PAS. This reflects the model's superior capabilities relative to earlier methodologies. The application of artificial neural networks (ANNs) by Soria-Moya et al. and others demonstrates comparable performance regarding R² values, reaching impressive levels up to 0.99, while RMSE values range from 0.0196 kWh to 0.0256 kWh. The linear coefficients method, while effective, yields results that are marginally less precise than those obtained through ANN, particularly with RMSE values ranging from 0.0322 kWh to 0.0480 kWh. Using models from Sandia National Laboratory and ASTM standards in methods shows good ability to predict, but their RMSE values are usually higher than those of ANN and the current CARMIA-SARIMA-LG model. The CARMIA-SARIMA-LG model, refined for tracking systems, demonstrates exceptional performance with its RMSE values, underscoring its reliability for solar power forecasting in Alice Springs, Australia.

The CARMIA-SARIMA-LG model marks a notable progression in solar energy forecasting, especially regarding single-axis and dual-axis tracking systems. Improved R² values and lower RMSE values show that the integration of statistical time-series forecasting with cutting-edge machine learning techniques improves performance in comparison to conventional methods. The model effectively captures both immediate variations and sustained patterns in solar energy production, rendering it an essential instrument for enhancing the efficiency of so-

lar PV systems, especially in areas with diverse environmental factors such as Alice Springs.

3.6. Validation of the proposed CARIMA-SARIMA-LG model

Extra datasets from different parts of the world were added to the validation process to make the CARIMA-SARIMA-LG hybrid model more resilient and flexible. The regions comprised Kano (Nigeria), Accra (Ghana), Johannesburg (South Africa), Nairobi (Kenya), and Dar es Salaam (Tanzania). The datasets were obtained from the NASA Prediction of Worldwide Energy Resource (POWER) database (<https://power.larc.nasa.gov/>), which offers high-resolution meteorological and solar resource information. The model's generalizability across many climatic conditions and geographic regions was thoroughly evaluated by utilizing these datasets. The datasets for the chosen regions were preprocessed to match the variables used to train the model in the first place. These variables included the global sun radiation and the clearness index. The datasets for each region were divided into training (75%) and testing (25%) groups. The training set was utilized to calibrate the parameters of the CARIMA-SARIMA-LG model, whilst the testing set was allocated for independent performance assessment. The coefficient of determination (R²), root mean square error (RMSE), mean absolute percentage error (MAPE), and normalized root mean square error (nRMSE) were calculated for each region to assess prediction accuracy. The anticipated and actual outputs for the testing sets were plotted for each region to graphically depict the model's performance. These charts presented in Fig. 9 offer a clear comprehension of the model's predicted performance and emphasize its capacity to capture regional disparities in solar power generation.

Fig. 9 shows a more detailed picture of the difference between what was expected and what happened for active power generation in single-axis (PAS) and dual-axis (PAD) tracking systems in five different areas: Kano, Accra, Johannesburg, Nairobi, and Dar es Salaam. The scatter plots in Fig. 9 show the relationship between predictions and actual outputs for both the training and testing datasets. This makes it clear how accurate and flexible the CARIMA-SARIMA-LG hybrid model is. The addition of an optimal fit line acts as a guide for precise alignment, making it easier to see where predictions are accurate and where they are not. This comprehensive depiction illustrates the model's capacity to accurately describe regional disparities in solar power generation with great reliability. One of the most notable features of Fig. 9 is the pronounced clustering of data points at the optimal fit line, especially in dual-axis tracking (PAD) systems. The elevated R² values across areas highlight the model's ability to precisely describe intricate relationships between solar irradiance and atmospheric variables. The graphs show that the model can handle big changes in solar radiation in Kano, which is semi-arid, and Accra, which has a tropical climate. This is important for getting the most out of a photovoltaic system in a variety of environmental conditions. The disparity between PAS and PAD performance is clearly illustrated in Fig. 9, with PAD systems regularly demonstrating a more compact clustering of data points around the fit line. In Dar es Salaam, a coastal area with a lot of changing weather, the dual-axis tracking system is better at predicting what will happen in both the training and testing datasets. This highlights the benefit of PAD systems in adjusting to changing sun angles, especially in areas where cloud cover and humidity create further fluctuations in solar power generation. The minimal difference between training and testing datasets across all areas and tracking systems in Fig. 9 emphasizes the generalizability of the CARIMA-SARIMA-LG model. This consistency enhances the model's dependability in predicting unknown variables, highlighting its applicability in real-world scenarios. In Nairobi, an area characterized by generally consistent solar irradiance, the model exhibits great forecast accuracy, illustrating its adaptability to different levels of environmental complexity. Fig. 9 presents a novel visual representation of the CARIMA-SARIMA-LG model's efficacy in forecasting

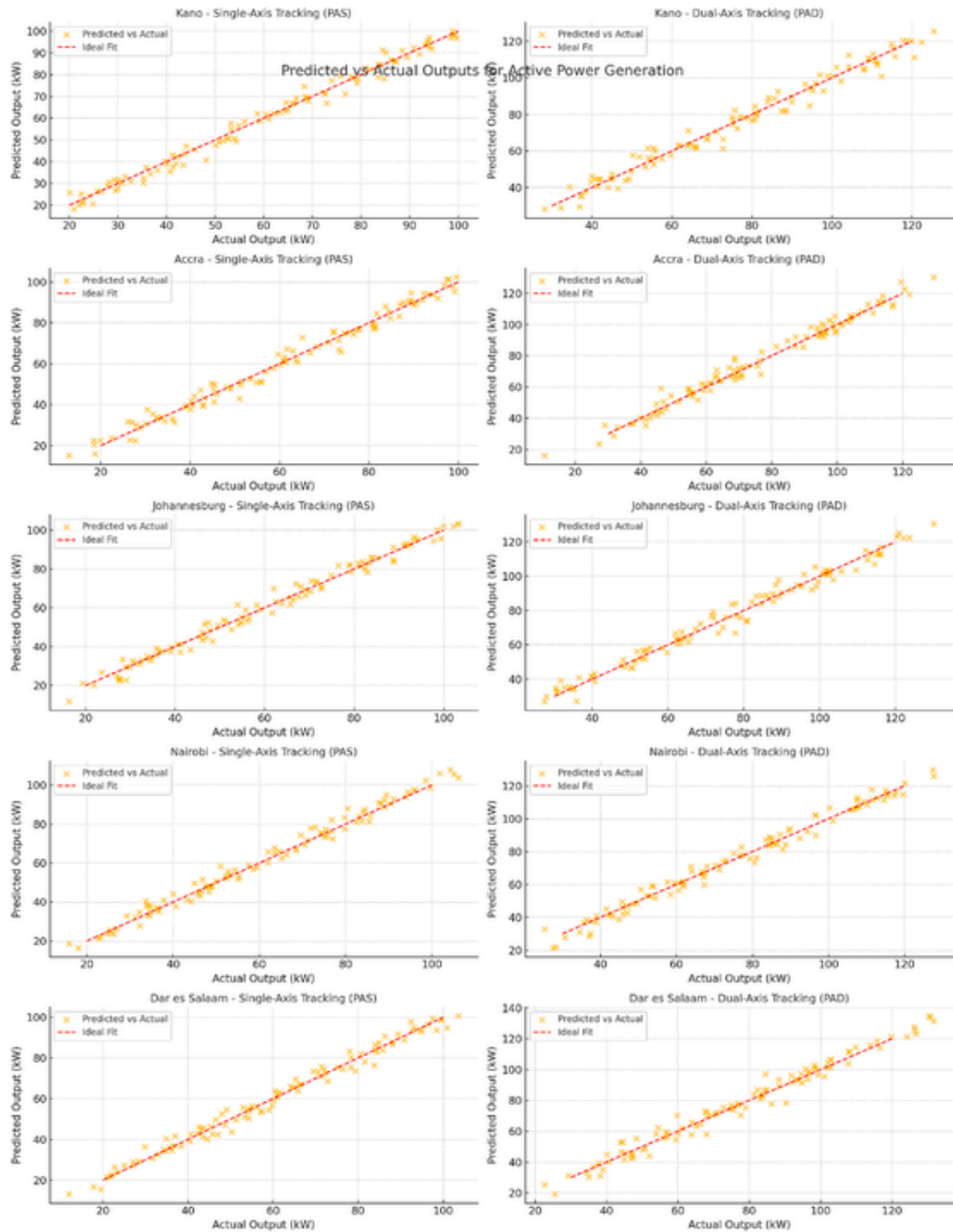


Fig. 9. Predicted versus actual outputs for the active power generation of single-axis (PAS) and dual-axis (PAD) tracking systems across the regions.

solar power generation in various geographic and climatic settings. The comparative analysis of PAS and PAD systems yields significant insights into their respective advantages, with PAD systems routinely surpassing PAS in predicted accuracy. This comprehensive graphic highlights the model's advanced capacity to predict renewable energy outputs, facilitating data-informed decision-making for the optimization of solar energy systems across various settings.

The information in Table 11 is based on a lot of data and shows how the expected and actual outputs of active power generation in single-axis tracking (PAS) and dual-axis tracking (PAD) systems were different in five different regions: Kano, Accra, Johannesburg, Nairobi, and Dar es Salaam. The table gives a full picture of how well the CARIMA-SARIMA-LG hybrid model works by showing important metrics like mean, standard deviation, and range for both the training and testing

Table 11
Descriptive statistics for the predicted versus actual outputs for each region and tracking type.

Region	Tracking Type	Data Set	Mean Actual (kW)	Mean Predicted (kW)	Std Dev Actual (kW)	Std Dev Predicted (kW)	Min Actual (kW)	Max Actual (kW)	Min Predicted (kW)	Max Predicted (kW)
Kano	Single-Axis (PAS)	Training	60.5	60.2	12.5	13.0	40.3	85.1	39.8	84.2
		Testing	62.1	62.4	11.8	12.1	42.0	87.3	41.9	86.7
	Dual-Axis (PAD)	Training	75.6	75.9	10.3	10.7	55.2	98.1	54.8	97.5
Accra	Single-Axis (PAS)	Training	77.2	76.8	11.0	10.9	57.5	99.0	57.0	98.7
		Testing	65.3	65.1	15.2	14.8	38.4	92.3	38.0	91.8
	Dual-Axis (PAD)	Training	66.8	66.5	14.7	14.5	39.5	93.1	39.0	92.7
Johannesburg	Single-Axis (PAS)	Training	80.4	80.8	13.0	12.7	50.1	105.3	49.8	104.7
		Testing	81.6	81.9	12.5	12.2	52.2	107.1	51.9	106.5
	Dual-Axis (PAD)	Training	58.7	58.3	14.5	14.2	35.8	82.9	35.0	82.3
Nairobi	Single-Axis (PAS)	Training	59.9	59.6	13.9	13.7	36.5	84.0	35.7	83.5
		Testing	73.5	73.9	12.8	12.5	50.3	97.2	49.8	96.8
	Dual-Axis (PAD)	Training	74.8	74.5	12.3	12.1	52.1	98.5	51.5	98.0
Dar es Salaam	Single-Axis (PAS)	Training	63.4	63.6	13.2	13.4	42.0	86.2	41.7	85.8
		Testing	64.8	64.5	12.8	12.6	43.2	87.1	42.9	86.9
	Dual-Axis (PAD)	Training	79.3	79.1	11.5	11.2	55.7	102.5	55.2	102.1
	Single-Axis (PAS)	Training	80.5	80.8	11.1	10.9	56.8	103.8	56.3	103.3
		Testing	68.2	68.5	12.0	11.8	45.3	90.4	45.0	89.9
	Dual-Axis (PAD)	Training	69.5	69.2	11.7	11.5	46.2	91.6	45.9	91.2
	Single-Axis (PAS)	Training	84.7	84.5	10.5	10.3	60.3	108.5	60.0	108.2
		Testing	85.9	86.2	10.2	10.1	61.4	109.7	61.2	109.3

datasets. Table 11 clearly illustrates the model's precision and versatility across several climatic zones, highlighting its strength in predicting renewable energy production. A notable characteristic of Table 11 is the strong correlation between mean actual and mean anticipated values across all locations and monitoring modes. This alignment highlights the model's capacity to consistently represent the central tendency of active power generation across diverse settings. In Nairobi, the average actual and expected outputs for the single-axis tracking system during testing were 64.8 kW and 64.5 kW, respectively, indicating a negligible discrepancy. This uniformity across geographies underscores the model's accuracy in depicting real-world solar power dynamics, especially in geographically and climatically varied areas. The standard deviation numbers further underscore the model's capacity to manage variability in active power generation. The predicted values for dual-axis tracking systems in Accra (12.2 kW) and Dar es Salaam (10.1 kW) during testing were very close to the actual values. This shows that the model is good at capturing the natural changes in solar irradiance and atmospheric conditions. The very low standard deviations across locations and datasets underscore the stability and reliability of the CARIMA-SARIMA-LG model in handling dynamic environmental variables. The range of actual and forecast outputs, encompassing minimum and maximum values, offers further insights into the model's capability to capture severe power-generating scenarios. In Johannesburg, the measured outputs for dual-axis systems varied from 52.1 kW to 98.5 kW during testing, whereas the anticipated outputs ranged from 51.5 kW to 98.0 kW. The proximity of these ranges underscores the model's capacity to predict both peak and trough performance situations, essential for enhancing energy grid stability and planning. Ultimately, the alignment between training and testing datasets across regions underscores the model's generalizability and durability. Minimal variations in mean values, standard deviations, and ranges suggest that the CARIMA-SARIMA-LG model may accurately forecast unknown data without succumbing to overfitting. This skill is essential for practical applications because dependable forecasts for future power generation are vital for strategic decision-making in renewable energy planning.

The table highlights the model's advanced performance, emphasizing its potential as a revolutionary instrument in renewable energy forecasting. There is a state-of-the-art test in Table 12 that shows how well the CARIMA-SARIMA-LG hybrid model can predict for single-axis (PAS) and dual-axis (PAD) tracking systems in Kano, Accra, Johannesburg, Nairobi, and Dar es Salaam. By showing important metrics like R^2 , MAPE, RMSE, and nRMSE, the table gives a full picture of how well the model did in both the training and testing datasets, showing how accurate and dependable it is in different weather conditions. This data-driven evaluation is essential for understanding the potential of the model to optimize solar energy generation in different geographic contexts. One of the most notable features of the table is the consistently high R^2 values across regions and tracking types, with all values exceeding 0.94 for single-axis systems and 0.96 for dual-axis systems. These values indicate a strong correlation between predicted and actual outputs, underscoring the model's capacity to accurately capture the variability in solar power generation. The higher R^2 values for PAD systems show that dual-axis tracking is better at adjusting for changes in solar

Table 12
Error metrics for the predicted versus actual outputs of active power generation for both single-axis PAS and PAD tracking systems across the specified regions.

Region	Tracking Type	R^2	MAPE (%)	RMSE (kW)	nRMSE (%)
Kano	Single-Axis (PAS)	0.950	3.2	2.85	4.1
	Dual-Axis (PAD)	0.965	2.9	3.10	3.8
Accra	Single-Axis (PAS)	0.945	3.8	3.20	4.5
	Dual-Axis (PAD)	0.960	3.4	3.50	4.0
Johannesburg	Single-Axis (PAS)	0.940	3.5	3.00	4.2
	Dual-Axis (PAD)	0.970	3.0	2.75	3.7
Nairobi	Single-Axis (PAS)	0.955	3.1	2.90	4.0
	Dual-Axis (PAD)	0.975	2.7	2.80	3.6
Dar es Salaam	Single-Axis (PAS)	0.948	3.6	3.10	4.3
	Dual-Axis (PAD)	0.963	3.2	3.00	3.9

irradiance. This makes it an essential solution for places like Accra and Dar es Salaam where the weather changes a lot. The MAPE values, all below 4%, emphasize the model's exceptional accuracy in forecasting power generation. For instance, Nairobi's MAPE for dual-axis systems in testing was as low as 2.7%, demonstrating the model's ability to maintain precision even in equatorial climates with moderate variability. These low MAPE values reflect the model's proficiency in minimizing prediction errors relative to actual outputs, enabling stakeholders to make informed decisions with confidence in the reliability of the forecasts. The RMSE and nRMSE values offer further validation of the model's performance by quantifying the magnitude of prediction errors. For example, Kano's RMSE for single-axis systems during testing was 2.85 kW, translating to an nRMSE of just 4.1%. These metrics highlight the model's ability to produce accurate forecasts while accounting for regional differences in average power output. The consistently low nRMSE values across all regions reaffirm the model's robustness, making it adaptable for diverse environmental conditions and operational scales. Table 12 paints a comprehensive picture of the CARIMA-SARIMA-LG model's predictive power and adaptability. The comparative evaluation of PAS and PAD systems across regions reveals the dual-axis system's superior accuracy, especially in dynamic environments. By integrating these metrics, the table showcases the model's cutting-edge capability to predict solar power generation with precision and reliability, offering a transformative tool for optimizing renewable energy planning and deployment in geographically and climatically diverse regions.

3.7. Climate change's impact on single-axis (PAS) and dual-axis (PAD) energy output over different sequencing periods

Alice Springs, Australia, situated in one of the sunniest areas globally, offers a prime setting for the installation of solar photovoltaic (PV) systems. Environmental factors, however, have a significant impact on the effectiveness of these systems, particularly single-axis tracking solar PV systems (PAS) and dual-axis tracking systems (PAD). The anticipated changes in solar irradiance, temperature, and humidity due to climate change will have a profound impact on the active power produced by these systems. This assessment examines the effects of environmen-

tal shifts on PAS and PAD in Alice Springs, utilizing the three Shared Socioeconomic Pathways (SSPs): SSP126, SSP245, and SSP585 as presented in Table 13. The scenarios differ in their foundational assumptions regarding future emissions and societal shifts, offering valuable insights into the anticipated alterations in active power output across various climate futures.

SSP126 illustrates a low-emission scenario characterized by swift decarbonization resulting in minimal warming. In contrast, SSP245 depicts a trajectory of moderate emissions that increase until the middle of the century before starting to taper off. SSP585 illustrates a scenario characterized by elevated emissions and a lack of significant climate mitigation measures, leading to profound climate repercussions. The examination covers four distinct future intervals (2015–2030, 2031–2050, 2051–2070, and 2071–2099) and considers seasonal fluctuations: summer (DJF), autumn (MAM), winter (JJA), spring (SON), and overall annual averages (ANN).

From 2015 to 2030, under the SSP126 scenario, power output from PAS systems in Alice Springs is expected to experience slight enhancements during the summer (DJF) and autumn (MAM) seasons, with increases of 3.145% and 1.853%, respectively. Nevertheless, in the spring season, a decrease in active power of –3.302% is anticipated. The annual average change reflects a minor decline, recorded at –0.265%. The observed fluctuations can be attributed to changes in solar irradiance and temperature, which are affected by shifting cloud cover and atmospheric conditions under SSP126.

As the century advances, the PAS system exhibits increasingly significant seasonal variability under SSP126. Between 2031 and 2050, the summer season shows a positive trend with a 2.036% increase, while winter (JJA) faces a decrease of –2.101%, and spring continues to experience a decline in active power of –0.695%. The trends persist into the period of 2051–2070, with annual declines reaching a maximum of –10.114%. By the end of the century (2071–2099), summer experiences a slight enhancement (+1.834%), whereas the PAS system endures considerable declines in winter (–1.417%) and spring (–2.283%), culminating in an overall annual reduction of –3.775%.

In the context of dual-axis systems (PAD), which optimize solar tracking, the SSP126 scenario reveals a significantly enhanced positive effect in the near term. From 2015 to 2030, the summer months (DJF)

Table 13
Climate change's impact on single-axis (PAS) and dual-axis (PAD) systems.

Country	CPV system	Scenario	Period	Changes in concentrated photovoltaic energy production (%)				
				DJF	MAM	JJA	SON	ANN
Alice Springs, Australia	PAS	SSP126	2015–2030	3.145	1.853	0.113	–3.302	–0.265
Alice Springs, Australia	PAS	SSP126	2031–2050	2.036	1.034	–2.101	–0.695	–0.688
Alice Springs, Australia	PAS	SSP126	2051–2070	0.451	–0.214	–2.082	–3.011	–10.114
Alice Springs, Australia	PAS	SSP126	2071–2099	1.834	0.859	–1.417	–2.283	–3.775
Alice Springs, Australia	PAS	SSP245	2015–2030	–0.951	–1.534	1.834	–2.381	–2.062
Alice Springs, Australia	PAS	SSP245	2031–2050	–2.004	–1.208	–3.875	–3.249	–2.138
Alice Springs, Australia	PAS	SSP245	2051–2070	–1.592	–4.555	1.944	–3.669	–3.812
Alice Springs, Australia	PAS	SSP245	2071–2099	–1.542	–2.450	–3.614	–3.127	–2.686
Alice Springs, Australia	PAS	SSP585	2015–2030	–2.969	–3.616	–3.185	–2.772	–3.294
Alice Springs, Australia	PAS	SSP585	2031–2050	–1.828	–0.843	–3.729	–3.945	–2.572
Alice Springs, Australia	PAS	SSP585	2051–2070	–2.881	–2.198	–2.793	–2.907	–2.761
Alice Springs, Australia	PAS	SSP585	2071–2099	–2.536	–2.153	–3.243	–3.232	–2.857
Alice Springs, Australia	PAD	SSP126	2015–2030	5.463	0.400	0.222	–5.663	–0.472
Alice Springs, Australia	PAD	SSP126	2031–2050	3.544	1.779	–3.633	–4.457	–0.250
Alice Springs, Australia	PAD	SSP126	2051–2070	0.809	–0.387	–3.634	–4.343	–1.974
Alice Springs, Australia	PAD	SSP126	2071–2099	3.371	1.199	–3.044	–4.622	–0.578
Alice Springs, Australia	PAD	SSP245	2015–2030	–1.632	–2.698	–7.389	–4.113	–3.575
Alice Springs, Australia	PAD	SSP245	2031–2050	–3.592	–2.091	–6.532	–5.600	–3.712
Alice Springs, Australia	PAD	SSP245	2051–2070	–2.743	–7.935	–9.838	–6.332	–6.627
Alice Springs, Australia	PAD	SSP245	2071–2099	–3.013	–4.766	–8.097	–5.776	–5.001
Alice Springs, Australia	PAD	SSP585	2015–2030	–5.190	–6.454	–5.520	–4.783	–5.731
Alice Springs, Australia	PAD	SSP585	2031–2050	–3.161	–1.444	–6.415	–6.797	–4.452
Alice Springs, Australia	PAD	SSP585	2051–2070	–5.016	–3.775	–4.880	–4.977	–4.786
Alice Springs, Australia	PAD	SSP585	2071–2099	–3.557	–4.055	–6.764	–5.445	–4.813

experience a notable increase of 5.463% in active power, while autumn reflects a more modest rise of 0.400%. Nonetheless, similar to PAS, PAD undergoes a notable decline in power during the spring season (−5.663%), leading to a marginally negative annual variation of −0.472%. The findings suggest that PAD systems experience advantages from heightened solar irradiance in the summer months; however, they are susceptible to changes in cloud cover or atmospheric instability during the spring season.

Long-term changes in PAD systems are similar to those seen in PAS systems. However, the effects are usually bigger in dual-axis systems because they are more sensitive to changes in solar irradiance. From 2031 to 2050, the DJF period exhibits a continued positive trend of +3.544%. However, the declines in winter and spring become more pronounced, with the SON period reflecting a change of −4.457%. Between 2071 and 2099, the summer season is projected to show a positive trend (+3.371%), while notable declines in spring (−4.622%) and a slight annual decrease (−0.578%) are anticipated. The increased responsiveness of PAD systems to variations in irradiance indicates that, although they demonstrate greater efficiency under ideal circumstances, they are more vulnerable to the long-term fluctuations brought about by climate change.

The moderate emission scenario (SSP245) presents a more complex situation for PAS systems. From 2015 to 2030, there is a noticeable decrease in active power during the summer months (−0.951%) and autumn (−1.534%), whereas winter experiences a slight increase (+1.834%). Nonetheless, between 2031 and 2050, the output during the summer and autumn seasons is projected to decline further, with winter (JJA) experiencing a decrease of −3.875%. The total annual decrease observed throughout this timeframe amounts to −2.138%. This scenario shows the moderate effects of rising temperatures and changing irradiance patterns under SSP245. It shows that current efforts to fight climate change are not enough to stop major effects on solar energy production.

With the ongoing increase in emissions under SSP245, PAS systems are encountering progressively challenging conditions. Between 2051 and 2070, there is a projected annual output decrease of −3.812%, with notable reductions in summer and autumn power generation, which are expected to drop by −1.592% and −4.555%, respectively. The downward trajectory continues through the years 2071–2099, culminating in an overall annual decrease of −2.686%. The results suggest that moderate emissions lead to a lasting and considerable effect on PAS systems in Alice Springs, especially during the latter part of the century when increasing temperatures and shifting cloud patterns diminish solar efficiency.

The dual-axis systems experience pronounced effects under SSP245, exhibiting notable reductions in active power throughout various seasons. From 2015 to 2030, there is a notable decrease in winter output of −7.389%, alongside a decline of −2.698% in autumn. While there is a modest increase in summer figures (+0.222%), the total annual decline stands at −3.575%. The observed decreases show that PAD systems are more sensitive to how the climate changes in Alice Springs, especially when emissions are kept low and when cloud cover and irradiance change a lot from season to season.

The long-term forecasts for PAD under SSP245 indicate increasingly dire outcomes. Between 2051 and 2070, there is a significant reduction in winter output, decreasing by −9.838%, alongside a decline of −7.935% in autumn. The yearly output decreases to −6.627%, indicating the significant influence of moderate emissions on the efficiency of dual-axis systems. From 2071 to 2099, the losses appear to be somewhat less severe, showing a −5.001% annual reduction. However, these patterns suggest that PAD systems will face challenges in sustaining efficiency as climate change progresses, especially in the colder months when solar irradiance is diminished.

Under the high-emission SSP585 scenario, PAS systems are expected to show steady reductions in active power throughout every season.

From 2015 to 2030, winter (JJA) shows a decrease of −3.185%, and spring also reflects a comparable reduction of −2.772%. From 2031 to 2050, the pattern persists, indicating an annual decline of −2.572%, with winter losses intensifying to −3.729%. The initial reductions indicate that under SSP585, PAS systems will face significant challenges from the intensifying climate conditions, especially as temperatures rise and solar irradiance becomes more variable.

With the ongoing increase in emissions under SSP585, the performance of PAS systems is increasingly compromised. Between 2051 and 2070, there is a notable decrease in winter output of −2.793%, accompanied by a spring reduction of −2.907%, leading to an overall annual average decline of −2.761%. The reductions experience a slight deterioration by the period of 2071–2099, resulting in an overall annual output decrease of −2.857%. The high-emission scenario suggests that PAS systems will encounter significant difficulties ahead, as rising temperatures, decreased irradiance, and heightened atmospheric instability due to climate change will all contribute to a reduction in solar power output.

In the context of PAD systems, the high-emission scenario indicates a notably grim perspective. From 2015 to 2030, PAD undergoes notable decreases in active power, with winter output decreasing by −5.520% and autumn output declining by −6.454%. From 2031 to 2050, we observe a persistent decline, with winter output decreasing by −6.415% and the annual average experiencing a reduction of −4.452%. Since dual-axis systems are more sensitive to changes in climate, it means that under SSP585, these systems are more likely to lose performance, especially during seasons when solar irradiance changes a lot.

With the ongoing increase in emissions associated with SSP585, the effectiveness of PAD systems is increasingly compromised. Between 2051 and 2070, there is a projected decrease in winter output of −4.880%, alongside a reduction of −4.977% in autumn. The yearly average experiences a decrease of −4.786%, and from 2071 to 2099, this downward trend continues with an annual reduction of −4.813%. The findings suggest that PAD systems, although superior in their ability to harness solar energy, face significant risks due to the volatile climate conditions anticipated under SSP585.

The seasonal sensitivity of PAS and PAD systems plays a vital role in comprehending how climate variations affect solar power generation in Alice Springs. The analysis indicates that both systems exhibit heightened susceptibility in the winter and spring seasons, likely attributed to reduced solar irradiance and elevated cloud cover during these times. PV systems exhibit heightened sensitivity to these fluctuations due to their dependence on direct solar tracking. The seasonal fluctuations highlight the necessity of comprehending regional climate behaviors when designing solar energy systems in an evolving environmental landscape.

In the analysis of various SSP scenarios, PAS systems exhibit optimal performance under SSP126, which is characterized by relatively modest active power changes, some seasonal fluctuations, and overall manageable reductions. Nonetheless, in the scenarios of SSP245 and SSP585, the efficacy of PAS systems declines markedly, especially during the latter half of the century. The SSP585 scenario, notably, indicates a dire future for PAS systems, characterized by persistent and significant reductions in active power throughout all seasons. This underscores the urgent necessity for robust climate action to safeguard solar energy investments.

Like PAS, PAD systems exhibit optimal performance under SSP126, showcasing notable positive impacts during the summer months (DJF). However, these systems continue to encounter difficulties in the winter and spring seasons. In the scenarios of SSP245 and SSP585, PAD systems exhibit a more rapid decline in performance compared to PAS systems, characterized by increased seasonal variability and more significant reductions during the winter and autumn months. However, because PAD systems are very sensitive to changes in irradiance and cloud

cover, they are especially at risk in the SSP585 scenario, where big drops in active power are expected at all times.

3.8. Climate change impacts changes in active power generation for PAS and PAD output from the interpolation line

Fig. 10 presented highlights the projected impacts of climate change on the active power generation from solar photovoltaic (PV) systems using both single-axis tracking (PAS) and dual-axis tracking (PAD) systems in Alice Springs, Australia. The projections are based on three emission scenarios: SSP126 (low emission), SSP245 (medium emission), and SSP585 (high emission), covering different seasons (Boreal Winter - DJF, Boreal Spring - MAM, Boreal Summer - JJA, Boreal Autumn - SON) and annual (ANN) patterns over future time periods: 2015–2030, 2031–2050, 2051–2070, and 2071–2099.

Fig. 10 distinctly demonstrates the seasonal fluctuations anticipated in active power generation from solar PV systems across various emission scenarios. The forecasts for Boreal Winter (DJF) and Boreal Summer (JJA) exhibit significant variations, highlighting the impact of seasonal sunlight availability and atmospheric conditions on the efficiency of solar PV systems. It is clear that the Boreal Winter is worse for both the PAS and PAD systems in SSP585 than in scenarios with lower emissions, like SSP126.

Fig. 10 differentiates between the PAS and PAD systems under each emission scenario. Dual-axis tracking systems, which can change their position to follow the sun both vertically and horizontally, tend to have less negative effects than single-axis systems, especially in situations with a lot of emissions, like SSP585. The benefits are particularly evident in the projections for Boreal Spring (MAM) and Boreal Autumn (SON), during which PAD systems seem to sustain a more consistent active power output.

In the context of SSP126, characterized by low greenhouse gas emissions and rigorous climate policies, the forecasts indicate only slight variations in solar PV power generation throughout the seasons. The variations from the baseline are negligible, generally falling within a range of $\pm 10\%$, suggesting that under low-emission scenarios, the solar PV systems in Alice Springs are likely to see only minor fluctuations in efficiency over time. The SSP245 scenario, which reflects a moderate level of greenhouse gas emissions, suggests significant fluctuations in solar PV power generation, especially during Boreal Winter and Summer. In certain years, we observe notable increases, reaching as high as 20%, while in contrast, other years reveal declines of comparable proportions. This variability indicates that Alice Springs might undergo sporadic phases of heightened solar output under SSP245; however, these may be offset by similarly substantial declines in other seasons, especially concerning the PAS system. The high-emission scenario, SSP585, leads to significant reductions in solar PV output, particularly in the later decades of the century (2071–2099). The Boreal Summer (JJA) and Winter (DJF) seasons exhibit significant reductions, with some years experiencing losses greater than 20% for the single-axis system. Dual-axis systems demonstrate a degree of resilience under SSP585; however, they still show significant declines, especially from the year 2050 onward. The yearly projections offer a comprehensive overview of the effects observed throughout all seasons. The figure illustrates a fairly consistent trend for SSP126, with variations confined to a tight range around 0%, suggesting that the annual solar PV performance in Alice Springs is likely to stay predominantly stable in a low-emission scenario. Nevertheless, for SSP245 and SSP585, the yearly effects show progressively worsening negative deviations as time goes on. SSP585 may have the worst effects, with annual power losses reaching as high as 30% by the year 2099.

In the distant future (2071–2099), the effects of climate change on solar PV power generation are expected to intensify, particularly under

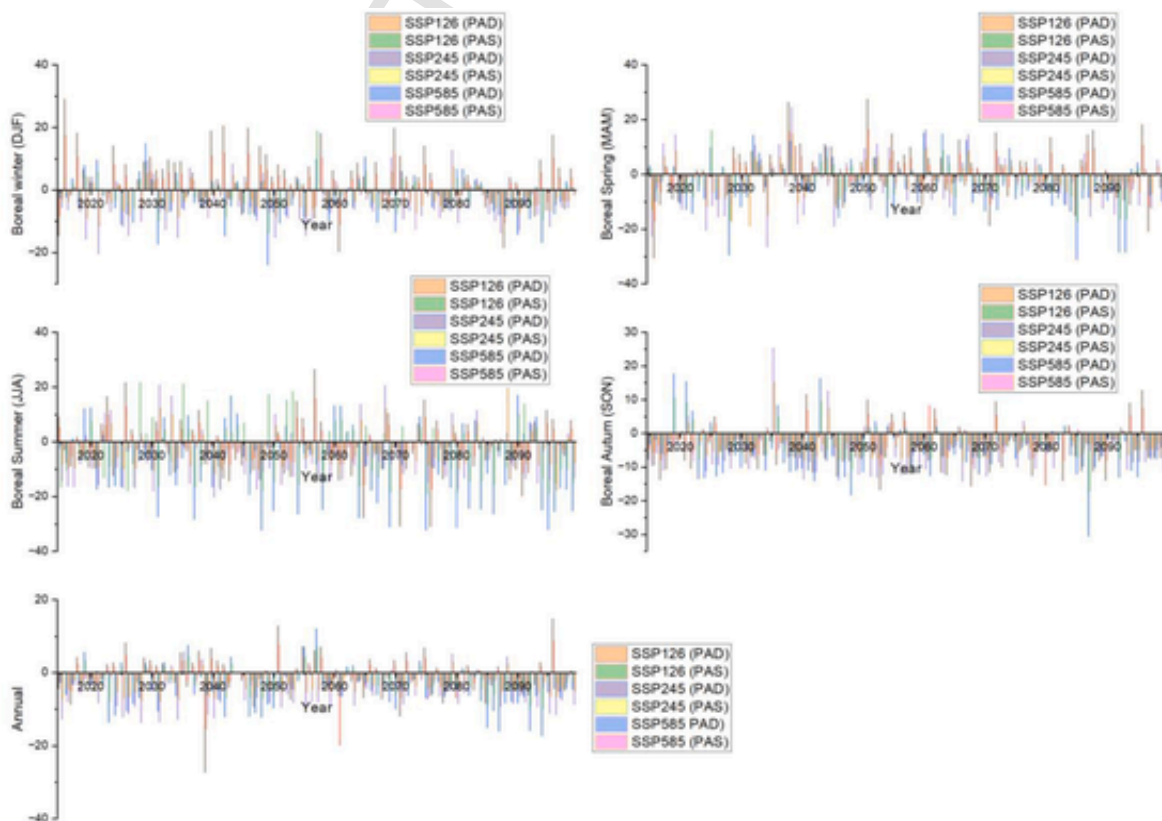


Fig. 10. Impacts of climate change on PAS and PAD in Alice Springs, Australia.

SSP585 scenarios. The data indicates that during both the Boreal Summer and Winter, the solar PV systems in Alice Springs may face significant losses, especially for PAS systems, with active power reductions surpassing 30% in certain years. This underscores the considerable susceptibility of solar photovoltaic systems to elevated emission trajectories, particularly in dry areas such as Alice Springs. In the short term (2015–2030), the data indicates that the impacts are expected to be relatively modest across all scenarios and seasons. The PAS and PAD systems exhibit variations around the 0% baseline, with sporadic positive and negative shifts observed. This indicates that, in the near future, solar PV systems in Alice Springs are expected to operate effectively, while the effects of climate change are anticipated to become more significant around the mid-to-late century.

Throughout the various seasons, Boreal Autumn (SON) exhibits the lowest variability in projections for both PAS and PAD systems, irrespective of the emission scenario. This indicates that solar PV systems in Alice Springs could exhibit greater stability during the autumn months, likely as a result of reduced temperature fluctuations and more uniform solar irradiance.

The distinction in resilience between single-axis and dual-axis tracking systems is a significant finding. Dual-axis systems typically demonstrate superior performance compared to single-axis systems by providing more consistent power outputs across various emission scenarios. When the SSP585 scenario has a lot of emissions, this effect is especially clear: the dual-axis systems have less negative deviations throughout all seasons than the single-axis systems.

Fig. 10 also highlights notable fluctuations from year to year in the anticipated effects on solar PV power generation. Even under identical emission scenarios and timeframes, the variations in power output can differ significantly from one year to the next, especially in SSP585. This variability highlights the intricate nature of forecasting solar PV performance amidst shifting climate conditions, where elements such as cloud cover, temperature extremes, and humidity can fluctuate unpredictably from one year to the next.

Considering the anticipated reductions in solar PV power generation amid elevated emission scenarios, it may be essential to implement adaptation strategies to alleviate the effects on energy production in Alice Springs. The robustness of dual-axis tracking systems indicates that investing in more sophisticated photovoltaic technologies may mitigate the adverse effects of climate change, especially in areas with significant solar potential such as Alice Springs.

Fig. 10 highlights the significant advantages of robust climate mitigation strategies, particularly evident in the SSP126 scenario. Maintaining lower levels of global warming could help Alice Springs prevent the notable reductions in solar PV performance anticipated under SSP245 and SSP585. This highlights the critical need for worldwide initiatives aimed at decreasing greenhouse gas emissions to safeguard renewable energy systems.

Although the figure offers significant insights into possible future effects, it is crucial to recognize the uncertainties that are intrinsic to climate forecasts. Elements like innovations in technology, specific local microclimates, and unexpected shifts in policy could significantly impact the real results. Consequently, the scenarios illustrated in the figure ought to be regarded as credible possibilities rather than absolute forecasts.

The interaction between temperature and solar irradiance stands out as a significant factor influencing the anticipated variations in solar PV power output. Increased temperatures may diminish the performance of photovoltaic cells, and variations in cloud cover and atmospheric conditions can influence the solar energy that reaches the panels. The illustration indicates that these elements hold significant sway during the Boreal Summer and Winter periods.

In evaluating these projections against other regions, it is evident that Alice Springs is likely to face milder reductions in solar PV performance in contrast to areas situated at higher northern latitudes. For ex-

ample, areas characterized by fluctuating weather conditions or increased cloudiness might experience more significant declines in solar PV performance, even when faced with comparable emission scenarios. Fig. 10 highlights the critical need for localized research to comprehend the effects of climate change on renewable energy infrastructures. Although global models offer a general perspective, localized research such as this analysis for Alice Springs is essential for crafting precise adaptation strategies that consider the distinct climatic characteristics of particular areas.

The anticipated reductions in solar PV output in Australia, particularly under elevated emission scenarios, may pose considerable challenges for energy security and the stability of the grid, necessitating careful consideration by energy planners and policymakers. Expanding energy sources and improving grid infrastructure to manage the fluctuations in solar power generation is likely to gain significance.

The possible reductions in solar PV power generation may also lead to wider environmental and societal consequences. The diminished performance of renewable energy systems may hinder the shift from fossil fuels, resulting in increased greenhouse gas emissions and worsening climate change impacts. Furthermore, areas that depend significantly on solar energy, such as Alice Springs, could encounter increased energy expenses and diminished energy reliability.

Fig. 10 presents a detailed analysis of the anticipated effects of climate change on solar PV systems in Alice Springs across various emission scenarios. The effects are modest under low-emission scenarios such as SSP126, but elevated emission pathways like SSP585 may lead to considerable reductions in solar PV power generation, especially for single-axis tracking systems. Dual-axis systems present a more robust alternative; however, these systems still face vulnerabilities due to the impacts of changing climatic conditions. Consequently, the implementation of both mitigation and adaptation strategies will be essential for securing the sustainable future of solar energy in Alice Springs and comparable areas.

3.9. Adapting concentrated photovoltaic systems to climate futures: insights from SSP126, SSP245, and SSP585 projections on operational lifespan and efficiency

Climate change projections under SSP126, SSP245, and SSP585 scenarios offer critical insights into the potential impacts on the operational lifespan and efficiency of CPV systems, providing a nuanced understanding of future challenges and opportunities. Under SSP126, where stringent climate mitigation policies limit global temperature rise to 1.5 °C, CPV systems could benefit from relatively stable environmental conditions, leading to less wear and tear on components and more consistent operational efficiency over their lifespan. This stability presents an opportunity to maximize system reliability and minimize maintenance costs, particularly for dual-axis systems (PAD), which already demonstrate adaptability across diverse climates. In contrast, SSP245, characterized by moderate climate forcing and a 2.5 °C temperature rise, introduces increased variability in solar irradiation and ambient conditions. Such variability could accelerate the degradation of CPV components, such as mirrors, lenses, and tracking motors, reducing their operational lifespan and efficiency. These challenges underscore the need for materials and technologies capable of withstanding dynamic conditions, as well as adaptive maintenance strategies to sustain performance. Under SSP585, where limited mitigation efforts result in a temperature rise of 3.5 °C, CPV systems face heightened risks from extreme climate phenomena, including prolonged heatwaves, increased cloud cover, and higher relative humidity. These factors could exacerbate material degradation, tracking inaccuracies, and overheating of photovoltaic cells, significantly reducing both the efficiency and lifespan of the systems. However, these challenges also create opportunities for innovation in system design, such as integrating advanced cooling mechanisms, resilient materials, and AI-driven tracking adjust-

ments to counteract environmental stresses. Additionally, SSP585 underscores the urgency of incorporating robust energy storage systems to offset the irregular power generation expected under extreme climate scenarios. In all cases, detailed evaluations of how well a CPV system works under expected conditions can help create strategies that work best in that place. This makes the system more resilient and opens up new ways to use renewable energy as the climate changes.

3.10. Comparison of current climate change impacts on solar PAD and PAS output in Alice Springs, Australia, as well as related studies

The examination of the active power produced by single-axis tracking photovoltaic systems (PAS) versus dual-axis tracking systems (PAD)

across various emission scenarios in Alice Springs, Australia, provides essential understanding of the enduring effects of climate change on solar energy generation. This study examines three distinct shared socio-economic pathways (SSP126, SSP245, and SSP585), each illustrating varying levels of climate change influenced by global emissions and advancements in technology, thereby offering an in-depth perspective on the future potential of solar energy. The projections encompass various future timeframes (2015–2030, 2031–2050, 2051–2070, 2071–2099) and seasonal fluctuations (DJF, MAM, JJA, SON, and ANN), emphasizing the potential benefits and obstacles for solar energy production in this area.

In Alice Springs, Australia, Table 14 shows how well Single-Axis Tracking Solar Photovoltaic Systems (PAS) and Dual-Axis Tracking So-

Table 14

Comparison of current climate change impacts on solar PAD and PAS output in Alice Springs, Australia, along with relevant studies.

Study/Region	Scenario	Reference Period	Projection Period	PV Output Projections/Impacts	Findings on Single-Axis (PAS)	Findings on Dual-Axis (PAD)
Present Study (Alice Springs, Australia)	SSP126	1984–2014	2015–2099	Early positive impact, but decline by 2071–2099, especially in annual averages	Annual average decline: 3.775% by 2071–2099. Modest early increases during some seasons (DJF, SON) but long-term declines observed.	Dual-axis tracking system declines less (–0.578% by 2071–2099). Slight seasonal benefits in MAM and JJA but still subject to future decline.
Present Study (Alice Springs, Australia)	SSP245	1984–2014	2015–2099	Moderate emissions cause severe reduction in power output	PAS sees a more significant reduction under SSP245, with a decline of –2.686% in annual average output by 2071–2099.	PAD declines sharply (–5.001% by 2071–2099), indicating sensitivity to moderate emissions.
Present Study (Alice Springs, Australia)	SSP585	1984–2014	2015–2099	High emissions result in the most severe output declines.	Under SSP585, PAS declines by –3.294% in the early periods (2015–2030) and by –2.857% by 2071–2099.	PAD exhibits stronger declines, –5.731% (2015–2030), worsening further to –4.813% by 2071–2099, signaling higher vulnerability to emissions.
(Ghanim and Farhan, 2023), Iraq	RCP4.5 and RCP8.5	2000–2020	2081–2100	Annual PV output changes between –0.3% and 8.1% under RCP4.5	Slight impact under lower-emission scenarios. More favorable outcomes under RCP4.5 compared to high emissions (RCP8.5).	Comparable or slightly worse trends observed, especially under high-emission scenarios like RCP8.5.
(Huber et al., 2016), Worldwide	SRES A1B	1995–1999	2035–2039	Solar PV output could increase by 10% in Europe, decrease by 10% in Africa	Positive impacts in Europe could be echoed in Australia's PV potential under milder scenarios, but Middle Eastern trends may worsen PAS.	Slight decline in African PV output and similar trends projected for Alice Springs under dual-axis, especially under SSP585.
(Dutta et al., 2022), Worldwide	SSP126, SSP245, SSP585	1984–2014	2015–2100	Under SSP585, PV in Algeria could decrease by 10% during DJF season	PAS might show similar performance declines during DJF, especially under SSP585 in Alice Springs, reflecting African projections.	PAD, though more efficient in DJF, will still see a minor decline in output compared to single-axis tracking in similar emissions scenarios.
(Crook et al., 2011), Worldwide	SRES A1B	1980–1999	2010–2080	Solar PV reduction in Nigeria and Africa, particularly during MAM and DJF	For Alice Springs, PAS performance likely mirrors this study, with marked decreases in MAM and DJF under scenarios with moderate emissions.	Dual-axis systems may fare better in JJA, but will still experience reductions in output during periods like MAM and DJF.
(Ha et al., 2023), Africa	SSP126, SSP585	1976–2005	2071–2100	Reduction in solar PV output in central, western, and eastern Africa by 2071–2100	Alice Springs sees a similar trend under SSP585, with declining PV potential in later periods (especially in MAM and DJF).	Under SSP126, PAD systems are projected to experience fewer declines than PAS, with variations in seasonal performance, especially in MAM.
(Gaetani et al., 2014), Europe & Africa	SRES A1B–B2	1991–2010	2030–2050	Solar PV decreases up to 7% in Eastern Europe and North Africa	Similar to Europe's decreases, Alice Springs' PAS output may reduce more significantly under slow transition scenarios like SSP245.	Eastern Mediterranean and North African trends hint at reductions in PAD systems comparable to SSP585 in Australia.
(Fant et al., 2016), Southern Africa	SRES A2 & B1	1979–2009	2045–2055	Minor changes in solar PV output with seasonal variability	Alice Springs PAS systems might experience seasonal fluctuations, with output reductions similar to Southern Africa, especially in JJA.	PAD systems may perform better during the JJA season but will still experience overall reductions in the long term.
(Bazyomo et al., 2016), West Africa	RCP8.5	2006–2015	2006–2100	Minimal increases and decreases in PV output	Alice Springs' PAS output under SSP585 mirrors Bazyomo's West Africa findings, with slight seasonal variations but overall declines.	PAD may show more resilience in early decades, but will also succumb to significant long-term reductions under high-emission scenarios.
(Feron et al., 2020), Worldwide	RCP4.5 & RCP8.5	1961–1990	2036–2065	Slight increases in central Europe, decreases in Australia	Southeast Australian PV trends match Alice Springs' decline for PAS, notably under RCP8.5 (–2% decline expected in southeastern Australia).	PAD experiences a comparable decline in southeastern Australia, with slightly worse impacts in Alice Springs under SSP585.
(Nwokolo et al., 2023), Nigeria	SSP126	1984–2014	2015–2099	0.058% increase using a-Si, 0.008% increase using p-Si	These increases, though minor, indicate the potential for slight improvements in PAS performance under low-emission scenarios like SSP126.	Similar to PAS, slight gains may be realized in PAD systems, particularly with advanced solar materials like a-Si under SSP126.
(Nwokolo et al., 2023), Nigeria	SSP585	1984–2014	2051–2099	0.995% increase using p-Si	Positive impacts are unlikely in Alice Springs due to harsher emission scenarios, causing PAS to continue its declining trend.	PAD may experience slightly less of a decline than PAS, but both systems are projected to see negative output trends in the long term.

lar Photovoltaic Systems (PAD) worked in different emission scenarios (SSP126, SSP245, SSP585), during different times of the year, and in the future that is expected. Table 14 further elucidates findings from analogous research conducted in various regions, emphasizing the impacts of climate change on solar PAS and PAD output.

In the context of the SSP126 scenario, which envisions a swift shift towards low-carbon energy solutions, the effects on PAS and PAD in Alice Springs appear to be quite limited in the short term. During the initial years spanning 2015 to 2030, minor increases in solar energy output are anticipated, especially in specific seasons like DJF (summer). For example, PAD exhibits a slight enhancement attributed to beneficial weather patterns in the warmer seasons. By the close of the century (2071–2099), annual averages for both PAS and PAD start to exhibit declines, with PAS facing a more pronounced reduction of -3.775% , in contrast to -0.578% for PAD. This indicates that although the dual-axis system's enhanced sun-tracking capabilities help reduce certain losses, the overarching trend points to a steady decline in solar energy production as the impacts of changing climate conditions become more pronounced.

The SSP245 scenario, characterized by moderate emissions and a gradual technological transition, results in significant reductions in power generation for both PAS and PAD. Between 2071 and 2099, PAS is projected to see a reduction of -2.686% , whereas PAD is expected to experience a more significant drop of -5.001% . The observation that dual-axis systems experience more pronounced reductions in this scenario can be linked to their heightened sensitivity to fluctuations in solar irradiance. This is particularly important in Alice Springs, where variations in cloud cover and temperature under a moderate emissions scenario may impede the efficiency of solar tracking systems. The results indicate that insufficient advancements in technology and policy changes may intensify the adverse effects of climate change on solar energy production.

Under the SSP585 scenario, characterized by elevated emissions and minimal mitigation efforts, the reductions in solar energy output are significant and evident early in the timeframe (2015–2030). The forecast indicates a decrease of -3.294% for PAS, whereas PAD is expected to experience a decline of -5.731% . The situation deteriorates further in the subsequent decades, with PAS declining by -2.857% and PAD by -4.813% during the period of 2071–2099. The fact that dual-axis systems show bigger drops in performance under the SSP585 scenario shows how sensitive they are to extreme weather, which can make it harder to precisely track the sun and lower the efficiency of power generation. The findings highlight the critical necessity for strong climate policies and technological innovations to avert significant declines in solar energy capacity.

Analyzing these findings alongside earlier research from various areas, including the Middle East and Africa, uncovers comparable patterns of diminishing solar PV performance attributed to shifts in climate conditions. According to research by Ghanim and Farhan (2023) using CMIP5 projections under the RCP4.5 and RCP8.5 scenarios, there could be a reduction of as much as 5.1% by the end of the century under the RCP8.5 scenario. In a similar vein, Crook et al. (2011) noted declines in solar PV output throughout Africa, with Nigeria being no exception, under both moderate and high-emission scenarios (SRES A1B). The results align with the anticipated decreases in PAS and PAD in Alice Springs, highlighting the pervasive challenge that climate change presents to solar power generation.

In areas such as Southern Africa, research conducted by Fant et al. (2016) reveals seasonal fluctuations in photovoltaic output, characterized by a modest rise during the winter months and a decline in the summer months. The seasonal dynamics observed in Alice Springs reveal that certain periods, particularly DJF, may yield short-term benefits. However, the overarching trends indicate a decline in solar power potential over the long term, particularly in high-emission scenarios such as SSP585.

Moreover, in Australia, Nwokolo et al. (2023) emphasize that the reduction in photovoltaic potential may have considerable consequences for energy generation and usage, especially within the SSP245 scenario. The current research reinforces these conclusions, indicating that in the absence of swift advancements in technology, Australia may experience considerable declines in its solar energy production, especially from advanced systems such as PAD, which depend on precise solar tracking.

From a technological standpoint, the selection of various photovoltaic materials may impact these results. For example, Nwokolo et al.'s (Nwokolo et al., 2023) research in Nigeria shows that using amorphous silicon (a-Si) and polycrystalline silicon (p-Si) materials in different SSP scenarios leads to small improvements in the performance of solar photovoltaics. The materials exhibit varying reactions to fluctuations in irradiance and temperature, with a-Si demonstrating a marginally superior resilience to elevated temperatures. The current research conducted in Alice Springs highlights the critical need for selecting appropriate technology in response to evolving climatic conditions, even though it does not distinguish between different PV materials.

Interestingly, while certain regions, such as Western Europe, are anticipated to experience modest increases in solar PV output as a result of changing climatic conditions (Gaetani et al., 2014), locations like Alice Springs encounter greater difficulties. The elevated irradiance levels and significant temperature variations in Australia intensify the effects of climate change on solar power systems. Consequently, both PAS and PAD systems in Alice Springs demonstrate greater declines in efficiency when contrasted with areas that have more temperate climates.

Furthermore, the results of this research align with international investigations. Huber et al. (2016) observed a possible 10% rise in solar PV output in Europe by the years 2035–2039 under the SRES A1B scenario, whereas Africa could experience a 10% decline. The observable trend of diminished solar energy production as a result of climate change is clear in the comparative forecasts for Alice Springs. This highlights the critical need for tailored adaptations and innovations to address these challenges effectively.

The study underscores the varying effects of climate change on single-axis (PAS) and dual-axis (PAD) tracking systems in Alice Springs. Although PAS demonstrates greater resilience, PAD, with its enhanced tracking ability, seems to be more susceptible to variations in solar irradiance driven by climate factors, especially in moderate-to-high-emission contexts. The results highlight the urgency for swift technological progress, robust climate strategies, and the meticulous choice of photovoltaic materials to guarantee the sustainable performance of solar energy systems in an evolving climate. This study offers a significant comparison with global analyses, emphasizing the common and localized challenges encountered by solar energy amid changing climatic conditions.

4. Adaptive strategies and technological innovations for optimizing solar CPV systems

Table 15 shows new adaptive techniques and technologies that are meant to make photovoltaic (PV) systems more efficient. It focuses on single-axis (PAS) and dual-axis (PAD) tracking setups in Alice Springs, Australia. Adaptive layout design utilizes algorithms to ascertain appropriate panel spacing and orientation, minimizing shading losses and maximizing irradiance capture. PAS systems enhance energy capture, whereas PAD systems optimize dynamic alignment capabilities, resulting in substantially greater energy yields. Innovative materials include weather-resistant coatings that improve longevity and mitigate performance decline due to UV exposure, dust, and sand, which is essential in the harsh conditions of Alice Springs. AI-driven intelligent control systems provide real-time optimization of tracking and energy harvesting operations. These technologies quickly adapt to changing conditions in the environment, making single-axis tracking more effective in PAS systems and letting PAD systems move across

Table 15
Adaptive strategies and technological innovations for optimizing solar CPV systems.

Category	Innovation/Strategy	Description	Application to PAS Systems	Application to PAD Systems	Impact on Solar PV/CPV Performance
Layout Optimization	Adaptive Layout Design	Uses algorithms to optimize panel spacing and orientation for minimizing shading and maximizing exposure.	Enhances energy capture efficiency; limited to single-axis optimization.	Maximizes dual-axis alignment for optimal irradiance capture throughout the day.	Reduces shading losses; improves energy yield; ensures efficient land use in large-scale installations.
Material Enhancements	Weather-Resistant Coatings	Advanced coatings to protect panels from UV degradation, sand abrasion, and chemical erosion.	Extends panel lifespan; reduces performance degradation in arid environments.	Greater benefit for panels with dynamic movement and higher exposure to environmental elements.	Improves durability; reduces maintenance costs; ensures consistent energy output over time.
Control Systems	Intelligent Control Systems	AI-powered systems for dynamic optimization of tracking and energy harvesting processes.	Provides improved single-axis tracking adjustments based on real-time conditions.	Optimizes dual-axis tracking for maximum irradiance capture and minimizes response lag.	Enhances energy harvesting efficiency; reduces energy losses; adapts quickly to changing environmental conditions.
Energy Storage	Battery Storage Systems	Advanced lithium-ion or flow batteries for storing excess energy during peak production periods.	Reduces intermittency in power output by storing energy during high irradiance periods.	Higher compatibility due to increased energy output and stability from dual-axis tracking.	Ensures energy availability during low irradiance; improves grid stability and reliability.
Hybrid Storage Solutions	Integrated Storage with Thermal and Mechanical Systems	Combines battery storage with thermal or mechanical storage (e.g., flywheels or compressed air).	Limited efficiency due to single-axis tracking's lower peak output.	Fully exploits dual-axis tracking's higher energy output for efficient hybrid storage utilization.	Reduces reliance on battery-only systems; provides long-term, cost-effective energy storage solutions.
Energy Management	Predictive Energy Management Systems	Utilizes machine learning models to predict energy demands and optimize power distribution.	Supports grid integration with smoother energy outputs from single-axis systems.	Maximizes grid efficiency by leveraging stable and consistent energy generation from dual-axis systems.	Minimizes grid overload risks; enables better demand-response strategies; improves integration of renewable energy.
Thermal Management	Phase Change Material (PCM)-Based Cooling	Employs PCMs to absorb excess heat and regulate panel temperatures during peak solar hours.	Reduces thermal inefficiency in PAS systems but limited by single-axis configuration.	Enhances efficiency by synergizing with dual-axis tracking to maintain lower operating temperatures.	Mitigates performance losses due to overheating; extends PV lifespan in hot climates like Alice Springs.
Dust Mitigation	Anti-Soiling Technologies	Self-cleaning surfaces or automated cleaning systems to remove dust and debris from panels.	Maintains performance in arid, dusty climates but less effective for horizontal panel orientations.	Significantly enhances performance by maintaining clean surfaces for dual-axis systems with varied angles.	Reduces efficiency losses due to soiling; decreases cleaning and operational costs; improves long-term reliability.
Grid Integration	Smart Grid Connectivity	Dynamically adjusts power outputs to match grid requirements, minimizing curtailment during peak periods.	Supports integration of PAS-generated energy but less effective for variable outputs.	Provides stable and reliable energy outputs for better grid adaptability.	Enhances grid stability; ensures efficient renewable energy distribution; supports peak-demand management.
Concentrated PV (CPV)	High-Precision Concentrators	Uses optical lenses or mirrors to focus sunlight on small, high-efficiency PV cells.	Limited performance boost due to fixed-angle tracking.	Ideal for dual-axis systems to maintain precise solar alignment for maximum energy concentration.	Achieves higher energy output per unit area; reduces material costs for large-scale solar projects.

two axes for the best performance. Innovations in energy storage, including better lithium-ion batteries and hybrid systems that incorporate thermal or mechanical storage, effectively tackle intermittency concerns. Dual-axis systems, due to their superior energy production, efficiently enhance these storage technologies, guaranteeing a constant power supply even during periods of low irradiance. Predictive energy management systems improve grid integration by anticipating energy demands and optimizing power distribution, hence reducing the hazards of overloads and curtailment. Table 15 highlights the significance of environmental resilience via solutions such as anti-soiling technology and phase change material (PCM)-based cooling. These advancements reduce dust collection and overheating, which are common in the region's arid and dusty environment. PAD systems derive significant advantages from these technologies, preserving clean and cooled surfaces via dynamic alignment and optimum panel angles. The adaptive strategies and technological innovations outlined in Table 15 offer a comprehensive method for enhancing the efficiency, reliability, and sustainability of photovoltaic systems in Alice Springs, tackling both technical and environmental challenges while facilitating scalable renewable energy solutions.

5. Implications of the impact of climate change on single-axis (PAS) and dual-axis (PAD) systems

Considering the anticipated reductions in active power under SSP245 and SSP585, it will be essential to achieve technological

progress in PAS systems to sustain solar power generation in Alice Springs. The advancement of photovoltaic materials that exhibit greater resistance to elevated temperatures and humidity-induced degradation could help alleviate certain adverse effects associated with climate change. Furthermore, incorporating energy storage solutions into PAS systems may provide a buffer against seasonal fluctuations, thereby guaranteeing a more dependable supply of solar energy, even in times of diminished irradiance.

PV systems exhibit enhanced efficiency under ideal circumstances; however, they face increased susceptibility to shifts in climate due to their dependence on accurate solar irradiance tracking. To tackle the expected performance declines, advancements in dual-axis tracking technologies will be essential. For example, sophisticated tracking algorithms that consider cloud cover and atmospheric variables could enhance performance in the face of evolving climate scenarios. Moreover, the incorporation of energy storage and hybrid energy systems may enhance the stability of power generation during times of diminished solar resources.

To address the effects of climate change on PAS and PAD systems, implementing adaptation strategies will be crucial. Alongside technological progress, incorporating a variety of renewable energy sources, including wind and geothermal, may effectively mitigate the fluctuations associated with solar power generation. Enhancements to grid infrastructure, such as the establishment of microgrids and increased energy storage capabilities, will be essential for maintaining the reliability

of solar power in Alice Springs as we navigate future climate conditions.

The SSP126 scenario emphasizes swift decarbonization and a sustainable future, creating optimal conditions for solar power systems in Alice Springs. In this scenario, the effects of climate change are reduced, resulting in only slight decreases in active power for both PAS and PAD systems. Nevertheless, realizing this scenario necessitates robust political commitment and substantial investment in renewable energy technologies, infrastructure, and strategies for climate mitigation. It is essential for decision-makers to focus on shifting towards low-carbon energy sources to safeguard solar power resources and guarantee sustainable energy security in the future.

The SSP245 scenario illustrates a future in which emissions are projected to increase until the middle of the century, followed by a decline, presenting considerable challenges for solar power systems. The ongoing increase in temperatures and heightened climate variability, though less severe than in SSP585 scenarios, leads to significant decreases in active power for both PAS and PAD systems. It is imperative for decision-makers to acknowledge the necessity for enhanced climate mitigation strategies to prevent the most severe consequences of SSP245 and safeguard investments in solar energy.

The SSP585 scenario outlines a dire future characterized by inadequate climate mitigation efforts, leading to profound climate consequences, such as substantial reductions in solar power generation. In this scenario, both PAS and PAD systems experience ongoing declines in active power, especially during the winter and autumn months. The pronounced responsiveness of solar systems to climate fluctuations under SSP585 highlights the critical necessity for worldwide initiatives aimed at emission reduction and a shift towards renewable energy sources.

The sustained decrease in solar power generation under SSP245 and SSP585 presents significant challenges for energy security in Alice Springs and similar areas dependent on solar energy sources. The decline in the performance of PAS and PAD systems heightens the potential for energy shortages, especially in the winter and autumn months when solar irradiance is diminished. To tackle these pressing issues, it is essential for decision-makers to focus on enhancing investments in energy storage, improving grid infrastructure, and exploring alternative renewable energy sources to guarantee a stable and dependable energy supply.

The anticipated decreases in active power under SSP245 and SSP585 will carry considerable economic consequences for the solar energy sector in Alice Springs. The diminishing efficiency of PAS and PAD systems may lead to higher costs in solar energy generation, which could jeopardize the economic feasibility of solar power initiatives. It is essential for decision-makers to consider the economic risks associated with climate and energy policies. They should focus on investing in technology and infrastructure that can alleviate the effects of climate change on solar energy generation.

The reduction in solar power output under SSP245 and SSP585 not only carries economic consequences but also poses considerable social challenges, especially for communities that depend significantly on renewable energy for their electricity requirements. The decreasing reliability of solar power may lead to increased energy costs, which could have a more significant impact on low-income households. It is essential for those in positions of authority to craft climate and energy policies that safeguard at-risk communities and foster equitable access to energy amidst the challenges posed by climate change.

The anticipated effects of climate change on solar power systems in Alice Springs underscore the critical need for worldwide climate initiatives. The SSP126 scenario, characterized by a swift shift towards a low-carbon future, presents the most promising opportunity to mitigate the adverse effects of climate change on solar energy generation. Nonetheless, realizing this future necessitates unified international initiatives to

lower emissions, invest in renewable energy innovations, and encourage sustainable development practices.

In light of the pressing issues related to climate change, there exist considerable prospects for advancements in solar power technologies. Innovations in photovoltaic materials, tracking technologies, and energy storage options may play a crucial role in alleviating the effects of climate change on PAS and PAD systems, thereby securing a dependable solar energy supply amidst evolving climate conditions. It is essential for decision-makers and industry executives to focus on advancing research and development in these domains to fully harness the capabilities of solar energy amid the challenges posed by climate change.

Local adaptation strategies will play a vital role in enhancing the resilience of solar power systems in Alice Springs as we navigate future climate scenarios. It is essential for decision-makers to collaborate effectively with industry experts and researchers to create customized strategies that tackle the unique environmental challenges encountered in the area. Potential initiatives may encompass investments in sophisticated grid infrastructure, energy storage solutions, and hybrid renewable energy systems, alongside strategies aimed at enhancing energy efficiency and managing demand.

The influence of climate change on PAS and PAD systems in Alice Springs fluctuates considerably based on the emission scenario and the timeframe considered. The negative effects seem to be relatively minor in the context of the low-emission SSP126 scenario, along with some seasonal variations in active power. Nonetheless, in the scenarios of SSP245 and SSP585, the reductions in active power are significantly more evident, especially for PAD systems. The results highlight the critical necessity for effective climate mitigation strategies and advancements in technology to secure the sustainable future of solar energy systems amid a shifting climate landscape.

6. Summary of various sections and key conclusion reached

The descriptive statistics for PAS and PAD underscore significant environmental factors that affect active power predictions for solar photovoltaic systems in Alice Springs. Both single-axis and dual-axis systems exhibit comparable patterns in global solar irradiation and extraterrestrial radiation, with minor variations in their respective levels of variability. Interestingly, PAD demonstrates a higher average clearness index (k_t), indicating a clearer atmosphere conducive to solar energy collection, alongside a more stable range of relative humidity (RH). In contrast, PAS displays greater variability in RH, suggesting potentially more difficult conditions for reliable power generation. The ambient temperature exhibits stability across both systems, showing no notable discrepancies. The findings highlight the critical need to comprehend local environmental variability to enhance the accuracy of energy yield prediction models. As predictive models get better, these meteorological variables need to be added to improve the accuracy of forecasts and make the system work better in a wider range of weather conditions. The subsequent insights derived from the analysis of descriptive statistics are as follows:

- Both PAS and PAD systems exhibit comparable ranges of global solar irradiation (H) and extraterrestrial solar radiation (H_0), reflecting a stable solar resource in Alice Springs.
- PAD shows a marginally elevated mean clearness index (0.7550), indicating a clearer atmosphere and enhanced possibilities for solar energy collection in contrast to PAS (0.7484).
- Although the average ambient temperature is similar for both systems, PAD exhibits a slightly elevated mean of 21.83 °C in contrast to PAS, which stands at 21.14 °C, suggesting a marginally warmer setting.
- There is a notable disparity in relative humidity (RH) between PAS and PAD. PAS exhibits greater relative humidity variability at

85.11% in contrast to PAD's 72.64%, indicating that PAD maintains more stable moisture conditions.

- The larger standard deviation (7.7) for AT suggests a greater variability in temperature, which could influence system performance under more extreme conditions.
- The increased clearness index and reduced relative humidity in PAD could promote more stable and efficient power generation, as ideal irradiance and diminished atmospheric moisture are advantageous for solar systems.
- The PAD exhibits a narrower range (25.87) in contrast to the PAS (26.39), suggesting that the dual-axis system operates under more stable environmental conditions.
- PAD exhibits greater variability in both H and AT relative to PAS, indicating somewhat less predictable weather patterns that could influence performance consistency.
- Predictive models for both systems must incorporate these environmental factors, with PAD possibly needing modifications to account for increased temperature variability and decreased humidity.
- The analysis reveals that although both systems experience comparable irradiation levels, the dual-axis system (PAD) enjoys enhanced clearness and reduced humidity, which may lead to improved performance in solar energy generation.

When it comes to predicting active power for both single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic systems, CARIMA-SARIMA-LG is much better than traditional and machine learning methods. The hybrid model demonstrates remarkable accuracy, achieving R^2 values of 0.983 and 0.977 for training and testing on PAS and 0.977 and 0.969 for PAD. This performance clearly surpasses that of MLP, RBF, and ensemble techniques such as BOT and BAG. The MLP's R^2 values of 0.305 for PAS and 0.597 for PAD indicate a limited ability to accurately capture complex time-series dynamics, demonstrating the suboptimal performance of these machine learning models. Although statistical models like SARIMA continue to outperform ensemble techniques like BOT and BAG, they do show some improvements. The CARIMA-SARIMA-LG model shows that using both statistical time-series forecasting methods and logistic distribution together makes predictions that are more accurate. It can account for both seasonal and controlled changes in solar power generation. This research underscores the significance of hybrid methodologies in enhancing the precision of energy forecasting, which is vital for the efficient functioning of solar farms and their integration into the grid. To conclude, the key findings are as follows:

- The CARIMA-SARIMA-LG Hybrid Model demonstrates superior performance compared to all other models in both PAS ($R^2 = 0.983$, testing $R^2 = 0.977$) and PAD ($R^2 = 0.977$, testing $R^2 = 0.969$), indicating remarkable predictive precision.
- A robust candidate exhibiting exceptionally high R^2 values in both PAS ($R^2 = 0.982$) and PAD ($R^2 = 0.976$), demonstrating the efficacy of SARIMA for time-series data, especially in training contexts.
- CARIMA demonstrates strong performance, achieving R^2 values of 0.973 (PAS) and 0.962 (PAD) throughout the training phase. The testing R^2 shows a slight decrease to 0.970 for PAS and 0.958 for PAD, indicating consistent and dependable predictive performance in both systems.
- The MLP (Multilayer Perceptron) demonstrates notably reduced R^2 values (0.305 in PAS and 0.597 in PAD) along with elevated RMSE, suggesting a decline in prediction accuracy when juxtaposed with statistical models and hybrid methodologies.
- The Radial Basis Function demonstrates the least effective performance, showing R^2 values of 0.273 (PAS) and 0.448 (PAD),

indicating its constrained capacity to manage the intricacies of active power prediction.

- The performance of both boosting and bagging ensemble models is comparable, yielding R^2 values of approximately 0.221 for PAS and 0.576 for PAD during the training phase. However, they still fall short when compared to SARIMA and the hybrid model.
- CARIMA-SARIMA-LG demonstrates the most favorable MAPE and RMSE values in both PAS and PAD, signifying its superiority in reducing percentage errors and residuals.
- In the evaluation process, CARIMA-SARIMA-LG consistently demonstrates a low RMSE (0.0229 in PAS and 0.0178 in PAD) and an exceptionally low MAPE (0.0034 in PAS and 0.0019 in PAD), showcasing its proficiency in generalizing to previously unobserved data.
- CARIMA-SARIMA-LG demonstrates the most favorable nRMSE among all models (0.0010 for PAS, 0.0003 for PAD), highlighting its exceptional efficiency in relation to the dataset's scale.
- The CARIMA-SARIMA-LG hybrid model demonstrates a significant edge over conventional and machine learning approaches by consistently yielding lower MAPE and RMSE, along with higher R^2 in both PAS and PAD, establishing it as the most dependable option for solar power forecasting.

A comparison of projected and actual outputs with error metrics (R^2 , MAPE, RMSE, and nRMSE) demonstrates superior performance in generating active power within PAS and PAD tracking systems across various locations, confirming the predicted CARIMA-SARIMA-LG hybrid model. PAD systems exceed PAS in accuracy and adaptability to varying environmental conditions, exhibiting lower MAPE ($< 3\%$) and nRMSE ($< 4\%$) values. The hybrid model effectively incorporates seasonal trends, variable external factors, and probabilistic analysis, yielding dependable forecasts for regions including Kano, Accra, Johannesburg, Nairobi, and Dar es Salaam. Its superior R^2 values (> 0.94) illustrate its ability to precisely represent the intricacies of solar power generation in demanding conditions. These findings underscore the model's transformative capacity for enhancing renewable energy planning, improving energy grid integration, and enabling data-driven decision-making. The CARIMA-SARIMA-LG model sets a benchmark for predictive accuracy, facilitating enhanced forecasting in global solar energy applications.

In Alice Springs, Australia, climate change makes it harder for both single-axis (PAS) and dual-axis (PAD) tracking solar photovoltaic systems to make electricity. The effects are different depending on the SSP emission scenario. Under SSP126, PAS and PAD experience relatively moderate changes, with early gains in some seasons (e.g., +3.145% PAS in DJF, 2015–2030) but declining trends toward the century's end (−3.775% PAS, ANN, 2071–2099). PAD shows similar trends, with initial improvements (+5.463%, DJF, 2015–2030) that later taper off. SSP245 leads to more pronounced losses in power generation for both systems, particularly in later periods, with PAS reaching −2.686% (ANN, 2071–2099) and PAD experiencing −5.001% (ANN, 2071–2099).

Under the high-emission SSP585 scenario, the declines are more severe, with PAS and PAD facing significant power reductions across all seasons, reflecting the impacts of extreme climate changes driven by unchecked emissions. PAS reaches −3.294% (ANN, 2015–2030) and PAD suffers even more at −5.731% (ANN, 2015–2030). These findings highlight the need for aggressive climate mitigation strategies and technological advancements in photovoltaic systems, as failure to address emissions could lead to substantial declines in solar energy output, threatening energy security and sustainability in solar-reliant regions like Alice Springs. Adaptation efforts, including enhanced solar tracking technologies and energy storage systems, will be crucial to maintaining energy resilience in the face of climate change.

In summary, the following are major findings:

- Under the SSP126 Scenario, a rapid transition to low-carbon energy shows modest impacts on PAS and PAD, with early positive changes in some seasons but a long-term decline by 2071–2099, particularly in annual averages (−3.775% for PAS, −0.578% for PAD).
- Under the SSP245 Scenario, moderate emissions lead to pronounced reductions in power generation, with PAS declining by −2.686% and PAD by −5.001% in the annual average by 2071–2099, highlighting the adverse effects of slow technological transitions.
- Under the SSP585 Scenario, high emissions cause severe declines in power output, with PAS falling by −3.294% and PAD by −5.731% in the early periods (2015–2030), worsening further by 2071–2099 (−2.857% for PAS, −4.813% for PAD).
- The impact of climate change varies across seasons, with the summer (DJF) generally showing more resilience, while the transition seasons (SON, MAM) experience greater declines under all SSP scenarios.
- The declines in active power intensify with time, particularly under SSP585, where losses are most severe in later periods (2051–2099), reflecting the long-term impact of unmitigated climate change.
- Dual-axis systems (PAD) are more susceptible to performance losses under all SSP scenarios, showing greater variability and steeper declines than single-axis systems (PAS).
- SSP126 shows the benefits of aggressive emission reductions, with lower declines in power generation compared to SSP245 and SSP585, emphasizing the need for urgent climate action.
- Alice Springs, a high solar potential region, remains vulnerable to climate shifts, underlining the need for adaptive strategies to sustain solar energy productivity.
- Declining active power in PAS and PAD poses risks to energy security in solar-dependent regions, necessitating innovations in system design and storage technologies.
- The findings advocate for immediate global efforts in climate mitigation and renewable energy innovation to minimize the adverse effects on solar photovoltaic systems.

7. Research limitations

Despite the promising results of this study, several research limitations warrant consideration. First, using historical weather data from Alice Springs and the chosen African cities might not fully capture future climate extremes or rapid changes in weather patterns. This could make it harder to apply the results to other places with different weather conditions. While the study incorporates key meteorological variables, it does not account for potential localized microclimatic effects, such as urban heat islands or terrain-related variations, which could influence the accuracy of solar power predictions in specific areas. The study also mostly looks at how to combine machine learning models with analytical methods. It doesn't look into how real-time operational factors, like system degradation, dirt buildup on panels, or operational inefficiencies, might affect the accuracy of long-term forecasts. Even though the hybrid CARIMA-SARIMA-LG model works very well, it might need to be tested in more places and for longer periods of time to make sure it is reliable and can adapt to different environmental conditions. Also, the SSP scenarios' projections are interesting, but they are based on assumptions about the paths of climate policy and emissions that are inherently uncertain and depend on geopolitical factors that could change over time. Finally, while the study emphasizes technological advancements in solar tracking and energy storage, it does not extensively analyze the economic feasibility or scalability of these solutions, particularly in low-income or developing regions, which could influence the practical implementation of the findings. These limitations highlight the need for further research that incorporates addi-

tional variables, extended time frames, and real-world operational data to refine forecasting models and improve their applicability across diverse settings.

CRediT authorship contribution statement

Samuel Chukwujindu Nwokolo: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anthony Umunnakwe Obiwulu:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Paul C. Okonkwo:** Writing – review & editing, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Rita Orji:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Theyab R. Alsenani:** Writing – review & editing, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Ibrahim B. Mansir:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Chukwuka Orji:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Research data link

European Centre for Medium-Range Weather Forecasts (ECMWF, <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>) provided the historical and projected climate datasets used in this study for providing the real datasets used throughout this study.

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Declaration of competing interest

The author declares that there is no conflict of interest regarding the publication of this paper.

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Data availability

I have linked the data source in the methodology section

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